Forecasting Intraday Realized Variance with Exogenous Features and Block Ridge Regression

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Abstract

We forecast thirty-minute-ahead S&P 500 realized variance on a near-24-hour panel of over a quarter million bars (1998–2024), asking what forty-one exogenous channels add to a HAR backbone, which estimator can extract it, and what structure the nonlinearity has. Every forecast is scored as its plain back-transform — the square-root-scale forecast squared and multiplied by its time-of-day baseline — with no recalibration. Two discipline findings precede the model findings: the representation of missing and stale data moves accuracy more than any modeling decision; and causal hyperparameter tuning on short validation tails is repeatedly worse than measured constants. The exogenous information is real but dense and weak: invisible to selection and unpenalized joint fits, recoverable only by shrinkage, and best handled, among the linear estimators, by a two-block ridge that shrinks the exogenous block far harder than the backbone. Gradient-boosted trees on the same design improve on the two-block ridge by more than the ridge improves on the benchmark. The two-block ridge — one exact rolling solve, two fixed penalties — improves on the benchmark in point estimate (Diebold–Mariano -1.7); with its two penalties re-selected causally it does so at DM -3.8.

1 Introduction

This paper forecasts thirty-minute-ahead realized variance for the S&P 500 complex on a near-twenty-four-hour panel running from January 1998 to April 2024 — over a quarter million scored half-hour bars, most of them outside regular trading hours. The question is not whether volatility is predictable; on this panel the target's own lag ladder carries the overwhelming share of the predictable variation; the lag ladder is the benchmark throughout. The questions are what, if anything, forty-one exogenous channels — higher-order moments, liquidity, cross-sectional aggregates, sentiment, implied volatility, and options order flow — plus four scheduled FOMC calendar channels overlaying that design, add to that backbone; which estimator class can extract it; and what structure the extractable nonlinearity has.

Every answer is measured out of sample, bar by bar, under one common evaluation protocol, and every claim is paired against the same per-bar least-squares benchmark (QLIKE 0.23806).

Four findings organize the paper. The first is about the data itself. On a mixed-frequency panel, the representation of missing and stale data is a larger determinant of forecast accuracy than any modeling decision we report: at any given bar, over a third of the exogenous design is forward-filled rather than freshly printed, and a construction that cannot distinguish the two produces single-window failures orders of magnitude larger than every model-class effect in the paper. The preparation pipeline — availability indicators computed before the fill, masked scaling statistics, neutral substitution — is therefore presented in full, with the incident record attached.

The second point is a convention of evaluation rather than a finding: models are fit on a variance-stabilized scale and scored on the raw variance scale, and every forecast is scored as its plain back-transform $\hat{y}^2 B$ (Section 4.2), with no smearing factor and no recalibration.

The third finding is about the exogenous information and the estimator jointly — they cannot be separated. The exogenous channels carry real, statistically significant information, but none of it is recoverable by an unshrunk fit at this width: selection and unpenalized joint fits fail, and only shrinkage recovers the signal. The head-to-head that follows locates the structure: under one causal tuning rule the penalty families are not separated on the full design (lasso minus ridge -0.00076 at DM -0.5), heavier ridge shrinkage lowers the loss steadily, the lasso's zeroes fall where zeroing is free, and no small set of columns or principal directions concentrates the signal. The coefficient structure is *dense but weak*, the regime of Giannone et al. [2021] and Shen and Xiu [2025] — many nonzero coefficients, none large — and the estimator built for it is ridge with a block-diagonal Tikhonov penalty: the strong low-dimensional backbone barely penalized, the wide exogenous block shrunk two orders of magnitude harder, with fixed penalties and no selection step anywhere.

The fourth finding is about nonlinearity. Gradient-boosted trees beat the paper's linear ladder: on the wide design the hedged tree bank improves on the two-block ridge by -0.01148 at DM -10.7, more than the ridge improves on the benchmark (-0.00203), and a decomposition of a tree fit locates most of its explainable variance in additive terms carried by the HAR block. The paper's linear ladder ends at the block-diagonal ridge — one solver, two penalties, refit exactly at every bar — which scores 0.23603 at DM -1.7 against the benchmark's 0.23806 with its two penalties fixed, and 0.23347 at DM -3.8 with each block's penalty re-selected causally. The paper closes by reading the forecasts against the last thirty minutes of the expiration-day chain. The last-30-min trade buys or sells one straddle at 15:30 ET on the sign of the gap between the forecast of the half hour's variance and the variance the straddle's quotes imply, and holds it to the cash settlement (Section 5.4). With every forecast used as its plain back-transform and scored on the same 866 days, every one of the paper's forecasts improves on selling the straddle every day, three of the eight significantly. Over the 112 forecasts of the closing study's master table, including forecasts fitted for the traded bar alone on two input sets, the trade does not order the models: against the paper's block-diagonal ridge 32 trade above it and 79 below, 0 and 0 of them significantly, and a better forecast of the half hour's variance is not by itself a better trade. Whether sizing on the magnitude of the gap improves on its sign turns on the sample, in point estimate (Appendix B.9); the trade's live form after the sample ends in April 2024 is not scored here.

A methodological theme runs through all four: what tuning buys is measured, not assumed. Every hyperparameter in the campaign is either a fixed constant or selected causally on a short forward validation tail, and the campaign measures what that selection buys. The answer depends on the estimator: weighting the tree experts beats selecting one (-0.00571 at DM -5.8), the causally tuned elastic net improves on causally tuned ridge (-0.00304 at DM -2.1), and re-selecting the two block penalties causally improves on fixing them (0.23347 against 0.23603). No constant is optimized on evaluation loss.

Roadmap. Section 2 places the paper in the literature. Section 3 defines the panel, the preparation pipeline, and the exogenous buckets. Section 4 fixes the evaluation framework — the benchmark, the testing machinery, and the back-transform — and then the methods: the estimators and the block constructions. Section 5 reports the results in the order summarized above. Section 6 concludes. The appendix carries the full arm-level master table and the protocol notes; Appendix B collects side analyses of the last-30-min trade, its P&L by cell, and the diagnostics parked from Section 5.4.

2 Related Works

Realized-volatility modeling. The paper's target and backbone descend from the literature on realized measures: realized variance as a sum of squared intraday returns [Andersen and Bollerslev, 1997, Barndorff-Nielsen and Shephard, 2004], its approximate long memory documented by Andersen et al. [2003], and the heterogeneous autoregressive model of Corsi [2009], whose daily/weekly/monthly averages we adapt to a base-2 intraday ladder. Refinements of the HAR backbone are numerous: measurement-error-dependent coefficients [Bollerslev et al., 2016], state-dependent time variation [Bekierman and Manner, 2018], decoupled short- and long-run behavior [Bennedsen et al., 2022], joint models of returns and realized measures [Hansen et al., 2012], and multiple-indicator intraday structures [Engle and Gallo, 2006]. We keep the backbone deliberately plain — per-bar least squares on the lag ladder — because the paper's questions concern what can be added to it, and its strength makes every addition a stringent test: on this panel the HAR backbone's coefficients are 94–97% directionally stable over two decades, so a predictor that survives next to it carries information the ladder does not.

Machine learning for volatility and return prediction. Tree ensembles are strong off-the-shelf estimators in the recent machine-learning literature on asset pricing and volatility [Gu et al., 2020, Christensen et al., 2023], typically in gradient-boosted form [Ke et al., 2017, Chen and Guestrin, 2016]. Interpretability machinery — functional ANOVA, additive and pairwise additive models [Lou et al., 2013], and Shapley-based attribution [Lundberg and Lee, 2017, Lundberg et al., 2018] — is usually applied *post hoc*. This paper uses the trees to measure what the linear design leaves unrecovered, and decomposes a tree fit to see where its explainable variance lies (Section 5.2).

Sparse selection versus dense shrinkage. The linear estimators are standard families — ridge [Hoerl and Kennard, 1970], lasso [Tibshirani, 1996], elastic net [Zou and Hastie, 2005], with the least-angle-regression path [Efron et al., 2004] behind our exact online lasso

solve — but the question they are put to has a now-substantial literature: whether economic signal is concentrated in a few predictors or spread thinly across many. Giannone et al. [2021] compare sparse and dense representations across macro, micro, and finance prediction problems and find that the posterior rarely concentrates on a sparse model — sparsity is largely an illusion, and dense methods (ridge, factor models) predict best. Kozak et al. [2020] reach the parallel conclusion for the cross-section of expected returns, where a dense, heavily shrunk stochastic discount factor dominates sparse characteristic selections. Shen and Xiu [2025] supply the estimation theory: in the weak-signal regime the zero predictor is asymptotically Bayes-optimal, ridge can improve on it while lasso cannot for any penalty, and the ranking extends to nonlinear methods (dense random forests over sparse boosted trees, ℓ_2- over ℓ_1-penalized networks); lasso's failure reflects signal *weakness*, not necessarily the absence of sparsity. The rare-and-weak detection literature [Donoho and Jin, 2004] formalizes the underlying phase structure, in which a dense regime of individually undetectable effects is separated from the sparse regime by a sharp boundary. Our head-to-head (Section 5.1) is the intraday-volatility instance of this phenomenon, measured under a stricter protocol than the literature's cross-validation standard: every penalty is selected causally on a short forward tail, and the comparison is scored on a consistent loss. That protocol isolates a mechanism the asymptotic results abstract from: on short validation tails the price of a hyperparameter mistake is asymmetric across families — benign for ridge, whose loss is flat in the penalty near the optimum, destructive for selectors, whose penalty is partly a selection dial — which is why the paper's final model carries fixed penalties rather than tuned ones.

Retransformation. A mean model estimated on a log or square-root scale and scored on the original scale is not inverted by the algebraic inverse: by Jensen's inequality the naïve back-transform is biased for the original-scale mean. The standard nonparametric correction is Duan [1983]'s smearing estimator. Manning [1998] and Manning and Mullahy [2001] place that correction inside the forecast: under heteroscedasticity a single smearing factor is itself biased, so the retransformation is a modeling choice rather than a reporting footnote. This paper fits on a diurnally-adjusted square-root scale and scores QLIKE on raw variance, so the same gap applies. Here the forecast is used uncorrected, as $\hat{y}^2 B$ (Section 4.2), so each arm carries its own retransformation bias into the QLIKE comparison.

The clock of the variance risk premium. Hizmeri and Bevilacqua [2025] identify *when* volatility uncertainty is incorporated into real-time option prices. Volatility of volatility measured at 10:00 EST, during the U.S.–European overlap, predicts next-day variance-asset returns; that content is gone by midday. Variance prices respond with a delay, consistent with underreaction and slow-moving beliefs about volatility. This paper's last-bar option reading (Section 4.5) is the 15:30 end of the same clock. If the morning is when the early bird gets the vol, 15:30 is when that bird has already flown: for the paper's eight forecasts in Section 5.4 quoted implied variance exceeds the forecast on 72 to 83% of expiration days, so a two-sided reading of the forecast is mostly a short. What the forecast adds is first the direction of the other days: the rule differs from always short only on the days it buys. Whether sizing on the gap adds to that turns on the sample (Appendix B.9). We do not restage their next-day VVIX regression; the object here is same-day remaining variance

against the expiration-day chain.

3 Data

3.1 The Panel

The raw data are 30-minute bars for the S&P 500 complex, gridded to a regular 30-minute frequency. The panel's extent is not set by a chosen date: **a bar exists in the panel if and only if the realized-variance source printed on it** — the same observed-print fact the availability indicators of Section 3.2 are built from, applied as the grid's own existence criterion. Under this rule the panel begins at the source's first print, in January 1998, and ends at its last, in April 2024. No weekend rule, holiday calendar, or session table is supplied anywhere: what is absent — weekends, and sessions that do not yet exist — is absent for that one reason, that the source did not print. The converse deserves saying, because a reader will assume otherwise: the source is a near-24-hour *futures* tape, so exchange holidays and early closes are not absent from it. A cash-market holiday is a shortened futures session and prints regular-hours bars ending at 13:00 or 11:30 with no 16:00 bar at all; an early close prints post-close bars right through 16:00, on which realized variance collapses to a few percent of what that slot normally carries. Both are legitimate rows of a futures panel and are kept. They are not legitimate rows of a *cash-session* exercise, and the last-bar option reading of Section 4.5 does not trade them: a cash-market holiday has no expiration, and an early close is dropped by an explicit test.

The panel's session structure therefore evolves as the market's did. The source prints around the clock from its first day — the panel's first bar is an overnight bar — but the early sessions are sparser: roughly 7,500 bars per year in 1998 (median 32 bars per trading day before 2004), growing through roughly 10,600 bars in 2003 to a stable $\approx$ 12,400 bars per year from 2005 onward, when the near-24-hour cycle (Sunday 18:30 through Friday 20:00, 48 bars per full trading day, roughly 34 of them outside regular hours) is fully live. The Friday boundary in that cycle is a grid convention and not a market fact: the futures week ends at 17:00 ET, and the bars the 20:00 cut keeps after it carry a thin tape — median thirty-minute volume about one and a quarter percent of the 17:00 bar's — which the panel treats like any other low-activity slot rather than dropping. A time-of-day slot that thickens or enters mid-sample behaves exactly like a feed going live: the rolling per-slot estimators of Section 3.2 have whatever history the slot has and accumulate it causally. Any statement about intraday structure in this panel refers to the full cycle as it stood at the time, not to the 09:30–16:00 window alone.

The existence rule replaced an earlier calendar-only filter (weekend trim plus forward fill) after that filter produced a measurable incident: bars following an early close existed on the calendar grid as forward-filled dead-session bars, and on one such bar — the only one in more than twenty-five thousand — two feeds' availability indicators disagreed for a single bar, which turned a nearly singular solve into forecasts orders of magnitude off target. On the overlapping 2005–2024 span the rule removes 4,925 such dead-session bars ($\approx$ 2% of the calendar grid). The accounting is: 303,442 print-bars on the grid, 300,317 panel rows after the construction burn-in, and 273,489 scored evaluation rows after the 24,000-bar training

window; the first walk-forward window of each arm is not scored.

The forecasting target is realized variance. For each 30-minute bar it is computed as the sum of squared one-minute log-returns within the bar:

$$RV_{d,j} = \sum_{i=1}^{M_{d,j}} r_{d,j,i}^2, \qquad M_{d,j} \leq 30, \tag{1}$$

where $r_{d,j,i}$ is the i-th one-minute log-return inside the j-th 30-minute bar of day d, and $M_{d,j}$ is the number of returns the tape printed in that bar. The vendor supplies that count beside every bar; it reaches exactly 30 and never exceeds it, which is how the sampling frequency is read off the file rather than assumed. Seventy-one percent of the panel's bars are full, and short bars are overnight and early-sample bars, not regular-hours ones: 99% of bars in the 10:00–16:00 window carry all thirty returns. Bars are indexed sequentially as RV_t. The forecast horizon is one bar ahead. The arrays carry that alignment the other way round: row t holds the target RV_t beside features every one of which is shifted by one bar, so a row's regressors are known at the end of bar $t-1$ and the model standing there forecasts the bar the row is labelled by. Throughout, "realized variance" denotes the target RV_t, and "volatility" denotes its square root; models are estimated on the square-root scale (Section 3.2). The construction in Equation (1) is the standard realized-measure definition. The heterogeneous autoregressive model of Corsi [2009] uses the trading day as its base unit; here both the target and the lag ladder are defined at 30-minute resolution.

The evaluation panel. Feature construction consumes a fixed burn-in of 3,125 bars at the start of the sample (January to June 1998), a constant of the panel construction that exceeds the longest feature lookback. This reduces the grid's 303,442 print-bars to a usable panel of 300,317 bars. The walk-forward protocol consumes a further 24,000 bars as the initial training window, and the first walk-forward window of each arm is not scored. Every arm reported in this paper is scored on the identical remaining rows; the paired bar-for-bar comparisons used throughout require this shared row index. The scored evaluation panel contains $n = 273{,}489$ bars.

Exogenous information. Beyond the target's own history, the design uses 41 exogenous channels. The channels are organised into thematic buckets that partition the columns exactly ($7 + 12 + 6 + 6 + 3 + 3 + 4 = 41$):

- **Moments** (7): own-asset higher-order return moments: bipower variation, realized semi-variance, autocovariance, absolute, cubic, and quartic returns.
- **Liquidity** (12): trading volume, turnover (total, buy-initiated, sell-initiated), effective and quoted spreads, number of trades.
- **Market, equal-weighted** (6) and **value-weighted** (6): cross-sectional stock-factor aggregates of return moments, bipower variation, semi-variance, and absolute returns.
- **Sentiment** (3): StockTwits attention, sentiment score, and sentiment count.

- **Implied volatility** (3): VIX, VVIX, VIX3M.
- **Volatility demand** (4): options order-flow indicators (SPX open/close, all options open/close, open only).

The buckets differ in column count; liquidity contributes twelve columns and sentiment three. A per-bucket comparison therefore mixes information content with dimensionality unless it is read per feature. Availability also differs across buckets: the cross-sectional aggregates are observed on 138,357 bars, the options order-flow bucket on 54,062 bars, and the VIX family does not exist for the earliest part of the sample. Row deletion would score each bucket on a different sub-sample. Section 3.2 describes the impute-and-indicate construction used instead; it scores every bucket on the same rows.

FOMC calendar channels. Four further columns enter as a scheduled public calendar overlay on top of the 41 exogenous channels. They are not part of the 7+12+6+6+3+3+4 = 41 partition above. The source is the `fomc release` flag: one 14:30 bar per scheduled FOMC statement (246 events; last live scheduled print 2023-11-01 14:30). `fomc_release` is 1 on that statement bar; `fomc_day` is 1 on every bar of a scheduled-statement day; `fomc_until_inv` and `fomc_since_inv` are the inverse bar-distance to the next and previous scheduled event. The unscheduled 2020-03-03 action is excluded from all four. After the last live scheduled print the four channels are missing (a dead tail), so the impute-and-indicate construction of Section 3.2 sees a stopped feed rather than a wall of zeros.

OLS on the HAR ladder plus this FOMC calendar bucket does not improve on the baseline (HAR + calendar OLS) on the shared evaluation panel: its QLIKE is higher, not significantly (Table 1). Joint OLS on all features, with the FOMC channels included among the rest, also loses to the baseline.

Table 1: FOMC calendar channels on the shared evaluation panel of Section 3.1, $n = 273{,}489$. QLIKE and Diebold–Mariano t versus the baseline (HAR + calendar OLS); negative t favors the arm.

Arm	QLIKE	DM t
baseline (HAR + calendar OLS)	0.23806	+0.0
FOMC calendar bucket (OLS)	0.24118	+0.5
all features (OLS)	0.32780	+1.6
ridge (joint)	0.25471	+1.4
lasso	0.23573	−1.4
block-diagonal ridge	0.23481	−2.4

3.2 Preparation

This subsection specifies the preparation pipeline. The pipeline is presented in full because, measured on this panel, the representation of missing and stale data has a larger effect on QLIKE than any model-class, basis, or penalty decision reported in this paper; the magnitudes are given under "Rolling robust scaling" below. At any given bar, 36% of the

exogenous design consists of forward-filled values rather than new measurements; on the sparsest channels the figure is 78%. Without availability information, an estimator cannot distinguish a filled value from a new measurement.

Overnight fills and availability indicators. The 41 exogenous channels are not all published on the near-24-hour grid. Most are US cash- or extended-session equity quantities; each prints only inside its own session. Averaged across channels, 0.638 of panel rows carry a print from the raw feed and the remaining 0.362 do not. Each channel is processed in the following order:

1. Compute a binary availability indicator from the raw feed: the indicator is 1 on bars where the feed printed and 0 otherwise.
2. Forward-fill the channel within its availability window.
3. Where the channel is structurally absent (no prior print exists), substitute a fixed in-distribution neutral value.

Every value column is accompanied by its indicator column. The indicator is computed before the fill. An earlier build of this pipeline computed the indicator after the fill, so every forward-filled bar was labelled observed. Under that build, the volatility-demand feed stopped publishing on 2023-08-31, the unlimited forward fill carried its final August print across the following eight months (8,453 bars, 3.44% of the sample), and the indicator read ≈ 1 throughout. Three further feeds (`vix`, `vix3m`, `vvix`) stopped publishing the same way on 2024-02-12. Every built cache is now gated on one check: for each channel, the mean of the constructed availability indicator must equal the fraction of rows on which the raw feed printed. The two quantities are estimates of the same number, so any gap indicates a construction error of this kind. On the uncorrected panel the check fails for 31 of 41 channels. The check requires no model, no held-out data, and no threshold tuning. We recommend it for any mixed-frequency panel in which exogenous series are filled.

The indicator also enters estimation as a regressor. Its linear coefficient absorbs any level shift between the pre- and post-availability eras, and the value column contributes only where the feed exists. Every bucket, including implied volatility and volatility demand, is therefore scored on the same 218,934-row index. The exact paired significance tests of the later sections require this shared index.

Stage 1: diurnal adjustment. Intraday volatility has a time-of-day profile. Let $s(t) \in \{1, \ldots, 48\}$ index the time-of-day slot of bar t, and let $\mathcal{W}_t = \{j : j < t,\ s(j) = s(t)\}$ be the set of the most recent $W = 20$ observations sharing that slot, with at least 5 required before the baseline is valid, and bars without a valid baseline are not scored. For non-negative series, the baseline is the same-slot rolling mean and the adjustment divides by it:

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \qquad \widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}, \tag{2}$$

Signed series (e.g. autocovariance) are divided by the same-slot rolling standard deviation instead. The window condition is $j < t$, not $j \leq t$, so the baseline at bar t uses no information

from bar t; the adjustment is non-anticipative. Warm-up bars pass through with baseline 1. They are consumed by the burn-in and never enter the evaluation panel.

Stage 2: the winsorized square-root target. After the diurnal adjustment, each variable class receives a fixed variance-stabilising map. The map is chosen in advance from the class's support and tail behaviour; no transform parameters are fitted. The maps are: $x \mapsto \sqrt{x}$ for squared-return and volatility-like measures, including RV itself, bipower variation, turnover, and effective spread; $x \mapsto \operatorname{sign}(x)\sqrt{|x|}$ for autocovariance; $x \mapsto x^{1/3}$ and $x \mapsto x^{1/4}$ for third and fourth moments; and $x \mapsto \log x$ for the implied-volatility family. Models are estimated on the adjusted $\sqrt{RV}$ scale. After transformation, each series is clipped to rolling 1st and 99th percentile bounds estimated on the trailing $W_{\text{clip}} = 240$ observations (about five trading days) ending at $t-1$. Winsorization is not inverted at evaluation time; only the diurnal division and the square root are inverted, by the plain back-transform $\widehat{RV}_t = \hat{y}_t^2 B_t$ of Equation (5). Because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$, squaring a forecast of $\sqrt{RV}$ gives a downward-biased forecast of RV; no correction is applied, and each forecast is scored with its own bias (Section 4.2).

HAR features. The target's own history enters through trailing rolling means of the adjusted series on a base-2 geometric ladder,

$$\mathcal{L} = \{1,\ 2,\ 4,\ \ldots,\ 2^{k_{\max}}\}, \qquad \overline{RV}_t^{(k)} = \frac{1}{k}\sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \tag{3}$$

The rungs span roughly 30 minutes, 2.5 hours, half a trading day, 2.6 days, 13 days, and 65 trading days. The sum starts at $t-1$, so the feature at bar t contains no contemporaneous information. The ladder adapts the daily/weekly/monthly averages of Corsi [2009] to six intraday horizons. The same ladder is applied to each included exogenous channel, contributing $|\mathcal{L}| = 6$ rolling-mean columns per channel.

Rolling robust scaling. Input features to the linear estimators are standardised by subtracting a rolling median and dividing by a rolling interquartile range. Both statistics are estimated from the current training buffer and updated at every step. Two rules govern the estimation.

First, the median and IQR are estimated from observed rows only, i.e. rows on which the availability indicator is 1. If the statistics are instead estimated over forward-filled values, a series whose feed has stopped contributes identical repeated values, its rolling IQR shrinks toward zero, and the standardising division divides by a value near zero. On the uncorrected panel, the stale volatility-demand columns reached 2,260 rolling IQRs by this mechanism; the raw series did not move, and the denominator shrank.

Second, a scale is used only after 512 observed values have accumulated, and the IQR is floored at 1% of the largest scale the channel has shown. This rule exists because masked estimation fails at the start of a feed's life: with few observed rows, a low-dispersion opening window would otherwise set the scale for all subsequent bars.

The three elements — indicators computed before the fill, masked estimation, and neutral substitution — are applied together. Each element applied alone does not remove the failure.

Applied together, they reduce the worst channel's maximum standardised value from 2,260.4 to 96.2 rolling IQRs. On the October 2023 window affected by the stale feed, the uncorrected panel scored QLIKE 6.37170 and the repaired panel scored QLIKE 0.12391. Every modelling effect reported in this paper lies between 0.0005 and 0.004 QLIKE. This difference in magnitude is the reason the preparation pipeline is presented before the modelling sections.

All stages are causal by construction, and the two places where the panel as shipped falls short of that are named here rather than left implicit. Every rolling statistic, fill, winsorization and scaling above uses only information dated strictly before t; the HAR ladder, the target, and the benchmark of Section 4.1 were re-derived under prefix truncation and reproduce bit for bit. Two stages of the exogenous build do not. The neutral value substituted where a channel is structurally absent (step 3 above) is the full-sample median of that column rather than a trailing one; and the availability indicators enter the design after the forward fill rather than before it, so on a filled bar they assert a live state the feed did not print. Neither stage touches the HAR block or the benchmark, which read no exogenous column. Their footprint on the design matrix is that 456 of 1,264 columns move under prefix truncation, by up to 0.8 standard deviations on the bars the last-bar reading of Section 5.4 uses and by much more at isolated points early in the sample. Both are one-line pipeline changes; both require the panel to be rebuilt and every exogenous arm re-scored, and that rebuild is pending.

3.3 Descriptive Analysis

4 Methods

The methods come in two halves. The first half fixes the evaluation framework before any comparison is made: the comparator each arm is measured against and the testing machinery that accompanies it (Section 4.1); and the back-transform that maps every forecast from the fitting scale to the variance scale before it is scored (Section 4.2). The second half defines what is compared: the estimators — one unpenalized member, the penalized linear family, and two tree ensembles — with the walk-forward protocol and the causal tuning rule every tuned arm obeys (Section 4.3); and ridge with a block-diagonal Tikhonov penalty (Section 4.4). Every estimator is fit on the shared panel of Section 3 and subject to the hyperparameter discipline of Section 4.1: no quantity is selected on evaluation loss.

4.1 The OLS–HAR Benchmark

Every result in the remainder of this paper is referenced to a single fixed comparator: ordinary least squares on the six HAR features of Equation (3) plus calendar dummies, with no exogenous variables, refit at every bar. The forecasting protocol is:

1. Walk-forward with a rolling origin. The training window is the most recent $T_{\text{train}} =$ 24,000 bars, which is the window's definition. Measured on the panel that is about 480 trading sessions, not the 500 a flat 48 bars per session would imply, because early-sample and shortened sessions print fewer. The window slides forward by one bar per prediction.

2. The model is refit at every bar. The refit is exact and computationally feasible because the sliding-window normal equations admit rank-1 up-dating (add the entering bar) and down-dating (remove the exiting bar).

3. The benchmark is read from the ridge path at $\alpha = 0$; it is therefore exact OLS.

4. Each forecast is mapped back to raw variance through Equation (5).

5. Forecasts are scored by QLIKE [Patton, 2011],

$$L_{\mathrm{QLIKE}} = \frac{1}{N}\sum_{t=1}^{N}\left[\, r_t - \log r_t - 1 \,\right], \qquad r_t = \frac{RV_t^{\mathrm{raw}}}{\widehat{RV}_t^{\mathrm{raw}}}, \tag{4}$$

computed on the full 273,489-bar evaluation panel. QLIKE is used for three reasons: it is scale-invariant in the level of volatility; it is strictly consistent for the conditional variance; and it penalises under-prediction more heavily than over-prediction, which matches the direction in which forecast errors are more costly for this target. QLIKE values are not comparable across preprocessing regimes: on the diurnally adjusted panel of the reference study, the same class of models scores several tenths of a QLIKE unit lower than on a raw, unadjusted panel. Every comparison in this paper is therefore made within one preprocessing regime, and every table is labelled with the panel it was computed on.

Table 2: The benchmark: per-bar OLS on the HAR ladder plus calendar dummies, scored on the frozen panel, its forecast used as the plain back-transform of Equation (5). QLIKE is computed on the raw variance scale; the secondary metrics are computed on the fitting (square-root) scale where the model is estimated, against the unwinsorized realization $\sqrt{RV_t/B_t}$ on the same scored bars.

Scored bars	273,489 (positivity exclusion removes 0.2%)
QLIKE (raw scale)	0.23806
Out-of-sample R^2 (fit scale)	0.413
Mincer–Zarnowitz α (fit scale)	−0.050
Mincer–Zarnowitz β (fit scale)	1.063
MSE (fit scale)	0.1415
MAE (fit scale)	0.2220

Squared-error metrics computed on the raw variance scale are uninformative on this panel: the median absolute error is 3.0×10^{-7} against a maximum of 1.7×10^{-3}, and the ten largest of the 272,907 scored errors carry 44% of the sum of squared errors, so a handful of volatility-spike bars dominates every squared-error aggregate — the asymmetry that motivates QLIKE as the primary loss (Section 4.1).

Differences in QLIKE at the fourth decimal place require a significance assessment. Every comparison against this benchmark is therefore accompanied by a Diebold–Mariano test [Diebold and Mariano, 1995] on the per-bar loss differential. The test procedure is:

1. Compute the per-bar QLIKE loss for each of the two arms on the shared row index, and take the per-bar difference.

2. Estimate the long-run variance of the loss differential with a Newey–West estimator.

3. Apply the small-sample correction of Harvey et al. [1997] to the test statistic.

Where several arms are compared simultaneously, we report the Model Confidence Set of Hansen et al. [2011]. Every arm is scored on the identical row index, so all of these tests are paired bar-for-bar comparisons.

Hyperparameter discipline. No quantity anywhere in this paper is selected using evaluation loss. Every hyperparameter that is tuned is tuned *causally*: the penalty (or penalty vector) is re-selected periodically from a grid fixed in advance, by minimizing validation error on a forward split of the current training window — fit block, embargo, validation tail — so that no forecast's configuration uses information from that forecast's future. Every hyperparameter that is not tuned is a fixed constant, set before the campaign was scored and reported with its arm: the fixed-penalty arms exist precisely as controls for what tuning contributes, and the structural constants of the design (training-window length, re-selection cadence, winsorization quantiles, block widths, estimation-window lengths) are conventions of the panel and estimator construction, disclosed where they are defined and never optimized against any loss.

4.2 The Back-Transform

Every model in this paper is fit on the preprocessed scale of Section 3.2: realized variance is divided by its time-of-day baseline and passed through the square root, so the regression target is $y_t = \sqrt{RV_t / B_t}$, where B_t is the diurnal baseline of Equation (2) at bar t, built from same-slot bars strictly before t. Evaluation is on the raw variance scale, because QLIKE is a strictly consistent scoring rule for the conditional variance and not for any transform of it. Each forecast $\hat{y}_t$ of the fitting-scale target is therefore mapped back to the variance scale by undoing the two preparation steps, and by nothing else:

$$\widehat{RV}_t = \hat{y}_t^{\,2}\, B_t. \tag{5}$$

The back-transform has no smearing factor, no recalibration map and no estimated quantity of its own: the scored object is the model's forecast squared and rescaled by the baseline, identical in form for every arm. The realization it is scored against is the raw realized variance RV_t itself, never one rebuilt from the winsorized fitting target. QLIKE (Equation (4)) is averaged over the bars at which both members of the pair are positive. Bars whose time-of-day baseline has fewer than five earlier same-slot observations are not scored. comparison — the linear-class comparisons of Section 5.1, and the tree and block-ridge results of Sections 5.2 and 5.3 — scores its arms on raw-scale pairs produced by exactly this map, on the identical row index.

Squaring does not commute with the conditional expectation. By Jensen's inequality the squared forecast falls short of the conditional mean of y_t^2 by the conditional variance of the fitting-scale error, so the forecast of Equation (5) is biased low by an amount that differs across arms with their own error variance. That shortfall is part of each forecast as scored: the comparisons below are comparisons of these forecasts, and QLIKE, which penalizes under-prediction more heavily than over-prediction, charges each arm for it.

4.3 Estimators and Causal Tuning

Minimum-norm least squares. The unpenalized estimator is *minimum-norm least squares*: among all coefficient vectors that minimize the training-window sum of squares, the unique one of minimal ℓ_2 norm, computed through the pseudoinverse of the window gram at the standard machine-precision cutoff. This is the $\alpha \to 0^+$ limit of ridge regression, it is defined at any design rank, and it carries no penalty and no tuned quantity of any kind — the strongest attribution claim an unpenalized estimator supports. The rank cutoff is not a tuned quantity either: the centered gram's spectrum is gapped by more than five orders of magnitude around the retention threshold (on the widest design, largest discarded eigenvalue 1.8×10^{-11} relative, weakest retained 6.2×10^{-6}, with no eigenvalue between them at any bar), so every tolerance across four decades retains the identical subspace and reproduces the reported forecasts bitwise. Each arm is refit at every bar over a rolling window of 24,000 bars (about 480 trading sessions). On a full-rank design (the benchmark's 52 columns) minimum-norm least squares coincides exactly with ordinary least squares. An earlier pass of this campaign used the plain normal-equations solve, whose solution is arbitrary in near-null directions of the design; two single-window incidents in the campaign record trace to that path, and its results are archived. Because the pseudoinverse is defined at any rank, the estimator requires no rank guards; the only design intervention retained is the go-live participation rule of Section 3.2, which is a statement about data existence, not numerics.

Estimators. The design matrix X has one row per bar in the current training window and one column per feature: the HAR ladder of the target, the calendar features, and the full exogenous block of 523 columns. Let y be the vector of targets over the same window, and let x_s denote row s of X. The two estimators solve, at each refit,

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \alpha \left\|\beta\right\|_2^2,$$

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \left\|\beta\right\|_1,$$

where the sums run over the bars of the training window and the penalties α and λ are held constant between the periodic re-selections described below. The ℓ_1 penalty produces a coefficient vector supported on a subset of columns; the ℓ_2 penalty produces a coefficient vector with all entries nonzero. The elastic net [Zou and Hastie, 2005] interpolates the two,

$$\hat{\beta}^{\text{enet}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \left(\tfrac{1-\rho}{2} \left\|\beta\right\|_2^2 + \rho \left\|\beta\right\|_1\right),$$

recovering ridge at $\rho = 0$ and the lasso at $\rho = 1$. Its role in this paper is diagnostic as much as predictive: in the causally tuned arms the mixing share ρ is *free* — re-selected from $\{0.25, 0.5, 0.75, 1.0\}$ jointly with the penalty strength at every retune — so the estimator, not the author, chooses how much selection to apply, and the selected trajectory is part of the evidence in Section 5.1. All three penalized families are parameterized in the units of `scikit-learn`'s solvers, against which every custom solver below was validated.

Walk-forward protocol. The panel is computed as 100 walk-forward chunks over a rolling 500-trading-day (24,000-bar) training window. Every arm is refit *every bar*: at each bar the oldest bar leaves the window, the newest enters, and the coefficients are re-estimated before the next forecast is issued. Each arm re-selects its own penalty periodically and causally: every 250 solves, the current training window is split forward into a fit block, a 25-bar embargo (covering the longest moving average entering the features), and a 125-bar validation tail, and the penalty value minimizing validation MSE on that tail is adopted. No forecast's hyperparameters ever see its own future. The lasso and elastic-net arms are solved as exact every-bar online homotopies [Garrigues and El Ghaoui, 2008]: the solution and active set at each bar are obtained by exactly updating the previous bar's solution for the one-row change in the window, not by a warm-started batch solver terminated at a tolerance, so the head-to-head is not confounded with optimization error. The implementation is a re-authored recursive elastic net (Gram-form active-set homotopy with rank-1 Cholesky maintenance, validated coefficientwise against `scikit-learn`). The ridge arm runs on an exact rank-one update path.

Tree ensembles. The nonlinear members of the estimator set are two implementations of gradient-boosted regression trees. Gradient boosting fits an additive ensemble forward-stagewise: starting from a constant, at stage m it adds a shallow regression tree f_m fit to the negative gradient of the loss at the current ensemble's predictions,

$$F_m(x) = F_{m-1}(x) + \eta\, f_m(x), \qquad f_m \approx \arg\min_{f \in \mathcal{T}} \sum_i \Big(-\frac{\partial \ell(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)} - f(x_i) \Big)^2,$$

where $\mathcal{T}$ is the space of depth-limited regression trees, $\eta \in (0, 1]$ is the shrinkage rate, and for the squared loss used throughout this paper the negative gradient is exactly the current residual, so each tree is a regression tree on the residuals of the ensemble so far.

XGBoost [Chen and Guestrin, 2016] fits each tree to a second-order Taylor expansion of the loss rather than the raw gradient. Writing $g_i = \partial \ell / \partial F_{m-1}(x_i)$ and $h_i = \partial^2 \ell / \partial F_{m-1}(x_i)^2$, the per-tree objective

$$\mathrm{Obj}_m \;\approx\; \sum_i \Big[g_i f_m(x_i) + \tfrac{1}{2} h_i f_m(x_i)^2 \Big] + \gamma T + \tfrac{1}{2} \lambda \sum_{j=1}^{T} w_j^2,$$

with T leaves and leaf weights w_j, has closed-form optimal weights $w_j^* = -G_j/(H_j + \lambda)$, where G_j and H_j are the gradient and Hessian sums over the rows in leaf j, and a split is accepted only if it improves $\frac{1}{2}\big[G_L^2/(H_L + \lambda) + G_R^2/(H_R + \lambda) - G^2/(H + \lambda)\big]$ by more than γ. The curvature term makes the leaf weights loss-aware (for squared error $h_i = 2$ and the scheme reduces to a regularized mean), and γ prices complexity per added leaf explicitly; row and column subsampling decorrelate successive trees. Our configurations use its histogram split finder (`tree_method="hist"`).

LightGBM [Ke et al., 2017] uses the first-order scheme with three engineering choices that matter at this design's width. First, feature values are binned into histograms, so split search costs depend on the bin count rather than the 24,000-row window length. Second, gradient-based one-side sampling keeps all large-$|g_i|$ rows and samples small-$|g_i|$ rows with

compensating weights, concentrating the fit on the hard bars. Third, exclusive feature bundling merges mutually exclusive columns — a property this design has in abundance, since an availability indicator and its siblings are zero whenever another feed's indicator is one — shrinking the effective width without changing the fitted function class. Trees grow leaf-wise (best-first) under a leaf-count cap rather than level-wise: deeper trees for the same budget, with the cap standing in for depth control.

Roles in the campaign. The two families enter twice, under different tuning disciplines. In the causal-tuning battery they are tuned arms like any other: their hyperparameters are re-selected on the same periodic forward-split schedule as every tuned arm, and the battery fully crosses the five estimators of this subsection with the nine feature buckets on the battery's shared panel (n = 218,934 thirty-minute bars, identical rows in every cell). In the frozen *expert bank* the configurations are fixed permanently before any scored bar: a dev-span study (rows before 2003, screen refits every 250 bars) searched each family's box — LightGBM: `num_leaves` 15–255 and `min_child_samples` 20–500 and `lambda_l1`/`l2` 10^{-8}–10 on log scales, `learning_rate` 0.01–0.3, `n_estimators` 50–500, feature and bagging fractions 0.5–1.0; XGBoost: `max_depth` 3–10, `min_child_weight` 1–100, `reg_alpha`/`reg_lambda` 10^{-8}–10, `gamma` 10^{-8}–1 on log scales, the same learning-rate and estimator ranges, subsample and column-subsample 0.5–1.0 — and the menu froze the top eight unique configurations plus two seeded Sobol points per family (2×10 experts). Each expert is then refit on the trailing 24,000-bar window at the disclosed refit cadence, on the identical wide design the linear arms read, with a fixed seed and threads following the slot count, so the tree class and the linear class are compared on the same information set with no per-arm tuning freedom. The shared comparator in both roles is the per-bar OLS–HAR incumbent (Section 4.1).

4.4 Block-Diagonal Tikhonov

A single ridge penalty cannot serve the HAR backbone and the wide exogenous design at once. The HAR-plus-calendar columns are few, well-conditioned, and individually strong, so shrinking them buys little variance and costs bias. The wide exogenous design requires heavy shrinkage — that is the penalty sweep's result (Section 5.1). A scalar α large enough for the weak exogenous columns over-shrinks the HAR columns; a value small enough for the HAR columns under-shrinks the exogenous columns.

The resolution is a **block-diagonal Tikhonov** penalty [Hoerl and Kennard, 1970, van Wieringen, 2023, van de Wiel et al., 2021]: one jointly solved least-squares problem, blocks ordered from strong and low-dimensional to weak and wide, each shrunk harder than the last. The B-block problem

$$\hat{\beta} = \arg\min_{\beta_1,\ldots,\beta_B} \Big\| y - \sum_{b=1}^{B} X_b \beta_b \Big\|_2^2 + \sum_{b=1}^{B} \alpha_b \|\beta_b\|_2^2, \qquad \alpha_1 < \alpha_2 < \cdots < \alpha_B,$$

reduces to ordinary unit-penalty ridge by column scaling. With

$$\tilde{X} = \big[\, X_1/\sqrt{\alpha_1} \;\cdots\; X_B/\sqrt{\alpha_B} \,\big] \quad \text{and} \quad \gamma_b = \sqrt{\alpha_b}\,\beta_b,$$

$$\Big\| y - \sum_b X_b \beta_b \Big\|_2^2 + \sum_b \alpha_b \|\beta_b\|_2^2 = \|y - \tilde{X}\gamma\|_2^2 + \|\gamma\|_2^2,$$

so $\hat{\beta}_b = \hat{\gamma}_b/\sqrt{\alpha_b}$. After scaling, every nested design runs in the same exact rank-one rolling least-squares path as every other linear model in this paper, refit at every bar. Setting a trailing block's coefficients to zero recovers the previous design exactly. There is no selection step and no tuned dial: the penalties are fixed constants.

Two blocks. The two-block design uses $B = 2$ and $(\alpha_1, \alpha_2) = (1, 100)$. Block 1 ($X_1$) is the HAR ladder of the target. Block 2 (X_2) is the same HAR averaging construction applied to every exogenous feature — HAR-of-`all_features`, the full wide basis. The two blocks are not fit separately and stacked. Setting $\beta_2 = 0$ reduces the model to ridge at $\alpha = 1$ on the backbone, which differs from the OLS–HAR incumbent of Section 3 only by that nominal penalty. The estimator matches the measured coefficient structure: block 2 keeps every column and shrinks them two orders of magnitude harder than block 1.

4.5 A Last-Bar Same-Day Option Reading

The one-bar forecast can be read against a listed price on the last thirty minutes of an expiration day. What follows fixes the instrument, the two measurements, the position rules, and the control; constants were set before the portfolio was scored.

Instrument. S&P 500 weekly (SPXW) options with zero days to expiry, cash-settled on the day's official close. The panel is the expiration-day chain at 15:30 ET (bid, ask, mid, underlying, quoted implied volatility), 2020 through 2025. At 15:30 the trade is the *straddle*: the nearest out-of-the-money call and the nearest out-of-the-money put of the same-day expiry, held as one position — the call at the smallest listed strike $K_c \geq S$ with a live mid and the put at the largest listed strike $K_p \leq S$ with a live mid, where S is the 15:30 underlying print. When S sits on a listed strike the two strikes coincide; otherwise they are one strike apart. Holding the straddle from 15:30 to the close is the *last-30-min trade*. Entry $P_{15:30}$ is the sum of the two mids. The position is held to the close and settled at intrinsic against the official S&P 500 close S_{cl},

$$\Pi_{\text{long}} \;=\; \big(S_{\text{cl}} - K_c\big)^+ + \big(K_p - S_{\text{cl}}\big)^+ \;-\; P_{15:30}, \tag{6}$$

with no delta hedge and no share unwinding. The long return is $R = \Pi_{\text{long}}/P_{15:30}$; a short earns $-R$. By construction $R \geq -1$, with equality on the days the close settles between K_p and K_c, either endpoint included, and both legs expire worthless. The 16:00 chain underlying is retained only as a check on the settlement print. Early-close sessions are dropped. The session ends at 13:00 on those days, so the 15:30 chain row is stamped after the close and carries stale quotes: the settlement is already known there and the short is a certain $+1$. The test is a non-positive time to expiry on the 15:30 row — the vendor's `early_close` column is true on every row of this file and is not used. Twelve such sessions appear in the chain. Six of them fall inside the forecast window, and the five of those that carry a 16:00 forecast row are the days the filter takes out of the scored frame; the remaining one never had a forecast row to lose.

Forecast remaining variance. Eight per-bar forecast series enter the same rules on the same contracts. Seven are estimators the body of the paper defines: the baseline (HAR + calendar OLS) forecast (Section 4.1), the block-diagonal ridge (HAR versus exogenous block, separate penalties; Section 4.4), LightGBM and XGBoost on the wide all-features design (Section 5.2, frozen expert menu), a causally tuned lasso on that same design under the selection protocol of the paper's tuned ridge (Section 5.1), a lasso at a fixed $\alpha = 10^{-4}$, and a causally tuned elastic net — the interpolating ℓ_1/ℓ_2 family — on the same wide all-features design (battery grid `logspace(-6,-2,5)` at `l1_ratio=0.5`, arm `b3_enet_tuned`). The eighth is not a competing model: it is the block-diagonal ridge again, refit on a design that carries no FOMC calendar columns, and it is carried as a diagnostic row — "block-diagonal ridge, without the FOMC columns" — because the gap between it and the ridge of record is what those calendar channels are worth on this trade, and nothing else in the table isolates them. All eight are stored on the estimation scale $y = \sqrt{RV/B}$. The 15:30 row is turned into a forecast of the raw variance of the 15:30–16:00 bar by the plain back-transform of Equation (7) below, the same for every forecast of this section. It is not the retransformation of the evaluation protocol of Section 4.2, which re-maps the forecast's level and adds a causal second moment: no number in this section is produced by that protocol, and none is carried between the two. Forecast files end on 30 April 2024. The chain itself runs to 31 December 2025, so the 413 expiration days from 1 May 2024 onward — the most recent twenty months of contiguous tape — carry no forecast row and are not scored in Section 5.4; the trade's live, long-only form has been run on them, and Appendix B.5 describes that record, which is not scored with the plain forecast and is not reported here.

One convention matters for honesty. *Alignment*: forecast rows are labelled by the end of the bar they predict, so the row read at 15:30 is the one stamped 16:00 — issued at 15:30 for the thirty minutes that remain — and not the stamp-15:30 row, which was issued at 15:00 for a bar already finished; every input to the signal below is therefore known at 15:30.

One related contamination is disclosed rather than repaired. The per-slot factor B that scales the target is a trailing twenty-session mean of the same slot, and it is built upstream of this section: on an early close it absorbs that day's post-close bars, so for the following twenty sessions the 16:00 slot's B sits a few percent low. 94 of the 866 scored days fall inside such a window. Repairing the factor on those days moves a forecast by up to 21% and changes its sign(s) Sharpe ratio, over the forecasts of Section 5.4, by a median of 0.08 and by at most 0.38 at the midpoint (0.38 crossed) — at most 0.21 for the paper's eight — and flips a median of 8 and at most 16 of its positions. Repairing it at the source means regenerating the forecast tables, and it is part of the pending campaign.

Which panel produced which row. The eight columns were not all produced on one design, and the tables say which each came from. The *FOMC panel* — the 1,264-column design that carries the FOMC calendar block — is the panel of record here: the block-diagonal ridge and the fixed lasso are fitted on it. The causally tuned lasso, the elastic net, LightGBM and XGBoost are still on the 1,144-column *earlier* panel, which has no such block; a run of those four on the panel of record is pending, and the tables mark them so. The baseline is invariant to the swap — it reads no exogenous column, and re-derived on both panels it agrees to 2×10^{-11} — so its row is unaffected. For an arm that does read

exogenous columns the swap is not cosmetic: on the ridge it moves the 16:00 forecast by a median of 1.1%, and by more than 1% on 53% of trading days. That is why the diagnostic row exists, and why a ranking that mixes rows from the two panels would not be a like-for-like ranking. Each table's file, panel, chunk tree, code fingerprint and refit contract is recorded in `results/spxw_pnl/MANIFEST.md`.

Refit contract, hyperparameters, and what is not recoverable. Every linear arm is refit at each bar on a rolling window of 24,000 bars (about 480 trading sessions); the tuned lasso and elastic net re-select their penalty from the fixed grid every 250 solves, on the embargoed forward tail of the current window. The two tree columns are reported as shipped and are less clean than that. Their settings are two entries of a frozen twenty-arm menu, the top LightGBM and the top XGBoost entry; each column's settings are stored with its forecast chunks, and where that menu comes from is set out with the forecasts fitted for the 16:00 bar alone below. Their refit cadence is mixed as well: one cluster refit at every bar and another every tenth bar, both contributed chunks to the same harvest, and neither the chunk metadata nor the shipped table records which cadence produced which stretch. A cadence coarser than one bar is stale but still causal — the fit still sees only strictly prior data, and an instrumented replay found no look-ahead at either cadence — so this is a provenance defect and not a leak. One caveat attaches to the ridge of record too: its table came from a cluster run identified by code fingerprint rather than by a commit. The same estimator specification reproduces the corresponding earlier-panel chunks to 2×10^{-13}, so the estimator is confirmed; what cannot be re-derived on a workstation is the panel build under that exact code. A campaign that regenerates all eight columns under one committed revision — tree menu re-frozen, and cadence, thread count, library versions and code hash written into every chunk — is the stated remedy, and it has not been run.

Two facts about the FOMC block itself belong with that. The release feed behind it ends on 1 November 2023, so 122 of the 866 scored expiration days — 14.1% — fall after the feed is dead, and the meetings of 13 December 2023, 31 January 2024 and 20 March 2024 carry no FOMC signal at all. And the release-*surprise* channel is a dead constant over the whole panel: it is logged and then winsorized on a rolling 240-bar window, which clips a single-bar event to the running bound every time. So "FOMC in the design" means, operationally, the meeting-day indicator and the bars-until and bars-since counters, and it means them only through 1 November 2023. The diagnostic row above measures the calendar channels as they actually are in the shipped panel, not as they would be with a feed that reaches the end of the sample.

What the 16:00 target is. The forecast series predict RV on the paper's own panel, which is built from a near-24-hour futures tape (Section 3.1). The row read at 15:30 is therefore the realized variance of the futures 15:30–16:00 bar; it is not a measurement of the cash market's settlement procedure, and it is nonzero on days the cash market is shut. The option, by contrast, settles at the official S&P 500 close. On a regular session the two describe the same half hour of the same index and the comparison is the intended one, and the sessions on which they are not aligned are not traded: a cash-market holiday has no expiration, and an early close is dropped by the filter of the instrument paragraph above.

Forecasts fitted for the 16:00 bar alone. The eight columns above fit one coefficient vector to all 48 half-hour bars of the day. The trade reads one bar, so a second family of forecasts is fitted for that bar alone — the *per-bar* forecasts. Their rows are the 15:30–16:00 bars (stamp 16:00) of the previous 2,000 sessions, about eight years; the target is the paper's $y = \sqrt{RV/B}$, winsorized as for every other arm; and the regression is re-solved every session. Every input is known at 15:30: for the target and for each exogenous series, its mean over the last $k \in \{1, 5, 25, 125, 625, 3125\}$ half-hour bars of the near-24-hour series, overnight bars included; an is-present flag per series; and calendar dummies (day of the week, option-expiry day and week, month and quarter end). Every exogenous input is divided by its time-of-day baseline (Section 3.2), and the HAR columns are means of the adjusted target. In the design these means are the columns `har_ma_`k (the target's own ladder) and x`_ma_`k (the ladder of series x). The design of all 48 bars also interacts the target's ladder with open- and close-of-session indicators; in a series with one bar per session those 12 columns are either zero or exact copies of `har_ma_`k, so the per-bar design leaves them out (the pooled twins below keep them). Two input sets are used, each named in the tables by its design name:

- `baseline` (HAR + calendar): no exogenous series;
- `all_features`: every exogenous series of the paper's design, the FOMC calendar channels included;

The linear estimators are OLS, ridge, lasso, and an elastic net with equal ℓ_1 and ℓ_2 weight. OLS has no penalty. The penalized ones are pre-tuned, as the trees are: the penalty is chosen once, before the trade period, from a fixed log-scale grid (ridge 10^{-2} to 10^{3}, lasso and elastic net 10^{-6} to 10^{-2}, one value per decade) by squared error on four forward folds of 250 sessions that tile the sessions from 6 January 2016 to 2 January 2020, each fitted on the 2,000 sessions before a 25-session embargo; the HAR + calendar columns are penalized at 1/100 of the other inputs' penalty (on `baseline`, where every column is HAR + calendar, there is one penalty). From the first trade day, at the start of every block of 250 sessions, the two neighbouring grid values challenge the penalty in force and replace it only when the validation gain clears a margin: the Newey–West t of the paired squared-error difference must fall below $-z_{0.05/2}$, the margin rule of the trees' re-tuning. Over the four re-tuning points of the trade period 4 of the 24 challenges cleared it. The causal re-choice used before — the penalty re-selected every 250 sessions on the last 125 sessions of the window, with no margin — switched the penalty at 24 of its 45 re-choices on the paper's input sets, of which 3 would have cleared that margin; 34 of its 54 picks sat at an edge of the grid, and a switch flipped 9 % of the positions in the block it governed. The per-bar trees — LightGBM, XGBoost and a random forest — are fitted on the same rows, inputs and window and refit either every ten sessions or every session; the two cadences are separate rungs. *Untuned*, each tree keeps fixed settings, the *shipped settings*, and all three come from one procedure: an Optuna search per tree on the 1998–2002 sessions alone, over the pooled all-bars `all_features` design and scored by a rolling backtest within those sessions, frozen as a menu of each tree's best entries before any later session is used, so it has no look-ahead into the trade period. The leaf minimum is scaled to the per-bar 2,000-row window (LightGBM's from 98 to 8, the random forest's from 12 to 1). LightGBM, XGBoost and the random forest each take their search's top entry. The random forest's search was run later, for this paper, by the same procedure

on the same sessions and design, over 400 trials. The trees are tuned in two ways, each on the same validation tail and by validation squared error. The *random search* re-selects the settings every 250 sessions from 32 fixed candidates of a grid centred on the shipped settings. *Optuna* runs 50 trials of a tree-structured Parzen estimator at each tuning point, the first trial being the shipped settings, and re-tunes every 1, 5, 25 or 250 sessions. The forecast, refit every session, uses the best of the first 10, 25 or all 50 trials of the latest tuning point. Each tuned path is also run with the settings chosen by validation QLIKE. Every per-bar fit — linear or tree — first drops the columns that are constant on its own training window and every exact copy of an earlier column, keeping first occurrences: the *per-window mask*. The linear estimators apply it at every penalty tune, as their identifiability rule; the trees at every refit and tuning point. Tree forecasts depend in their last bits on the processor's vector instruction class, and a tuning choice can flip on such a difference, so every tree table is computed on one processor model. Each per-bar linear forecast has a *pooled twin*: the identical specification fitted on all 48 bars of the same 2,000 sessions with one coefficient vector, so that the difference between the two is the per-bar fit alone. The pooled twins are pre-tuned in the same way, on all 48 bars of the same folds; there the pre-tuning neither helps nor hurts significantly: against the causal re-choice, QLIKE is worse in 5 of the 6 pairs, by $+0.9\,\%$ on average, and no Diebold–Mariano statistic or Newey–West t of the trade exceeds 1.96 in absolute value (largest 1.36 and 1.29). A likely reason is the pooled models' larger validation tail — all 48 bars of each session — which leaves causal re-tuning less noisy there: under the margin rule their penalty is replaced 11 to 40 times per model over 217 re-tuning points.

The variance forecast. Every forecast of this section, the paper's eight and the per-bar ones alike, is a forecast f_t of $y = \sqrt{RV/B}$ for the 16:00 bar, issued at 15:30, and it is used as its plain back-transform,

$$\widehat{RV}_t \;=\; f_t^2\, B_t, \tag{7}$$

where B_t is the time-of-day baseline of the 16:00 slot, the factor that scales the target. Nothing is estimated on top of it — no level map and no second-moment term — and the same $\widehat{RV}_t$ is the forecast the sign(s) rule reads and the forecast every loss scores. Because $E[\sqrt{X}]^2 \le E[X]$, squaring a forecast of y tends to forecast RV low, which tilts the comparison with implied variance below toward selling; the tilt is left in rather than estimated. QLIKE is scored against the per-bar specification's 16:00 target, the same target for every forecast, and two forecasts are compared on the same days by the Diebold–Mariano statistic of the daily QLIKE difference [Diebold and Mariano, 1995], with a Newey–West variance (Bartlett kernel; lag 6 on 866 days) and the correction of Harvey et al. [1997]. Sharpe ratios and their paired differences are reported as point estimates; no interval is computed.

Quoted implied volatility. Implied volatility is the chain's quoted value on each of the two legs at 15:30; the straddle's quote is the mean of those two. The vendor quote is an *hourly* volatility while the window that remains is thirty minutes, so the quote is rescaled before it is compared with anything,

$$\mathrm{IV}_{30,t} \;=\; \mathrm{IV}_t^{\mathrm{hourly}}/\sqrt{2}, \tag{8}$$

and $\mathrm{IV}^2_{30,t}$ is the implied variance of the remaining bar — the same window, and the same units, as $\widehat{RV}_t$. The vendor field is not always a quote: it is the output of a bisection on [0.0005, 0.025] that returns a bracket node when it fails to converge, so the two bounds and the first interior nodes of that bisection are censored values rather than prices, and they are masked here. On the five scored days where a nearest leg carries one, the implied volatility is recovered instead by inverting the Black–76 price of the straddle — call plus put at zero rate on the 15:30 index level, the vendor's own convention, over the remaining half hour — for the total volatility that reproduces the 15:30 midpoint. On uncensored legs that same inversion reproduces the quoted midpoint to quote precision (median model-to-quote price ratio 1.0000, 5th to 95th percentile 0.9969 to 1.0062), so "quoted" and "inverted" implied volatility coincide there. Dropping the 5 censored days outright instead of recovering them moves the sign(s) Sharpe ratios of Section 5.4 by a median of 0.04 and by at most 0.06 at the midpoint and 0.08 at the crossed spread.

Signal. The two measurements are compared in variance space,

$$s_t \;=\; \widehat{RV}_t - \mathrm{IV}^2_{30,t}, \qquad \mathrm{VRP}_t \;=\; \mathrm{IV}^2_{30,t} - \widehat{RV}_t \;=\; -\,s_t, \tag{9}$$

so $s_t > 0$ says the forecast is above the price of the remaining variance and $s_t < 0$ says the premium is positive. The daily portfolio return is $R'_t = q_t R_t$ for a position q_t fixed at 15:30 from information dated t or earlier.

Position rules. Two rules are scored on the same R_t and the same days; only q_t changes.

- *Always short*: $q_t = -1$ every day. The rule reads no forecast — it is the variance-risk-premium portfolio with nothing in it — and it is the control for the other. It is called the always-short control throughout, and it is one row of every table because it is the same portfolio whichever forecast sits beside it.
- sign(s): $q_t = +1$ when $s_t > 0$ and $q_t = -1$ otherwise — long the straddle when the forecast exceeds implied variance, short when it does not, with the tie at $s_t = 0$ resolved short so that the position is never flat. Direction only, at unit size. This is the rule the section studies.

Compounding at a fixed fraction of wealth. The tables above price one unit of premium per day and add P&L up. A reinvesting trader bets a fraction f of current wealth instead, so wealth compounds as $W_T = \prod_t (1 + f\, q_t R_t)$ and the natural objective is the annualized mean of $\log(1 + f R'_t)$. Ruin is one bad day: any $f \geq 1/|\min_t R'_t|$ puts wealth at or below zero on the worst day, so the unhedged short portfolio, with a worst day near -10 premium units, can only ever bet a small fraction of wealth no matter how good its mean. The fraction is fixed at $f = 0.03$ — three percent of wealth per day, a chosen round number, the same on every day and for both rules; nothing is estimated — and every realized wealth factor is asserted positive. Here f is a share of wealth deployed as body premium, the unit of the tables above. A defined-risk variant, which would bound the worst day by construction, is parked and not reported here. The exercise is reported for every forecast of the model set in Appendix B.15.

Scoring and control. The unit of account is the expiration day. For a portfolio of n daily returns, $t = \sqrt{n}\,\overline{R'}/\hat{s}(R')$, and the annualized Sharpe ratio is $\overline{R'}/\hat{s}(R') \times \sqrt{252}$; the two carry the same information at fixed n, and both are reported so the tables read either way. The $\sqrt{252}$ is a per-*trade*-day convention: one row of the daily series is one trading period, and 252 of them make a year. This frame does not trade 252 days a year. SPXW listed Monday, Wednesday and Friday expirations before June 2022 and every session after, so over 2020–2024 it trades 200.3 days a year and reaches 252 only from mid-2022 on. A calendar-time annualization — fill the non-expiration sessions with a zero return and keep $\sqrt{252}$ — multiplies every annualized figure in this section by 0.892 and leaves every comparison between rules and columns unchanged. The convention is stated rather than switched, because the per-trade-day figure is the one a reader can recompute from a table row's own mean and standard deviation. Fills are at the 15:30 midpoint throughout, and no borrow or market-impact model is applied. The one alternative fill that is reported is the *crossed spread*: both legs bought at the ask when the rule is long and both sold at the bid when it is short, at entry only, since the position is held into a cash settlement and pays no exit spread. It is quoted beside the midpoint figure in Section 5.4 rather than replacing it, because the midpoint figure is the one every other number in this section is computed at.

The always-short control is the 15:30 end of the clock in Section 2: Hizmeri and Bevilacqua [2025] locate the predictive content of volatility uncertainty at 10:00 EST and gone by midday; by the last bar the quoted implied already embeds a premium over remaining realized variance. On the scaled comparison of Equation (8) and with the plain forecast of Equation (7), implied variance exceeds the forecast on 72 to 83% of days for the paper's eight forecasts, so the portfolio leans short more often than long. Over all 112 forecasts of that section the share runs from 66 to 83% (median 70%). What the forecast contributes at this hour is the *sign* of the day — which side of the quoted premium to stand on. Sizing beyond the sign was tested rather than assumed (Appendix B.9), with two sizes set from earlier days only and kept on the side of $\operatorname{sign}(s)$: the *robust clip*, s_t over the median of $|s|$ on earlier days clipped to $[-1, 1]$, and the *rank of* $|s|$, $\operatorname{sign}(s_t)$ times the share of earlier days with a smaller $|s|$. In point estimate the answer turns on the sample: after a year of history (614 days) the robust clip trades above $\operatorname{sign}(s)$ for 8 of the 12 forecasts (median difference +0.08) and the rank of $|s|$ for 7 (median +0.02); scored from the second day (865 days) they do so for 5 and 4 (medians −0.10 and −0.22). The rule studied here stays $\operatorname{sign}(s)$. The compounding exercise above bets a fixed fraction of wealth and reads nothing from the signal.

4.6 Implied Volatility by Root Finding

Where an option price is read as a volatility, the map is the Black–Scholes–Merton formula [Black and Scholes, 1973, Merton, 1973] at zero rate on a forward-like underlying — the futures form, and the form the option vendor's own quoted field uses. Only the product $\sigma\sqrt{\tau}$ enters that form, so it is carried here as a single *total volatility* $s = \sigma\sqrt{\tau}$ over the window that remains. For a strike K on an underlying level F,

$$V^{\mathrm{BSM}}(s; F, K) \;=\; \phi\big[\, F\,\Phi(\phi d_1) - K\,\Phi(\phi d_2)\,\big], \qquad d_1 \;=\; \frac{\log(F/K) + \tfrac{1}{2}s^2}{s}, \quad d_2 \;=\; d_1 - s, \tag{10}$$

with $\phi = +1$ for a call and $\phi = -1$ for a put and Φ the standard normal distribution function.

What is inverted is the *package*, not a leg. The recovery of Section 4.5 needs the one volatility that reprices the nearest out-of-the-money call and put together, so the root is taken of

$$V^{\mathrm{pkg}}(s) \;=\; V^{\mathrm{BSM}}(s; F, K_c) + V^{\mathrm{BSM}}(s; F, K_p) \;-\; P^*, \tag{11}$$

where P^* is the quoted package midpoint. The map $s \mapsto V^{\mathrm{pkg}}(s) + P^*$ is continuous and strictly increasing on $s > 0$ — its derivative is the sum of two positive vegas — and rises from the intrinsic value $(F - K_c)^+ + (K_p - F)^+$ at $s \to 0$ to $F + K_p$ as $s \to \infty$, so a root exists and is unique exactly when P^* lies inside that interval, and there is no closed form for it.

It is found by bisection alone, on the bracket $[10^{-8}, 1]$ in total volatility: up to one hundred halvings, stopped when the bracket is narrower than 10^{-12}, which a unit bracket reaches in the first forty. Monotonicity makes the bracket sufficient and no Newton polish is applied; a guaranteed bracket is worth more at this precision than the last digits. Package midpoints at or below intrinsic — and any quote at $\tau = 0$ — have no root, and are returned as missing rather than clipped to a bound.

Two conventions are inherited rather than chosen. First, the underlying passed in is the 15:30 index level itself and not a put–call-parity forward: on legs where the vendor's field is a genuine quote, an inversion on the index level reproduces that field to 10^{-4} and a parity-forward inversion does not, so a re-inverted leg stays on the same footing as the legs around it. Second, the vendor publishes an *hourly* standard deviation, so the total volatility returned here is converted as $\mathrm{IV}^{\mathrm{hourly}} = s/\sqrt{h}$ over the h hours that remain — at 15:30, $s\sqrt{2}$ — before it is compared with anything.

5 Results

The results proceed from estimator comparisons to nonlinear structure. Section 5.1 establishes the coefficient structure of the exogenous signal — dense but weak. Section 5.2 locates the recoverable nonlinearity. Section 5.3 reports the block-diagonal Tikhonov.

5.1 Lasso versus Ridge

The exogenous panel carries information beyond the HAR lag structure of Corsi [2009]; whether any of it is recovered is decided by the estimator. This subsection holds the information set fixed and varies only the penalty of the linear estimator. Two estimators are compared: the lasso [Tibshirani, 1996], which applies an ℓ_1 penalty and sets a subset of coefficients to exactly zero, and ridge [Hoerl and Kennard, 1970], which applies an ℓ_2 penalty and shrinks all coefficients toward zero without setting any to zero. The result of the comparison is a property of how the predictive coefficients are distributed across the columns of the design. In this section that property is measured from two sides: on loss, the causally tuned penalty families are not separated on the full design, while heavier fixed ridge shrinkage lowers the loss steadily (Section 5.1); on the coefficients, the lasso zeroes 304 of 523 exogenous columns on the full basis, of which 235 are dead or constant availability indicators

(Section 5.1). The coefficient structure is *dense but weak*, in the sense of the sparse-versus-dense prediction literature [Giannone et al., 2021, Kozak et al., 2020, Shen and Xiu, 2025]: many nonzero coefficients, none individually large. Section 4.4 defines the estimator built for this structure: ridge with a block-diagonal Tikhonov penalty whose first design uses $\alpha = 1$ on the HAR-of-target block and $\alpha = 100$ on the HAR-of-all-features block, with no selection step.

The head-to-head.

Design. The penalized linear estimators are run on the full feature design — every exogenous column at once — on the evaluation panel of Section 3, with identical rows in every cell; nothing varies between the arms except the penalty.

Comparison protocol. Both arms are compared against the shared incumbent of Section 3: per-bar OLS on HAR plus calendar features with no exogenous inputs, at the benchmark QLIKE of 0.23806. For each arm, let ℓ_t^{arm} and ℓ_t^{inc} be the per-bar QLIKE losses [Patton, 2011] of the arm and the incumbent at bar t, and let $d_t = \ell_t^{\text{arm}} - \ell_t^{\text{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the sample mean $\bar{d}$ divided by its Newey–West HAC standard error, with the small-sample correction of Harvey et al. [1997] applied; a negative t favors the arm.

Result. The comparison is run on the full design — the HAR ladder, the calendar features, and the full exogenous block — with every estimator under the same causal tuning protocol, so that no family carries an informed penalty choice the others lack (Table 3). With every forecast used as its plain back-transform, neither causally tuned ridge (0.24023, DM +1.0) nor causally tuned lasso (0.23947, DM +0.6) beats the incumbent, and the two are not separated from each other: the paired lasso-minus-ridge differential is -0.00076 at DM -0.5. The elastic net, which re-selects its mixing share together with the penalty, scores 0.23719 (DM -0.4 against the incumbent), below both, and improves on tuned ridge by -0.00304 at DM -2.1. Under one causal tuning rule, the loss ordering of the penalty families on the full design therefore does not separate dense from sparse shrinkage; the coefficient structure is read from the lasso's own solution path below.

Table 3: Ridge versus lasso versus elastic net on the full design, every forecast used as its plain back-transform: out-of-sample QLIKE with the Diebold–Mariano t-statistic against the OLS–HAR incumbent in parentheses; the second row is the paired per-bar differential against causally tuned ridge, with its own Diebold–Mariano t. Panel: the frozen panel of Section 3, $n = 273{,}489$ identical rows. Ridge is the fine-grid causal tune (`b1_ridge_tuned_fine`); lasso is `b2_lasso_tuned`; elastic net is `b3_enet_tuned`. Negative t favors the arm.

	Ridge	Lasso	Elastic net
Causally tuned	0.24023 (+1.0)	0.23947 (+0.6)	0.23719 (-0.4)
Paired vs. tuned ridge	—	-0.00076 (-0.5)	-0.00304 (-2.1)
Incumbent (per-bar OLS–HAR)		0.23806	

A penalty-level sweep locates where the tuned arms operate. Holding the ridge family fixed and varying only the constant, QLIKE falls monotonically in α from 0.28677 at $\alpha = 0.1$ through 0.26467 at 0.3, 0.25121 at 1, 0.24417 at 3, 0.24023 at 10 and 0.23820 at 30 to 0.23764 at $\alpha = 100$, and rises again at 300 (0.23922). Only the bottom of the curve reaches the incumbent: $\alpha = 100$ is the one fixed point below it, and not significantly (DM -0.2). The tuned-ridge arm, which re-selects from a grid reaching $\alpha = 1000$ on a 125-bar validation tail, lands at 0.24023, no better than the fixed $\alpha = 10$ point and short of the bottom of the curve. The penalty *level* moves the loss far more than the choice of penalty family does under a common tuning rule, and the wide design rewards much heavier shrinkage than the $\alpha = 1$ default — the direction the block penalties of Section 4.4 take further.

The diagnosis: dense, not sparse. The lasso's own solution path shows what ℓ_1 selection does on this design. On the full basis the lasso zeroes 304 of 523 exogenous columns. Of those 304, 235 — 77% — are dead or constant availability indicators. Most of the lasso's selection therefore removes columns that carry no signal, and ℓ_2 shrinkage makes the same columns small without making them zero. No small subset of live columns concentrates the signal: the selection the ℓ_1 penalty performs is, for the most part, the removal of columns that never carried any. The coefficient structure has many small nonzero entries in the coordinate basis. This is the dense-but-weak structure of Giannone et al. [2021], read on the solution path rather than inferred from the loss ordering.

The block-diagonal Tikhonov of Section 4.4 is the estimator built for this structure; its scores are in Section 5.3.

5.2 Gradient-Boosted Trees

The linear results are now in place: the exogenous signal is dense and weak (Section 5.1). The remaining question is what nonlinear structure adds. On the identical wide design, a bank of gradient-boosted trees beats the paper's two-block ridge by a margin larger than the ridge's own improvement on the benchmark; a functional-ANOVA decomposition of a tree fit locates most of its explainable variance in additive terms.

The expert bank. The twenty frozen configurations of Section 4.3 were walked per bar on the full panel, every forecast used as its plain back-transform; all 20 harvests are complete. 18 of the experts beat the benchmark's 0.23806, each at strong paired significance. Every one of the 16 Optuna-selected experts is among them (QLIKE up to 0.22886, DM between -15.3 and -9.8). Of the 4 seeded Sobol points, included deliberately as unoptimized controls, 2 beat the benchmark — one of them is the best expert in the bank, at 0.22393 — and the worst scores 0.26791: the Optuna selection buys reliability rather than the best level. The bank aggregates below are computed on the experts' full common row set (263,732 bars, 2003 onward).

Weighting beats selecting. The bank is aggregated two ways on the same trailing-125-bar tail with a 25-bar embargo, refreshed every 250 bars (the identical schedule to the linear tuners): a *selection* arm, which fills the next block from the lowest-tail-loss expert, and a *hedged* arm, which applies exponential weights $\propto \exp(-\eta \sum \ell)$ with $\eta = \sqrt{8 \ln K / L}$ — the standard weighted-average forecaster rate, no free constant. All tree rows are restricted to 2003 onward, since the menu was tuned on the pre-2003 dev span. The selection arm scores 0.22540 (DM -6.1 against the benchmark); the hedged arm scores 0.21969 (DM -17.8), and improves on the selection arm by -0.00571 at DM -5.8. That is the paper's recurring tuning lesson in its purest form: causal selection on a short tail is expensive, and weighting is cheaper than selecting. Neither aggregate beats the best individual expert identified after the fact: on their common rows the hedge trails it by +0.00106 (DM +1.1). The selection trajectory splits across families (592 XGBoost to 464 LightGBM of 1056 boundaries), and no expert dominates: the most-selected fills 28% of boundaries.

Table 4: The tree expert bank, every forecast used as its plain back-transform (Section 4.2). Individual experts are scored on the full panel; the bank aggregates are computed on their common row set (263,732 bars, 2003 onward). DM t is against the OLS–HAR benchmark (0.23806); negative favours the arm.

	QLIKE	DM t
Best individual expert (a Sobol control)	0.22393	-11.1
Median expert	0.22724	—
Weakest Optuna-selected expert	0.22886	-11.9
Sobol controls (4, unoptimized), best and worst	0.22393, 0.26791	—
`tree_tuned` (causal best-expert selection)	0.22540	-6.1
`tree_hedge` (exponential weights)	0.21969	-17.8

Comparison against the linear ladder. The tree rows above are 2003+ by construction (263,732 bars); the linear arms are scored on 248,686 to 273,489 bars under their own row rules, so pooled levels across these row sets are not compared. The comparison is paired on common rows instead: against the two-block ridge of Section 5.3, on the same wide design, the hedged bank improves by -0.01148 at DM -10.7 — more than the two-block ridge's own improvement on the benchmark (-0.00203). Against the benchmark itself the hedged bank

improves by -0.01349 at DM -17.8.

Where the nonlinearity lives. A decomposition of the tree fits was computed in an earlier phase of this project.

Interaction order. A functional-ANOVA decomposition of a fitted depth-eight boosted tree attributes approximately 55% of its explainable variance to additive terms, 9% to pairwise interactions, and 36% to a higher-order remainder concentrated in no identifiable interaction. Within the additive part, the HAR block accounts for 96%.

The decomposition locates most of a tree fit's explainable variance in additive terms carried by the HAR block; what part of the remaining margin over the linear ladder sits at pairwise order, and what part higher, is not settled by it.

5.3 Block-Diagonal Tikhonov

The block-diagonal Tikhonov of Section 4.4 is one solver with nested block penalties at a 24,000-bar window. The design is two blocks, $(\alpha_1, \alpha_2) = (1, 100)$: HAR barely penalized, the wide exogenous block shrunk hard. It scores QLIKE 0.23603, Diebold–Mariano -1.7 against the OLS–HAR incumbent: an improvement in point estimate, short of conventional significance. The same two-block design with each block's penalty re-selected causally scores 0.23347 at DM -3.8 (Appendix Table 8).

Table 5: Block-diagonal Tikhonov, every forecast used as its plain back-transform (Section 4.2), 24,000-bar window. DM t is against the OLS–HAR benchmark (0.23806); negative DM favours the arm.

	QLIKE	DM t	n
OLS on HAR + calendar (benchmark)	0.23806	—	273,489
Two-block $(1, 100)$	0.23603	-1.7	272,907

What the block-diagonal ridge carries runs in a linear estimator — one solver, two penalties, refit every bar. The next subsection reads that one-bar forecast against the last thirty minutes of the expiration-day option chain.

5.4 A Last-Bar Same-Day Option Reading

The forecasts of Table 5 are one-bar forecasts of realized variance. This subsection reads them against the last thirty minutes of the expiration-day chain, under the rules of Section 4.5: the straddle bought or sold at 15:30 ET and cash-settled at the close, with the position fixed by the sign of $s_t = \widehat{RV}_t - \mathrm{IV}^2_{30,t}$. Joining the chain to the forecasts leaves 866 expiration days common to every forecast (3 January 2020 to 30 April 2024). The 413 expiration days after that date — the most recent twenty months of chain — carry no forecast row and are not scored here; the trade's live, long-only form has been run on them (Appendix B.5), a record not scored with the plain forecast and not reported here. Every forecast is turned into a variance by one rule, the plain back-transform $\widehat{RV}_t = f_t^2 B_t$ of Equation (7), so every

row of Table 6 is scored the same way on the same days: the paper's eight forecasts, fitted on all 48 bars of the day (panel A); the pooled twins of the per-bar forecasts (panel B); and the per-bar forecasts fitted for the 16:00 bar alone, linear (panel C), and trees (panel D, one row for each of their five rungs).

Table 6: The last-30-min trade under one scorer: sign(s) on the straddle, 15:30 ET to the close, on the 866 expiration days common to every row (3 January 2020 to 30 April 2024). Every forecast is used as its plain back-transform f^2B (Equation (7)). QLIKE: that forecast's mean QLIKE at the 16:00 bar; vs R: its percent difference from the reference R, the block-diagonal ridge (negative: more accurate), and the Diebold–Mariano statistic of that difference. $\text{Sharpe}_{\text{ann}}$: the portfolio's Sharpe ratio annualized by $\sqrt{252}$ per trade day, at the 15:30 midpoint and at the crossed spread (bought at the ask, sold at the bid, at entry only; the position settles in cash). ΔSharpe: the midpoint Sharpe ratio minus that of R and of always short on the same days, a point estimate. Buys: the share of days the rule is long. Inputs: the input set of Section 4.5 by its design name; panel A reads the paper's full design. Panel D holds one row per rung of the per-bar trees of Section 4.5, so that consecutive rows differ in one thing; the other trees and input sets of every rung are counted in the text. †: the four columns still fitted on the earlier design panel (Section 4.5). The rows are selected from the closing-strategy master table, which scores 112 forecasts the same way on the same days.

			vs R		$\text{Sharpe}_{\text{ann}}$		ΔSharpe, mid		
forecast	inputs	QLIKE	%	DM	mid	crossed	vs R	vs always short	buys, %
always short (no forecast)					0.20	−0.27	−1.08		0.0
A. The paper's forecasts: one regression for all 48 bars of the day									
baseline (HAR + calendar OLS)		0.1232	+15.4	+2.55	0.43	−0.05	−0.85	+0.22	17.1
block-diagonal ridge^{R}		0.1068			1.28	0.80		+1.08	27.0
block-diagonal ridge, without the FOMC columns		0.1057	−1.0	−1.67	1.36	0.88	+0.08	+1.15	27.3
LightGBM†		0.1060	−0.7	−0.18	1.43	0.93	+0.15	+1.22	19.1
XGBoost†		0.1063	−0.4	−0.11	1.26	0.77	−0.02	+1.06	18.7
lasso (causally tuned)†		0.1090	+2.1	+0.67	1.50	1.01	+0.22	+1.30	21.2
lasso ($\alpha = 10^{-4}$)		0.1076	+0.8	+0.76	1.06	0.58	−0.22	+0.85	28.4
elastic net (causally tuned)†		0.1110	+3.9	+1.29	0.81	0.32	−0.48	+0.60	21.0
B. Pooled twins: the per-bar specification fitted on all 48 bars									
ridge	`baseline`	0.1067	−0.0	−0.01	0.91	0.43	−0.37	+0.71	26.7
ridge	`all_features`	0.1021	−4.3	−1.75	1.44	0.96	+0.16	+1.24	27.1
C. Per-bar linear: one regression for the 16:00 bar alone									
OLS	`baseline`	0.0946	−11.4	−2.38	1.37	0.89	+0.09	+1.17	27.9
ridge	`baseline`	0.0946	−11.4	−2.38	1.37	0.89	+0.09	+1.17	27.9
lasso	`baseline`	0.0946	−11.4	−2.38	1.37	0.89	+0.09	+1.17	27.9
elastic net	`baseline`	0.0946	−11.4	−2.39	1.37	0.89	+0.09	+1.17	27.9
ridge	`all_features`	0.0958	−10.3	−2.28	1.42	0.93	+0.14	+1.21	27.1
lasso	`all_features`	0.0914	−14.3	−3.72	1.88	1.40	+0.60	+1.67	29.6
elastic net	`all_features`	0.0942	−11.8	−2.93	1.92	1.43	+0.64	+1.72	28.3
D. Per-bar trees: one row per rung, the same rows, target and window									
LightGBM, shipped settings, refit every 10	`all_features`	0.0962	−9.8	−2.32	1.37	0.88	+0.09	+1.16	28.5
LightGBM, shipped settings, refit daily	`all_features`	0.0939	−12.0	−3.08	1.33	0.84	+0.05	+1.13	28.9
LightGBM, random search, refit every 10	`all_features`	0.0982	−8.0	−1.72	1.81	1.33	+0.53	+1.61	30.5
LightGBM, random search, refit daily	`all_features`	0.0966	−9.5	−2.17	1.61	1.12	+0.33	+1.40	28.8
LightGBM, Optuna tuned daily, refit daily	`all_features`	0.1010	−5.4	−1.41	1.23	0.75	−0.05	+1.03	28.4

Read the control first. Selling the straddle every day earns a mean of 0.014 premium units per day, an annualized Sharpe ratio of 0.20 at the midpoint and −0.27 at the crossed

spread: a small premium at the midpoint, and one the spread consumes.

The paper's forecasts under one scorer. Putting the paper's forecasts in charge of *direction* lifts every one of them above the control: the eight sign(s) portfolios trade at Sharpe 0.43 to 1.50 at the midpoint and −0.05 to 1.01 at the crossed spread. The trade turns on the paired comparison — both rules trade the same contracts on the same days — and paired against always short the eight improvements run from +0.22 to +1.30, three of them with a significant HAC t (LightGBM, XGBoost, lasso (causally tuned)). The paper's forecast of record, the block-diagonal ridge, trades at 1.28 (0.80 crossed), +1.08 above always short (HAC t +1.71). With no level map a forecast fitted on all 48 bars keeps whatever level it has at the 16:00 bar (Section 4.5). The same estimator without the FOMC calendar columns differs from it by +0.08 in Sharpe ratio (HAC t +0.44). The eight columns take the same side as the block-diagonal ridge on 82 to 96 percent of days. The column furthest from it, the baseline (HAR + calendar OLS), trades −0.85 against it (HAC t −1.49), and the highest of the eight, the lasso (causally tuned), trades at 1.50.

The per-bar tables, rebuilt. Every per-bar row of Table 6 comes from one re-run of the 16:00 bar that changed four things, each defined in Section 4.5 and reported in full in Appendix B.16. The per-bar design drops the 12 session-edge columns that, in a series with one bar per session, are either zero or exact copies of `har_ma_`k (34 and 640 columns become 22 and 628 on `baseline` and `all_features`). Every tree fit now drops, as every linear fit already did, the columns that are constant on its own training window or copy an earlier one; it keeps a median of 17 of 22 and 369 of 628 columns. Trees are refit every session as well as every ten, and they are tuned by Optuna as well as by the random search. Tree forecasts depend on the processor's instruction class in their last bits and, through the tuning, beyond them: re-run on a second class, LightGBM's best-of-50 choice changed on 18 of 20 tuning points and, in a whole-run check on a LightGBM forecast outside the input sets of this paper, the forecast moved by up to 5.5% of its value, while re-run on the same class every one of those points was bit-identical for all three trees. Every one of the 505 chunks behind the tree tables was therefore computed on one processor class. The linear forecasts barely move: their identifiability mask had already removed the duplicated columns, so over the master table's 37 per-bar linear forecasts QLIKE changes by at most 0.06% and at most one positions.

The forecasts built for the traded bar. Panels B to D hold the forecasts of Section 4.5 that read the 16:00 bar on their own design: the pooled twins trade at 0.91 to 1.44, the per-bar linear forecasts at 1.37 to 1.92, and the rungs of the trees at 1.23 to 1.81. Against the block-diagonal ridge, 15 of the 21 other forecasts in the table trade above it at the midpoint (0 of them with a significant HAC t) and 6 below it (0); over all 111 other forecasts of the master table, 32 trade above it (0 significantly) and 79 below (0 significantly). A table of this many forecasts on 866 days is a search, and its Sharpe ratios rank the forecasts by point estimate, not by an order the sample establishes: the highest of the 112, the per-bar elastic net on `all_features` at 1.92, does not trade significantly above the block-diagonal ridge, and taking the maximum of that many Sharpe ratios would select on the trade's own noise.

The model set as a whole. The master table scores 112 forecasts the same way on the same days. Their sign(s) Sharpe ratios have a median of 1.08 at the midpoint and 0.59 at the crossed spread and run from 0.29 to 1.92; 112 of the 112 trade above always short (median difference +0.87), 20 of them significantly. On accuracy, 105 have a lower QLIKE than the block-diagonal ridge (24 significantly) and 6 a higher one (1 significantly), at a median QLIKE of 0.1003. Across the model set the rank correlation of QLIKE with the midpoint Sharpe ratio is −0.18 (a negative value: a lower loss goes with a higher Sharpe ratio). By input set the median Sharpe ratio is 0.73 over the 52 forecasts on `baseline` and 1.27 over the 51 on `all_features`, at median QLIKE 0.1007 and 0.0993. Figure 1 draws every model of the set that is fitted on both `baseline` and `all_features`, joined from its HAR + calendar end to its all-features end.

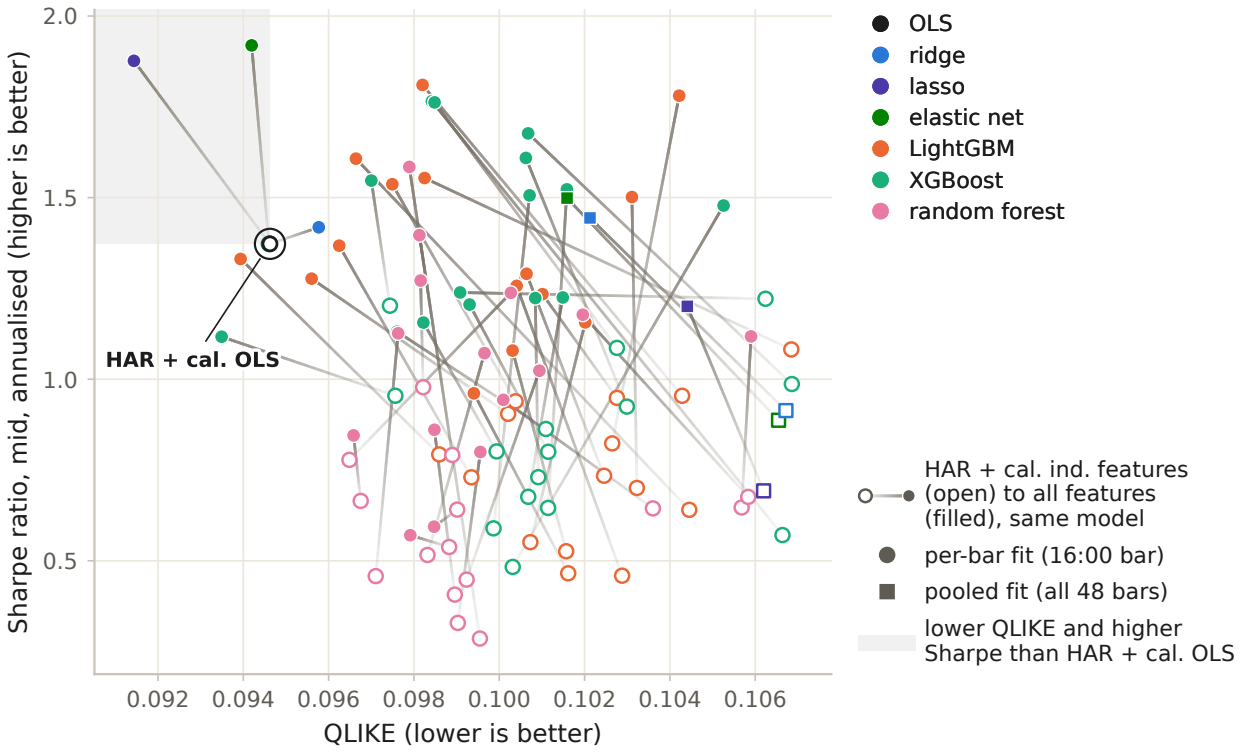


Figure 1: QLIKE against the sign(s) Sharpe ratio (midpoint) for every forecast of the model set fitted on both `baseline` (HAR + calendar) and `all_features`: 51 pairs (3 per-bar linear, 3 pooled 48-bar linear and 45 per-bar trees), on the same 866 days, every forecast its plain back-transform. Open markers are the HAR + calendar forecasts and filled markers the all-features ones; each pair is joined from its HAR + calendar end to its all-features end; circles are per-bar fits, squares pooled fits, and the colour marks the estimator. The HAR + calendar OLS (QLIKE 0.0946, Sharpe 1.37) is the reference point, and the shaded quadrant holds lower QLIKE and higher Sharpe than it (2 all-features and 0 HAR + calendar points). Adding the exogenous inputs raises the Sharpe ratio for 50 of the 51 models (median change +0.54; 4 significant by the Newey–West t of the daily midpoint-return difference) and lowers QLIKE for 38 (median change −1.7 %; 2 significant by Diebold–Mariano); 0 of the QLIKE increases and 0 of the Sharpe decreases are significant, so where QLIKE does not improve its change is within noise (significance at 5 %). The trade uses only which side of the implied variance the forecast falls on: across the pairs the change in the elementary score at that threshold correlates at −0.26 with the t of the P&L change (a lower score with a higher P&L), the QLIKE change at 0.02. Few of the gains are significant on their own, and the counts are not independent evidence: the models share the data.

One change at a time. Table 7 pairs forecasts of Table 6 that differ in one thing, every step on the same days. Replacing the block-diagonal ridge by the per-bar specification's ridge fitted on all 48 bars — design, window and penalty change together — moves the trade by +0.16 (HAC t +0.34). Fitting the 16:00 bar alone moves the trade by −0.03 on `all_features` (HAC t −0.07) while QLIKE changes by −6.2 percent; on HAR + calendar the same step changes QLIKE by −11.3 percent, a significant gain (DM −3.42), and moves

the trade by +0.46 (HAC t +1.05): a better forecast of the bar's variance is not by itself a better trade. Adding the exogenous inputs to the per-bar ridge moves the trade by +0.05 (HAC t +0.09) while QLIKE changes by +1.2 percent, and on `all_features` changing only the estimator moves the Sharpe ratio by between −0.15 (the random forest with the shipped settings) and +0.50 (the elastic net). The campaign blocks of the table walk the rungs of the 16:00 re-run for LightGBM on `all_features`: the per-window mask moves the shipped settings refit every ten sessions by −1.0% on QLIKE (DM −1.01) and −0.21 on the trade; refitting every session instead of every ten changes QLIKE by −2.4% (DM −1.23) and the trade by −0.04; against the shipped settings, both refit daily, the random search changes QLIKE by +2.9% and Optuna tuned every session by +7.5% (DM +2.87), and tuning Optuna every session rather than every 250 moves its trade by −0.32. Of all 21 steps of the table, the Sharpe ratio moves with a significant HAC t in none and QLIKE with a significant Diebold–Mariano statistic in two.

Table 7: One change at a time: paired differences between two forecasts of Table 6 that differ in one thing (the step's "to" forecast minus its "from" forecast), on the same 866 days, every forecast its plain back-transform f^2B. QLIKE: percent difference in mean QLIKE at the 16:00 bar (negative: the "to" forecast is more accurate), and its Diebold–Mariano statistic. ΔSharpe: difference in the sign(s) portfolio's annualized Sharpe ratio at the midpoint and at the crossed spread, a point estimate, with the Newey–West (HAC) t of the daily midpoint-return difference. Same side: the share of days the two forecasts take the same position. The first step changes the design, the training window and the penalty together; every other step changes one thing. Steps the master table also stores are reproduced by it exactly; the campaign blocks are the rungs of the 16:00 re-run for LightGBM on `all_features`. DM —: not stored for that pair.

step (to minus from)	QLIKE, %	DM	ΔSharpe, mid	HAC t	ΔSharpe, crossed	same side, %
From the paper's forecast to the per-bar specification						
block-diagonal ridge → pooled ridge, `all_features`	−4.3	−1.75	+0.16	+0.34	+0.16	88
Fit the 16:00 bar alone: pooled twin → per-bar ridge						
`baseline`	−11.3	−3.42	+0.46	+1.05	+0.46	88
`all_features`	−6.2	−1.56	−0.03	−0.07	−0.02	86
Inputs of the per-bar ridge						
`baseline` → `all_features`	+1.2	+0.26	+0.05	+0.09	+0.04	86
Estimator on `all_features`: per-bar ridge → alternative						
lasso	−4.5	−1.60	+0.46	+1.28	+0.47	92
elastic net	−1.6	−1.04	+0.50	+1.59	+0.50	93
LightGBM, shipped settings, refit every 10	+0.5	+0.14	−0.05	−0.13	−0.05	85
XGBoost, shipped settings, refit every 10	+1.3	+0.37	+0.13	+0.35	+0.13	85
random forest, shipped settings, refit every 10	+2.5	+0.56	−0.15	−0.34	−0.13	85
LightGBM, random search, refit every 10	+2.5	+0.56	+0.39	+0.93	+0.40	85
XGBoost, random search, refit every 10	+5.1	+1.14	+0.26	+0.46	+0.26	84
random forest, random search, refit every 10	+10.6	+1.85	−0.30	−0.81	−0.29	86
Campaign, design: the 12 session-edge duplicates dropped						
per-bar ridge, one penalty (the campaign's)	+0.0	—	+0.00	—	+0.00	100
Campaign, per-window mask: masked minus unmasked						
LightGBM, shipped settings, refit every 10	−1.0	−1.01	−0.21	−0.69	−0.21	96
LightGBM, shipped settings, refit daily	+1.4	+1.39	−0.23	−0.55	−0.23	94
LightGBM, random search, refit every 10	+2.6	+1.34	−0.08	−0.32	−0.08	94
LightGBM, random search, refit daily	+0.1	+0.03	−0.05	−0.17	−0.05	94
Campaign, refit cadence: every 10 sessions → every session						
LightGBM, shipped settings	−2.4	−1.23	−0.04	−0.08	−0.04	93
Campaign, tuning (LightGBM, refit daily)						
shipped settings → random search	+2.9	+1.05	+0.28	+0.58	+0.28	91
shipped settings → Optuna, tuned daily	+7.5	+2.87	−0.10	−0.31	−0.10	91
Optuna: tuned every 250 sessions → daily	+2.8	+0.81	−0.32	−0.65	−0.33	90

Trees, every rung. Over every tree, input set and rung of the re-run the reading of Table 7 holds. Against the per-bar ridge on the same inputs, of 48 paired QLIKE differences (three trees, two input sets, the four rungs with the shipped settings or the random search and Optuna at four tuning cadences) 46 lie above zero, 18 of them significantly, and 41 of the trees' trades lie below the ridge's, one significantly. Refitting every session lowers the QLIKE of five of the six tree forecasts (three trees, two input sets) with the shipped settings, one of them significantly. Tuning mostly makes a tree refit daily less accurate than the shipped settings refit daily: the random search is less accurate in 6 of 6 cases (one significantly) and

Optuna in 24 of 24 (12). Tuning Optuna every session, every 5 or every 25 sessions rather than every 250 is more accurate in 12 of 18 cases (none significantly). On the trial budget, the best validation score falls in the last ten of the 50 trials at 27 to 37% of tuning points, against 20% if the order did not matter, yet out of sample the best of 50 is less accurate than the best of the first 10 or 25 in 18 of 24 cases (two significantly) and more accurate in 6. The mask itself changes the trees little: a median of 0.4% in QLIKE, the masked tree less accurate in 10 of 16 cases (none significantly) and trading below the unmasked one in 14 (one significantly); XGBoost and the random forest were re-run unmasked with the shipped settings only, so for them the comparison leaves out the random search.

A tail trade. Summed without compounding, the 20 best days of a forecast's sign(s) portfolio are buys for 112 of the 112 forecasts and its 20 worst days are sells for 112, and the 20 best days carry a median of 99% of a forecast's summed P&L (59% to 338%). Against always short the summed difference is positive for 112 forecasts, at a median of $+53.5$ premium units; without its 10 best days it keeps less than half of itself for 112 of them and turns negative for 102 (median -30.0). The edge is a tail trade: a handful of long days carries it, and 866 days hold few of them.

At the crossed spread. Every Sharpe ratio of Table 6 is given at both fills. Charging the spread at entry — both legs bought at the ask when the rule is long, both sold at the bid when it is short, with nothing charged at the exit because the position settles in cash — costs every forecast's portfolio between 0.48 and 0.50 of annualized Sharpe and takes always short from 0.20 to -0.27. It is one crossing per day at the quoted touch, with no size, no queue and no model of impact — a bound on the fill rather than a fill model — but it is the crossing the venue charges.

Other structures. The straddle is one choice of instrument for sign(s). Appendix B.11 writes down 31 alternatives — strangles at several distances from the index, iron butterflies and iron condors at three wing widths, put and call verticals, call and put butterflies, 1×2 ratio spreads, and the straddle with a wing hedge on the days the rule sells — and here each is scored under every one of the 112 forecasts, per unit of straddle premium (3472 pairs). At the midpoint 755 pairs trade above the straddle, 29 of them significantly, and the straddle ranks first for 4 of the forecasts; the alternative with the highest median difference is the call vertical, w50 (median $+0.37$, above the straddle for 94 forecasts, 0 of them significantly). At the crossed spread 410 pairs trade above it (0 significantly) and the straddle ranks first for 3 forecasts. Wings cost at every width once the spread is paid: 672 of the 672 iron-butterfly and wing-hedge pairs lie below the straddle at crossed fills, against 668 at the midpoint; the median entry cost of crossing is 2.9% of the straddle's own premium and 4.0% to 6.1% of an iron butterfly's.

Diagnostics parked in Appendix D. An earlier version of this subsection read the paper's forecast under a different back-transform, since removed, and qualified it by two sample splits, the share of settlement pins, the correlation of the eight forecast columns, a decomposition of QLIKE into threshold decisions with its Murphy diagram, a same-day

regression of the settlement return on the signal, a dating (shift) test of the forecast row, and compounding at a fixed fraction of wealth. Those paragraphs are kept, verbatim, in Appendix B.12; they were computed under that other back-transform and are not set beside Table 6. The shift test, the scoring at the trade's threshold with its sample splits, and the compounding are repeated for the model set in Appendices B.13–B.15. Sizing the position by the size of the gap (Appendix B.9) turns on the sample in point estimate: after a year of history the robust clip and the rank of $|s|$ trade above $\operatorname{sign}(s)$ for 8 and 7 of the 12 forecasts, and from the second day for 5 and 4.

The trade as a whole is a demonstration under the stated constants of Section 4.5, not a trading claim: there is no delta hedge, no borrow and no market-impact model, and the crossed spread is a bound on the fill rather than a fill model. What survives is worth stating plainly. With every forecast used as its plain back-transform and on the same 866 days, the sign of the gap between forecast and quoted implied variance lifts each of the paper's eight forecasts above always short (three of them significantly), and 112 of the 112 forecasts of the master table (20 significantly). Across those forecasts the trade does not order the models: against the block-diagonal ridge 32 trade above it and 79 below, 0 and 0 of them significantly, and no step of Table 7 moves the Sharpe ratio significantly. A better forecast of the bar's variance is not by itself a better trade. The sample ends with the forecast panel in April 2024; the trade's live form after it (Appendix B.5) is not scored with the plain forecast and is not reported here.

6 Conclusion

On a quarter-million-bar intraday panel, we asked what exogenous information adds to a HAR backbone, what estimator can extract it, and what the nonlinearity is made of. The answers form a single chain. The exogenous panel carries real, statistically significant information — but the information is spread densely and weakly across hundreds of columns, so it is invisible to selection and to unpenalized joint fits, and recoverable only by shrinkage. The appropriate shrinkage is block-structured: almost none on the strong low-dimensional backbone, heavy and uniform on the wide exogenous block. The recoverable nonlinearity is not small: on the wide design the hedged tree bank improves on the two-block ridge by -0.01148 at DM -10.7, more than the ridge improves on the benchmark, and a decomposition of a tree fit places most of its explainable variance in additive terms carried by the HAR block. The linear model is ridge with a block-diagonal Tikhonov penalty, linear in its parameters throughout: two nested blocks and one exact rolling solve. With its two penalties fixed it improves on the benchmark by -0.00203 at DM -1.7, short of conventional significance; with each block's penalty re-selected causally, by -0.00459 at DM -3.8. The paper closes by reading the forecasts against the last thirty minutes of the expiration-day chain (Section 5.4). With every forecast used as its plain back-transform and on the same days, the sign of the gap between forecast and quoted implied variance carries a direction edge on the straddle: each of the paper's eight forecasts improves on always short. Across the many forecasts of the closing study's master table, including those fitted for the 15:30–16:00 bar alone, the differences in the trade's Sharpe ratio are mostly not significant and do not order the models, no single step from one forecast to another moves the trade significantly, and a better forecast

of the bar's variance is not by itself a better trade. Whether sizing on the size of the gap improves on its sign turns on the sample, in point estimate; the trade's live form after the sample ends is not yet scored with the plain forecast.

One finding sits beside the model rather than inside it, and may matter more broadly: what causal hyperparameter tuning on short validation tails buys depends on the estimator. Weighting the tree experts beats selecting one (-0.00571 at DM -5.8), and re-selecting the two block penalties causally improves on fixing them. It is measured, not assumed.

The limitations are stated where they bind. Results are computed on one asset complex and one loss, and levels are comparable only within one preprocessing regime. The natural extension is equally concrete: model confidence sets over the feature sets once per-bar losses are persisted. It would not change the chain above; it would tighten a link in it.

A Master Table of Campaign Arms

Table 8 lists every arm of the unification campaign: one frozen panel (Section 3), every forecast used as its plain back-transform (Section 4.2). QLIKE is the pooled loss of that forecast; Δ is the *paired* mean per-bar loss differential against the benchmark, and DM its Diebold–Mariano t-statistic, both computed on the row intersection of the two arms. **Row-set note.** Two row sets appear in this table: arms scored on the full evaluation panel and arms run under the `oos_mult=2` protocol (chunks 9–99, 248,686 rows), which excludes the earliest out-of-sample period. Pooled QLIKE *levels* are not comparable across protocols — the excluded rows carry above-average loss, so `oos_mult=2` arms read mechanically lower. The Δ and DM columns are paired on common rows and are comparable throughout; level comparisons should be made only within a protocol. The gated two-block ridge and the *trail*-family arms are the `oos_mult=2` panel.

Table 8: All campaign arms on the frozen panel, every forecast used as its plain back-transform.

Model	QLIKE	Δ	DM t
Benchmark			
OLS on HAR + calendar (benchmark)	0.23806	—	—
Single-bucket attribution (minimum-norm least squares)			
HAR + moments bucket	0.23753	-0.00053	-0.2
HAR + liquidity bucket	0.28502	+0.04696	+1.1
HAR + market bucket (equal-weighted)	0.23823	+0.00016	+0.3
HAR + market bucket (value-weighted)	0.23773	-0.00033	-0.5
HAR + sentiment bucket	0.23884	+0.00078	+0.6
HAR + implied-volatility bucket	0.23896	+0.00090	+1.5
HAR + volatility-demand bucket	0.24455	+0.00649	+1.0
HAR + all exogenous columns (joint)	0.92707	+0.68901	+1.2
Bucket designs under causally tuned ridge			
HAR + moments bucket	0.23581	-0.00225	-2.2

continued

Table 8 (continued)

Model	QLIKE	Δ	DM t
HAR + liquidity bucket	0.24129	+0.00323	+2.1
HAR + market bucket (equal-weighted)	0.23788	-0.00018	-0.2
HAR + market bucket (value-weighted)	0.23728	-0.00078	-0.9
HAR + sentiment bucket	0.23980	+0.00174	+1.1
HAR + implied-volatility bucket	0.23845	+0.00039	+0.5
HAR + volatility-demand bucket	0.23800	-0.00006	-0.1
HAR + all exogenous columns (joint)	0.24022	+0.00216	+1.0
Bucket designs under causally tuned elastic net (free mixing)			
HAR + moments bucket	1.49021	+1.25215	+1.2
HAR + liquidity bucket	0.24101	+0.00295	+1.6
HAR + market bucket (equal-weighted)	0.23765	-0.00041	-0.4
HAR + market bucket (value-weighted)	0.23752	-0.00054	-0.5
HAR + sentiment bucket	0.23825	+0.00019	+0.2
HAR + implied-volatility bucket	0.23957	+0.00151	+1.0
HAR + volatility-demand bucket	0.23796	-0.00010	-0.1
HAR + all exogenous columns (joint)	0.23916	+0.00110	+0.5
Penalized linear, full design			
Ridge, fixed $\alpha = 0.1$	0.28677	+0.04871	+2.5
Ridge, fixed $\alpha = 0.3$	0.26467	+0.02661	+2.4
Ridge, fixed $\alpha = 1$	0.25121	+0.01315	+2.5
Ridge, fixed $\alpha = 3$	0.24417	+0.00611	+2.4
Ridge, fixed $\alpha = 10$	0.24023	+0.00217	+1.4
Lasso, fixed $\alpha = 10^{-4}$	0.23734	-0.00073	-0.4
Ridge, causally tuned	0.24022	+0.00216	+1.0
Elastic net, causally tuned	0.23719	-0.00087	-0.4
Lasso, causally tuned	0.23947	+0.00141	+0.6
Block ridges			
Two-block ridge (stated penalties)	0.23603	-0.00203	-1.7
Two-block ridge (documented penalties)	0.23799	-0.00007	-0.0
Two-block ridge (per-block causal tuning)	0.23347	-0.00459	-3.8
Block ridges, `oos_mult=2` protocol (248,686 rows)			
Two-block ridge (gated rows, per-block causal tuning)	0.22695	-0.00471	-3.6

B Running Appendix: Side Analyses

This appendix collects analyses run alongside the paper that are not part of its argument: other entry clocks and holding periods for the same-day option trade, other forecast targets, and several checks on the last-30-min trade itself. Each subsection states the question, gives the table, states the finding, and says why it is kept out of the main text.

Conventions. Unless a subsection says otherwise: the instrument is the straddle of Section 4.5 (the nearest out-of-the-money call and put of the same-day expiry, one position); R is the return of the long straddle per unit of midpoint premium and a short earns $-R$; Sharpe ratios are annualized with $\sqrt{252}$ per trade day, as in Table 6; a crossed fill buys at the ask and sells at the bid at entry, and a position that settles in cash pays no exit spread; the forecast is the block-diagonal ridge; and the scored frame is the 866 expiration days of Table 6. Every forecast, at every bar, is used as its plain back-transform of Equation (7): the forecast f_t of $\sqrt{RV/B}$ squared, times the bar's time-of-day baseline, $\widehat{RV}_t = f_t^2 B_t$, with nothing estimated on top. Every figure is a point estimate; where a test is run, its Diebold–Mariano or t statistic or its Holm-adjusted p value is reported. The last-30-min trade is the trade of Section 5.4: entry at 15:30, cash settlement at the close. Panel stamps are bar-end labelled, so the forecast used at a clock t is the one issued at t for the bar $[t, t+30]$.

B.1 Every Thirty Minutes: A Re-Picked Straddle

Question. Does the forecast trade at every thirty-minute bar of the session, not only the last? At each clock t from 10:00 to 15:00 the nearest out-of-the-money straddle is bought or sold at its midpoint and closed at the next stamp's midpoint of the same strikes, a thirty-minute hold; the 15:30 straddle is held to settlement, so that leg is the last-30-min trade. The strikes are picked afresh at every clock. The chain carries only the same-day expiry, so no thirty-minute implied variance is quoted. It is built from the quoted variance to the close, $\mathrm{IV}^2_{\mathrm{hr},t} h_t$ (the vendor's hourly implied volatility squared, times the hours left), multiplied by w_t, the bar's average share of the rest of the day's realized variance on earlier days. The signal is $s^{\mathrm{m}}_t = \widehat{RV}_t - \mathrm{IV}^2_{\mathrm{hr},t}\, h_t\, w_t$, and a day's return is the sum of its twelve legs. Table 9 scores four rules: always short; always short with the 15:30 leg left flat; $\mathrm{sign}(s^{\mathrm{m}})$ at every clock; and always short before 15:30 with $\mathrm{sign}(s^{\mathrm{m}})$ on the 15:30 leg.

Table 9: The re-picked straddle, pooled over the twelve clocks 10:00–15:30 on the 866 expiration days. mean/day and t belong to the daily sum of the twelve legs, and the Sharpe ratio is that sum's mean over its standard deviation times $\sqrt{252}$. Crossings per day count the trades that cross the spread (a re-pick onto the same strikes with the same sign is a hold); the break-even half-spread is the mean daily midpoint profit over the mean crossings per day, in percent of premium. Panel (a) leaves flat the 13 bars whose vendor implied volatility is a censored solver value; panel (b) re-inverts it from the straddle midpoint, as the main text does, which moves the block-diagonal ridge from 2.57 to 2.55.

	mean/day	t	Sharpe (mid)	Sharpe (crossed)	crossings per day	break-even half-spread (% prem.)	% bars bought
(a) four rules, block-diagonal ridge							
always short	0.160	3.53	1.90	−1.92	16.4	0.98	0.0
always short, flat at 15:30	0.146	6.36	3.43	−3.62	16.0	0.91	0.0
sign(s^{m})	0.213	4.76	2.57	−1.69	17.8	1.19	28.4
always short, sign(s^{m}) at 15:30	0.239	5.48	2.96	−1.12	16.6	1.44	2.2
(b) sign(s^{m}) *for each forecast*							
baseline (HAR + calendar OLS)	0.161	3.58	1.93	−2.36	—	—	—
block-diagonal ridge	0.212	4.72	2.55	−1.72	—	—	—
block-diagonal ridge, without the FOMC columns	0.219	4.88	2.63	−1.64	—	—	—
LightGBM	0.223	5.07	2.73	−1.62	—	—	—
XGBoost	0.216	4.84	2.61	−1.71	—	—	—
lasso (causally tuned)	0.216	4.73	2.55	−1.64	—	—	—
lasso ($\alpha = 10^{-4}$)	0.193	4.33	2.34	−1.94	—	—	—
elastic net (causally tuned)	0.169	3.78	2.04	−2.23	—	—	—

Finding. At the midpoint every rule earns (Sharpe 1.90 to 3.43) and sign(s^{m}) is ahead of always short (2.57 against 1.90; the eight forecasts 1.93 to 2.73). At the crossed spread every rule loses (−3.62 to −1.12; the eight forecasts −2.36 to −1.62). The book crosses the spread 16.0 to 17.8 times a day, and the median half-spread, 1.69% of premium, is above every rule's break-even half-spread (0.91% to 1.44%). The least negative rule at the crossed spread is the one that uses the forecast only on the settling 15:30 leg (−1.12).

Why it is not in the main text. No rule survives the spread, and the implied side of the daytime signal is a construction — a profile share of a quote for the rest of the day — not a price.

B.2 Time to the Close: Holding from Every Clock, Delta-Hedged

Question. Enter at any clock and hold to the close: does a forecast of the rest of the day, read against the quoted variance for the rest of the day, pick the side? At each clock t from 10:00 to 15:30 the nearest out-of-the-money straddle is bought or sold at its midpoint and held to cash settlement, twelve overlapping trades a day. Every position is delta-hedged in the index: at the entry stamp and at every later thirty-minute stamp to 15:30, the straddle's Black–Scholes delta (entry strikes, that stamp's spot, and that stamp's implied volatility over the hours left) is offset by index units held to the next stamp, the last from 15:30 to the official close. The hedge trades at the vendor spot at no cost, and R_t adds its profit to the settlement value. The signal stretches the forecast to the rest of the day, $s_t = \widehat{RV}_t/w_t - \mathrm{IV}^2_{\mathrm{hr},t} h_t$, which

has the sign of s_t^{m} above; at 15:30, $w_t = 1$ and $h_t = \frac{1}{2}$, and s_t is the signal of the last-30-min trade.

Table 10: Holding to the close from every clock, delta-hedged, pooled over the twelve entry clocks 10:00–15:30 (10,387 trades); columns as in Table 9. Each trade crosses the spread once, at entry. The 15:30 leg column scores that leg alone at the crossed spread.

	days	mean/day	t	Sharpe (mid)	Sharpe (crossed)	mean/day (crossed)	15:30 leg Sharpe (crossed)	% bars bought
always short	866	1.014	5.44	2.93	2.21	0.766	−0.23	0.0
sign(s)	866	0.723	5.52	2.98	1.96	0.475	0.72	28.4

Table 11: The position against the outcome, by entry clock (midpoint, hedged R). Accuracy is the share of days on which the position had the sign of R: a buy with $R > 0$ or a sale with $R \leq 0$. The accuracy of always short is $1 - P(R > 0)$. The bars with a censored implied volatility are left out.

entry	days	accuracy sign(s)	accuracy always short	buy share	$P(R > 0)$	$P(R > 0 \mid \text{buy})$	$P(R \leq 0 \mid \text{sell})$
10:00	864	0.601	0.660	0.253	0.340	0.384	0.674
10:30	865	0.612	0.686	0.243	0.314	0.348	0.696
11:00	865	0.572	0.679	0.319	0.321	0.333	0.684
11:30	864	0.596	0.675	0.292	0.325	0.365	0.691
12:00	866	0.576	0.630	0.311	0.370	0.413	0.650
12:30	866	0.567	0.648	0.321	0.352	0.374	0.658
13:00	866	0.559	0.642	0.386	0.358	0.392	0.664
13:30	866	0.600	0.648	0.237	0.352	0.400	0.663
14:00	864	0.549	0.634	0.375	0.366	0.386	0.646
14:30	865	0.551	0.609	0.280	0.391	0.397	0.612
15:00	862	0.581	0.618	0.125	0.382	0.352	0.614
15:30	861	0.574	0.627	0.269	0.373	0.401	0.638

A forecast fitted on the rest of the day. The stretch $\widehat{RV}_t/w_t$ reaches the rest of the day from a one-bar forecast. Three maps fitted on the rest-of-day variance itself replace it, one regression per entry clock for $\log RV_t^{\mathrm{rem}}$ on sessions strictly before the day: M1 on the log of the next-bar forecast, M2 adding the realized variance from the open to t, and M3 adding the previous session's rest-of-day variance (the constructions of Section B.8). Each fitted value is back-transformed by the exponential alone. The 15:30 clock is left out because there the rest of the day is one bar.

Table 12: Rest-of-day signals on the eleven entry clocks 10:00–15:00 (9,521 trades, the 866 days). QLIKE scores each forecast against the realized variance from t to the close. sign(s^{rem}) trades $F_t^{\text{rem}} - \text{IV}_{\text{hr},t}^2 h_t$ with the fitted map in place of the stretch. ΔSharpe is the paired difference from the profile-stretch row at the midpoint.

	QLIKE, rest of day	Sharpe (mid)	Sharpe (crossed)	% bars bought	% bars sign differs	ΔSharpe (mid)
always short (no forecast)	—	3.38	2.65	0.0	—	—
sign(s), profile stretch $\widehat{RV}_t/w_t$	0.2629	3.00	2.00	28.5	—	—
sign(s^{rem}), M1	0.2393	2.91	1.93	32.4	5.7	−0.09
sign(s^{rem}), M2	0.2378	2.30	1.41	33.4	10.9	−0.70
sign(s^{rem}), M3	0.2235	2.26	1.34	32.9	13.2	−0.74

Finding. Held to the close and hedged, both rules pay their spread (Table 10): always short 2.93 at the midpoint and 2.21 crossed, sign(s) 2.98 and 1.96. The forecast is ahead of the forecast-free control at the midpoint and behind it at the crossed spread. The direction rule is right on 0.55 to 0.61 of days at each clock; always selling is right on 0.61 to 0.69, more often at 12 of the 12 clocks (Table 11). Where it buys, the straddle wins more often than on an average day at 11 of the 12 clocks. The fitted rest-of-day maps forecast the rest of the day better than the stretch (pooled QLIKE 0.2393, 0.2378 and 0.2235 against 0.2629), buy more often (32.4% of bars for M1 against 28.5%), and trade worse: Sharpe 2.91, 2.30 and 2.26 against 3.00 at the midpoint, with 6 of the 6 paired differences (two fills, three maps) below zero, from −0.09 for M1 to −0.74 for M3 at the midpoint.

Why it is not in the main text. Clock by clock before 15:30 the forecast does not improve on the forecast-free control (Section B.3), the daytime trades rest on a frictionless hedge the main text does not model, and the twelve trades of a day overlap.

B.3 The Sharpe Ratio Through the Day

Question. How does the Sharpe ratio of each rule change with the clock at which the position is taken? Table 13 keeps one clock at a time, one trade per expiration day, for the two books above: the thirty-minute re-pick (unhedged) and the delta-hedged hold to the close. The 09:35 row is the first stamp of the chain; it is scored with the forecast issued at 09:30 for the 09:30–10:00 bar, and only in the hold-to-close book.

Table 13: Annualized Sharpe ratio by entry clock, one trade per expiration day. Re-pick: the thirty-minute hold of Section B.1 (its 15:30 row is the last-30-min trade). To close: the delta-hedged hold of Section B.2. sign(s) is the block-diagonal ridge; the last column is the range over the eight forecasts of Table 6. Midpoint fills unless marked crossed. The censored implied volatility is re-inverted as in the main text; the hold-to-close notebook, which leaves those bars flat, differs from the sign(s) column by at most 0.04 (15:30: 1.18).

entry	days	re-pick always short	re-pick sign(s)	to close always short	to close always short, crossed	to close sign(s)	to close sign(s), crossed	to close, sign(s) eight forecasts
09:35	863	—	—	3.11	2.44	1.33	0.67	1.12 to 2.14
10:00	864	0.26	0.07	3.32	2.29	2.44	1.44	1.90 to 2.48
10:30	865	0.64	0.52	4.18	3.51	2.80	2.15	2.35 to 3.14
11:00	865	0.46	0.44	4.20	3.58	2.28	1.68	2.28 to 2.79
11:30	865	2.19	2.42	3.95	3.33	2.86	2.25	2.21 to 2.92
12:00	866	1.12	1.20	3.17	2.58	1.73	1.15	1.17 to 1.96
12:30	866	1.78	0.82	3.30	2.72	1.69	1.12	1.25 to 2.09
13:00	866	0.23	1.13	2.66	2.09	1.68	1.12	1.33 to 1.94
13:30	866	0.79	0.82	2.51	1.97	2.06	1.53	1.47 to 2.09
14:00	866	0.50	0.72	2.04	1.36	1.67	1.00	1.24 to 1.86
14:30	866	1.00	0.59	2.00	1.50	0.81	0.31	0.61 to 1.01
15:00	866	2.51	1.56	1.93	1.40	1.22	0.69	0.94 to 1.51
15:30	866	0.20	1.28	0.24	−0.23	1.15	0.68	0.39 to 1.48

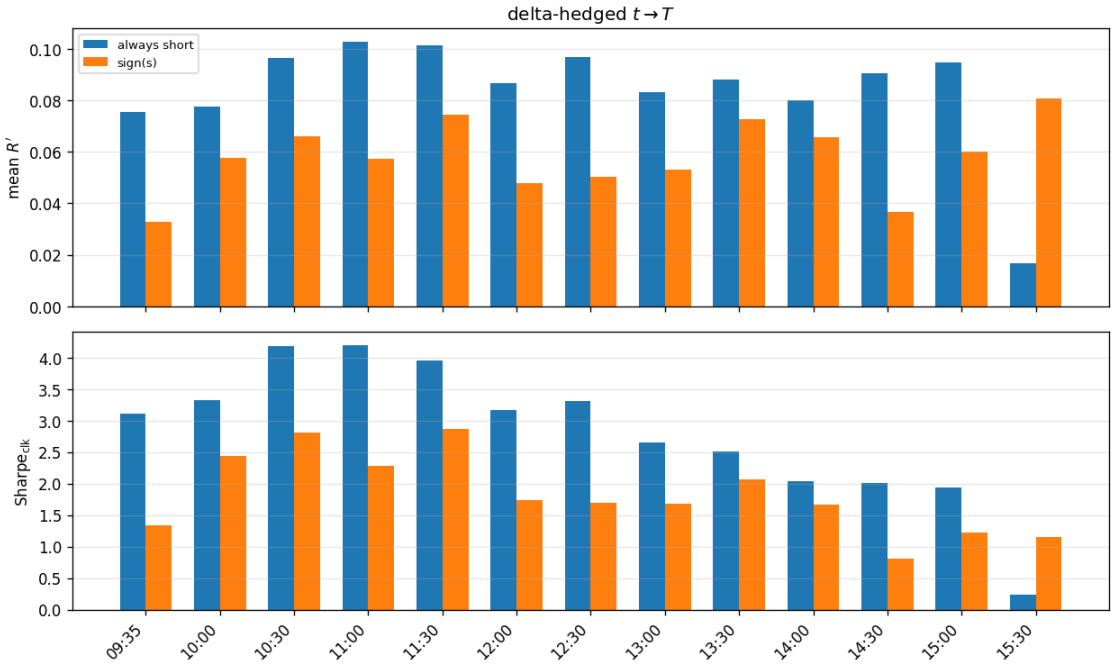


Figure 2: Delta-hedged hold to the close by entry clock, block-diagonal ridge: mean return (top) and annualized Sharpe ratio (bottom) of always short and sign(s) (holdclose notebook, where the bars with a censored implied volatility sit flat).

Finding. Hedged and held to the close, always short is strongest in the late morning (4.20 at 11:00, 4.18 at 10:30) and falls through the afternoon to 1.93 at 15:00 and 0.24 at 15:30

(−0.23 crossed). At 12 of the 12 clocks from 09:35 to 15:00 it is ahead of the sign(s) rule of every one of the eight forecasts at the midpoint; at the crossed spread it is ahead of the ridge's at 12 of them. The order reverses only at 15:30: sign(s) 1.15 (0.39 to 1.48 across the eight), 0.68 crossed. At the 09:35 open always short earns 3.11 (2.44 crossed) and the ridge's sign(s) 1.33 (0.67). In the unhedged re-pick the always-short Sharpe ratio moves between 0.20 and 2.51 from clock to clock, and sign(s) is ahead at 6 of 12 clocks, all before the spread.

Why it is not in the main text. What varies through the day is the premium earned by the forecast-free control, a different object from the one-bar forecast the paper studies; the forecast is ahead of that control only at the last bar, which is the main text's trade.

B.4 A Forecast Built for the Rest of the Day

Question. The rest-of-day signals above reach the close from a one-bar forecast. Does a model fitted directly on the rest of the day forecast it better? For every entry clock c from 09:30 to 15:30 the per-bar linear estimators (ridge, lasso, elastic net; 2000-session rolling window, causal penalty tuning) are fitted with the realized variance from c to the 16:00 close as the target, on two column sets of the per-bar models, HAR + calendar and all features. Each is compared on identical rows with the one-step construction of the same estimator and column set: the forecast of the bar starting at c, $f_c^2 B_c$, divided by that bar's prior-days share of the rest of the day. The direct forecast is back-transformed the same plain way — the squared prediction times the arm's causal per-clock profile of the rest-of-day variance.

Table 14: Direct rest-of-day forecasts against the one-step construction, 1,404–1,406 sessions from 2018-09-24 to 2024-04-30. ΔQLIKE is $100\,(\mathrm{QLIKE}_{\mathrm{direct}}/\mathrm{QLIKE}_{\text{one-step}} - 1)$ at one entry clock; the median is over the twelve clocks 09:30–15:00. Clocks better (worse) count the clocks at which the direct forecast's loss is lower (higher) with a Diebold–Mariano p below 0.05; clocks higher counts point estimates above zero. At 15:30 the two constructions are the same forecast.

column set	estimator	sessions	median ΔQLIKE (%)	clocks better	clocks worse	clocks higher (point)	15:30 |Δ| (%)
HAR + calendar	ridge	1,404–1,406	−18.8	11	0	0	0.000
HAR + calendar	lasso	1,404–1,406	−18.0	11	0	0	0.000
HAR + calendar	elastic net	1,404–1,406	−17.8	11	0	0	0.000
all features	ridge	1,404–1,406	−39.6	8	0	1	0.000
all features	lasso	1,404–1,406	−36.3	9	0	1	0.000
all features	elastic net	1,404–1,406	−44.5	8	0	1	0.000

Finding. The direct forecast has the lower loss at the median clock for every estimator and column set: −19% to −18% on HAR + calendar and −44% to −36% on all features. It is significantly better at 8 to 11 of the twelve clocks and never significantly worse; at 15:30 the two are the same forecast (largest difference 0.000%). The plain back-transform has rare extreme values: 4 of the 72 clock cells carry a QLIKE above 1 on one side or the other, the largest the all features ridge at 14:00 (132.0 against 0.197 for the one-step construction), and

the same kind of value gives the all-features ridge a plain QLIKE of 2.65 at 15:30. None of those cells is a significant comparison, which is why the table reports the median clock and per-clock tests rather than a pooled mean.

Trading test. The direct forecast replaces the stretch in the delta-hedged hold-to-close signal of Section B.2, $s_t = F_t^{\text{rem}} - \text{IV}_{\text{hr},t}^2 h_t$, on the same days, bars, fills and hedge (Table 15; per-bar ridge, plain back-transform on both sides). Pooled over the entry clocks 10:00–15:30 it trades better by a point estimate: on all features from 2.23 to 2.54 at the midpoint (+0.31) and from 1.13 to 1.58 crossed (+0.45); on HAR + calendar by +0.12 at the midpoint and +0.20 crossed. Over all 6 estimator and column-set combinations the direct forecast is ahead in 6 at the midpoint and 6 at the crossed spread. It buys less often (37.9% of bars against 44.2% on all features), the mirror image of the fitted maps of Table 12, and it changes the QLIKE on these bars by −62.1%; both signals stay below always short (direct minus always short −0.40, all features, midpoint).

Table 15: Trading test of the direct rest-of-day forecast in the delta-hedged hold-to-close book, per-bar ridge, on the 866 days (the generator's table, read as generated). Top: pooled over the entry clocks 10:00–15:30 (daily sums), Sharpe ratios of the current signal ($\widehat{RV}_t/w_t$) and the direct one (F_t^{rem}) and their paired difference, plain back-transform on both sides. Bottom: by entry clock, midpoint.

		midpoint			crossed spread			buy %	
bucket	F	current	direct	direct − current	current	direct	direct − current	cur.	dir.
No rest-of-day forecast: always short 2.93 / 2.21; block-diagonal ridge sign(s) *2.98 / 1.96 (midpoint / crossed)*									
all features	plain	2.23	2.54	+0.31	1.13	1.58	+0.45	44	38
HAR + calendar	plain	1.33	1.45	+0.12	0.20	0.40	+0.20	46	45

	always	all features			HAR + calendar		
entry	short	current	direct	direct − current	current	direct	direct − current
09:35	3.11	1.35	1.06	-0.29	1.04	1.48	+0.44
10:00	3.32	0.64	1.60	+0.96	-0.07	0.85	+0.92
10:30	4.18	0.53	1.49	+0.96	0.15	0.91	+0.76
11:00	4.20	0.83	1.78	+0.95	0.76	0.93	+0.16
11:30	3.95	0.11	1.52	+1.41	0.25	0.35	+0.10
12:00	3.17	0.85	1.04	+0.18	0.50	0.26	-0.24
12:30	3.30	1.54	1.16	-0.39	0.84	0.76	-0.08
13:00	2.66	1.48	1.19	-0.30	1.47	0.42	-1.04
13:30	2.51	1.42	1.76	+0.35	0.71	0.71	-0.00
14:00	2.04	1.74	1.95	+0.21	1.15	1.10	-0.05
14:30	2.00	1.06	1.22	+0.15	0.34	0.90	+0.57
15:00	1.93	0.74	1.53	+0.79	-0.00	0.45	+0.45
15:30	0.24	1.85	1.85	same forecast	1.31	1.31	same forecast

Why it is not in the main text. The paper's target is the one-bar variance and its rest-of-session sections are parked. A better rest-of-day loss has also not been shown to be a better trade: the fitted maps of Table 12 forecast the rest of the day better than the stretch and trade worse.

B.5 After the Research Sample: May 2024 to September 2026

The trade's live, long-only form has been run forward after the forecast panel ends. That record scores the live forecast pipeline, which is not the plain back-transform used in the rest of this paper, and it has not yet been re-scored with it; its tables and findings are held out of this version.

B.6 The Month-End Long Straddle

Question. On the last session of each month the long-only form of the trade used live buys the straddle at any price, with no forecast. Is 15:30 the right clock for that purchase? Study 83 buys, at each clock, the straddle nearest the spot at the ask and settles it at the close.

Table 16: The month-end long straddle by entry clock, 70 month-ends from 2020-01-31 to 2025-12-31, bought at the ask. The spread is the median relative spread at entry; minus 15:30 is the paired difference in mean R, with its paired t and the Holm-adjusted p over the twelve earlier clocks.

entry	median ask	median spread (%)	mean R	t	hit (%)	minus 15:30	paired t	Holm p
09:35	23.80	1.9	0.152	1.41	46	−0.281	−1.54	1.00
10:00	22.75	3.2	0.211	1.84	47	−0.222	−1.25	1.00
10:30	20.10	2.2	0.190	1.61	54	−0.242	−1.39	1.00
11:00	18.80	2.2	0.190	1.58	53	−0.242	−1.51	1.00
11:30	18.30	2.3	0.202	1.76	51	−0.230	−1.46	1.00
12:00	16.40	2.4	0.238	1.98	54	−0.194	−1.28	1.00
12:30	15.05	2.5	0.309	2.27	54	−0.124	−0.75	1.00
13:00	15.15	2.6	0.269	1.98	51	−0.163	−1.04	1.00
13:30	14.25	2.9	0.301	2.35	51	−0.132	−0.96	1.00
14:00	13.15	2.9	0.422	2.99	53	−0.011	−0.08	1.00
14:30	11.70	2.9	0.355	2.50	56	−0.077	−0.67	1.00
15:00	9.95	4.0	0.267	1.82	53	−0.166	−1.56	1.00
15:30	8.30	4.9	0.432	2.62	54	—	—	—

Finding. The purchase pays at every clock and most at 15:30 (mean R 0.432, $t = 2.62$); the earlier clocks earn 0.152 to 0.422, and no difference from 15:30 survives the Holm adjustment (smallest p 1.00). The 15:30 return is 0.189 over the first half of the month-ends and 0.676 over the second, from 2022-12-30. On the 40 SPX month-ends that study 86 buys at every clock from 15:00 to 15:50, the 15:30 purchase earns 0.356 at the ask ($t = 1.45$), and entries from 15:40 to 15:50 are 0.11 to 0.13 higher, with Holm p of 1.00.

Why it is not in the main text. It is a calendar trade that reads no forecast.

B.7 Per-Bar Against Pooled Forecasts

Question. A per-bar forecast fits one regression for each thirty-minute bar; its pooled twin fits one regression for all bars. Which is the better variance forecast? QLIKE is computed bar by bar for the 13 session bars on the 866 days, each forecast its plain back-transform. The per-bar forecasts are the ridge on HAR + calendar and the ridge, lasso and elastic net on all features.

Table 17: Per-bar forecast against its pooled twin: the percent difference in mean QLIKE (negative: the per-bar forecast is better), with the Diebold–Mariano t of the mean daily loss difference in parentheses, over the session (each day's mean over the 13 bars) and on three bars labelled by bar end. Bars better (worse) count the bars whose DM t is below -1.96 (above 1.96).

per-bar forecast	session mean (%) (DM t)	14:30 bar (%) (DM t)	15:00 bar (%) (DM t)	16:00 bar (%) (DM t)	bars better	bars worse
ridge (HAR + calendar)	−10.0 (−6.73)	−29.1 (−6.26)	−13.6 (−4.43)	−11.3 (−3.88)	5	0
ridge (all features)	+99.8 (1.00)	−35.4 (−3.68)	+20.4 (1.25)	+12.7 (0.90)	1	3
lasso (all features)	+15.9 (0.62)	−37.0 (−3.45)	−8.7 (−1.49)	−5.0 (−1.98)	2	0
elastic net (all features)	+0.3 (0.03)	−39.1 (−3.63)	−11.5 (−2.40)	−1.3 (−0.34)	2	2

Finding. Over the session 1 of the 4 per-bar forecasts has the lower loss, the ridge on HAR + calendar (−10.0%); those on all features are higher, up to 99.8% for the ridge. Most of that excess sits at one bar: at the bar ending 15:30 the per-bar ridge on all features has a QLIKE of 2.32 against 0.134 for its pooled twin (1625%, DM t 1.01), the rare extreme values of the plain back-transform met in Section B.4. Every per-bar forecast gains at one bar, 14:00–14:30 (−39.1% to −29.1%, DM t −6.3 to −3.5), the bar that holds the FOMC statement: on the 33 statement days the difference there is −72% to −41%, those days carry 0.49 to 0.56 of the pooled forecasts' loss at that bar, and on the other 833 days it is −14% to −1%. On the traded 16:00 bar 2 of the 4 have a DM t below −1.96; the largest gain is that of the ridge (HAR + calendar) (−11.3%, DM t −3.88).

Why it is not in the main text. The per-bar gain is a calendar effect at a bar the trade does not use, and on the wide column set the plain per-bar forecasts give it back on other bars.

B.8 Remaining-Variance Forecasts Across the Session

Question. How well does each one-bar model extend to the variance remaining until the close, at every entry clock, using only what is known at the clock? Three regressions for the log of the remaining variance, M1 to M3 of Table 18, are fitted per entry clock on strictly earlier days (at least 63) and back-transformed by the exponential alone; M0, the per-clock

mean of past log rest-of-day variance, uses no model. Each is built for the HAR OLS baseline and for the block-diagonal ridge, on the full panel: 5,645 days from 2001-10-18 to 2024-04-30.

Table 18: QLIKE of the rest-of-day forecasts, pooled over the entry clocks 10:00–15:30 (the target runs to the 16:00 close), on all 5,645 days and on the 2,089 days from 2016. DM t compares the block-diagonal ridge with the HAR OLS baseline in the same construction (negative: the ridge is better).

construction	all days HAR OLS	all days block ridge	all days DM t	from 2016 HAR OLS	from 2016 block ridge	from 2016 DM t
M0: per-clock mean of past $\log RV^{\text{rem}}$ (no model)	1.0892	—	—	1.1433	—	—
M1: log of the next-bar forecast	0.1728	0.1555	−4.70	0.2386	0.2101	−3.01
M2: M1 + realized variance so far	0.1672	0.1506	−6.19	0.2322	0.2049	−3.96
M2, one fit with clock dummies	0.1675	0.1511	−6.15	0.2317	0.2051	−3.90
M3: M2 + yesterday's RV^{rem}	0.1612	0.1463	−6.59	0.2200	0.1967	−4.07

Finding. The block-diagonal ridge beats the baseline in every construction and era ($|t|$ from 3.0 to 6.6); M1 already has 0.1555 against 0.1728, and M3 is the best, 0.1463, against 1.089 for the model-free clock mean. The loss drops between the 14:00 and 14:30 entries (M1, block ridge: 0.221 to 0.143) because the 14:00–14:30 bar, the FOMC statement bar, leaves the window. On the option days the same maps do not trade better (Table 12): M3, the best here, trades worst there.

Why it is not in the main text. The paper's rest-of-session sections are parked.

B.9 Sizing by the Size of the Forecast Gap

Question. sign(s) bets the same amount every day. Does a position that grows with the gap do better? Two sizes, each set from earlier days only and each on the side of sign(s): the robust clip, $q_d = \text{clip}(s_d/\tilde{m}_d, -1, 1)$ with $\tilde{m}_d$ the median of $|s|$ over the earlier days, a full position once the gap reaches its usual size and a smaller one below it; and the rank, sign(s_d) times the share of earlier days with a smaller $|s|$. The first block scores the days after a year of history, the second every day from the second. The forecasts are the paper's eight and the per-bar ridge, lasso and elastic net (HAR + calendar and all features).

Table 19: Sized rules against sign(s) on the same days. The ridge columns give the block-diagonal ridge's mean absolute position, its sized Sharpe ratio and the difference from sign(s). The count columns report, over the 12 forecasts, how many differences are positive, and their median.

size	sample	days	ridge mean $\|q\|$	ridge Sharpe (mid)	ridge: minus sign(s) (mid)	12 forecasts (mid): number > 0	median (mid)	12 forecasts (crossed): number > 0	median (crossed)
robust clip, $\text{clip}(s_d/\tilde{m}_d, -1, 1)$	after 252 days	614	0.60	1.64	+0.09	8	+0.08	8	+0.09
rank of $\|s_d\|$, sign kept	after 252 days	614	0.41	1.59	+0.04	7	+0.02	7	+0.03
robust clip, $\text{clip}(s_d/\tilde{m}_d, -1, 1)$	from day 2	865	0.61	1.22	−0.08	5	−0.10	5	−0.10
rank of $\|s_d\|$, sign kept	from day 2	865	0.42	1.06	−0.24	4	−0.22	3	−0.23

Finding. After a year of history both sizes are ahead of sign(s) for most forecasts, by small amounts: the robust clip for 8 of the 12 (median +0.08 at the midpoint and +0.09 crossed, from −0.19 to +0.74), the rank for 7 (median +0.02 and +0.03). For the block-diagonal ridge sign(s) earns 1.55 on those 614 days, the clip 1.64 (+0.09) and the rank 1.59 (+0.04). From the second day, when the first sizes rest on a few days of history, the order turns: the clip is ahead for 5 and the rank for 4 of the 12 (medians −0.10 and −0.22).

Why it is not in the main text. The gains are small and do not hold from the second day; the main text states only that the position uses the sign.

B.10 The Trade Beside the S&P 500 and Other Markets

Question. An investor already holds the S&P 500. How much of the sign(s) trade belongs beside it, and is the trade a disguised bet on another market? The weight is the maximum-Sharpe mix of the two risk-scaled daily returns, re-estimated every day from earlier days only (252-day warm-up, then expanding; no short positions), at the midpoint.

Table 20: The trade beside the S&P 500, 614 days from 2021-08-11. Correlations are those of the trade's daily return with the S&P 500 over the trade's own window (15:30 to the close), with the size of that move, and close to close. The risk share is the trade's share of the risk budget; the dollar column is the median premium held per $100 of index. The last column is the combination's Sharpe ratio minus that of the S&P 500 alone (0.40).

forecast	corr. S&P 15:30→close	corr. |S&P| 15:30→close	corr. S&P close-to-close	mean risk share	premium $ per $100 S&P	Sharpe combined	minus S&P alone
baseline (HAR + calendar OLS)	0.12	−0.49	0.04	0.26	0.13	0.32	−0.09
block-diagonal ridge	−0.02	−0.28	−0.01	0.70	3.93	1.44	+1.03
block-diagonal ridge, without the FOMC columns	−0.02	−0.27	−0.01	0.71	4.20	1.56	+1.16
LightGBM	−0.03	−0.29	0.02	0.80	6.37	1.33	+0.92
XGBoost	0.04	−0.39	0.03	0.76	5.05	1.04	+0.64
lasso (causally tuned)	0.01	−0.31	0.01	0.75	5.03	1.73	+1.32
lasso ($\alpha = 10^{-4}$)	−0.01	−0.25	−0.01	0.50	2.35	1.04	+0.64
elastic net (causally tuned)	0.00	−0.31	−0.01	0.59	2.65	0.82	+0.42
per-bar ridge (HAR + calendar)	0.12	−0.20	0.05	0.76	5.57	1.47	+1.07
per-bar ridge (all features)	−0.01	−0.26	−0.01	0.81	6.41	1.38	+0.97
per-bar lasso (all features)	−0.02	−0.28	0.01	0.86	9.69	1.59	+1.18
per-bar elastic net (all features)	−0.03	−0.27	0.01	0.84	8.05	1.67	+1.26

Table 21: Correlation of the trade's daily return (midpoint, block-diagonal ridge) with same-day moves of other markets, on the 866 days wherever both exist.

same-day move	days	block-diagonal ridge
S&P 500 return, 15:30→close	866	−0.040
S&P 500 |return|, 15:30→close	866	−0.203
S&P 500 return, close-to-close	866	−0.002
S&P 500 |return|, close-to-close	866	−0.065
VIX change, close-to-close (points)	861	−0.003
VIX change, 15:30→16:00 (panel)	795	0.047
Nasdaq-100 return, close-to-close	866	−0.022
Russell 2000 return, close-to-close	866	−0.001
10-year Treasury yield change, close-to-close (points)	856	0.044
long Treasuries (TLT) total return, close-to-close	864	−0.048
high-yield credit (HYG) total return, close-to-close	858	−0.015
dollar index return, close-to-close	856	−0.020
realized minus implied variance, 15:30→16:00	866	−0.032

Finding. The trade barely moves with the index's direction (close-to-close correlation −0.01 to +0.05 across the 12 forecasts) and moves against the size of the last half hour's move (−0.49 to −0.20), since the rule sells on most days. Earlier days put most of the risk budget in it: for the block-diagonal ridge a mean risk share of 0.70, a median $3.93 of premium per $100 of index, and a combined Sharpe ratio of 1.44 against 0.40 — a gain of 1.03. Across the forecasts the gain runs from −0.09 to 1.32, below zero for 1 of them. Of the 13 same-day series, the block-diagonal ridge's trade moves most with the size of the S&P 500's move from 15:30 to the close (−0.20), and with no other by more than 0.07 in absolute value.

Why it is not in the main text. The paper is about the forecast, not portfolio construction.

B.11 Strategy variations of the last-30-min trade

The straddle — the nearest out-of-the-money call and the nearest out-of-the-money put, same-day expiry, one position — is one choice of instrument for the sign(s) rule. Table 22 writes down the standard alternatives and the way the rule maps onto each: a variation is written as the structure L bought when $s > 0$, and it is sold (the position is $-L$) when $s < 0$; always short sells it every day. The straddle with a wing hedge is the one exception: it holds the plain straddle on buying days and the short iron butterfly on selling days. All structures are same-day expiry SPXW options entered at the 15:30 quotes and cash-settled at the official close, on the same 866 days as the straddle. Strikes are the live 15:30 strikes (K_c the smallest call strike at or above the index S, K_p the largest put strike at or below it); a wing is the nearest live strike at least $w \in \{10, 25, 50\}$ index points beyond the body strike. Fills are the midpoint of every leg, or crossed (every leg bought at the ask and sold at the bid). The return is the day's profit and loss in index points divided by that day's straddle

premium at the same fill, so every row shares the straddle row's denominator and a paired difference isolates the structure; the straddle row is the straddle trade's return. The forecast is used as its plain back-transform, the forecast of $\sqrt{RV/B}$ squared times the time-of-day baseline B. Every figure is a point estimate; a difference is paired against the straddle on the same days, rule and fill.

Table 22: The strategy variations. $C(K)$, $P(K)$: a call, a put at strike K; + bought, − sold in the buy-side form L. K_c^+, K_p^-: the next live strike beyond K_c, K_p.

variation	bought when $s > 0$ (L)	sold when $s < 0$ ($-L$)	own notional	purpose
straddle (baseline)	$+C(K_c) + P(K_p)$	short straddle	premium	convexity both ways at the money
strangle, one strike wide	$+C(K_c^+) + P(K_p^-)$	short strangle	premium	all time value; cheaper, less gamma at the money
strangle x OTM, $x \in \{0.5, 1, 2\}\%$	$+C(\geq S(1+x))+P(\leq S(1-x))$	short strangle	premium	tail-only exposure; sold, a small premium for the tails
iron butterfly, w	$+C(K_c) + P(K_p) - C(\geq K_c + w) - P(\leq K_p - w)$	short straddle + long wings (credit)	larger wing gap	tail cap on the selling side
iron condor, w	body: the strangle one strike wide; wings w beyond it	short strangle + long wings (credit)	larger wing gap	cheaper body, tail cap
put vertical, w	$+P(K_p) - P(\leq K_p - w)$	credit put vertical	wing gap	premium from the put side alone, capped
call vertical, w	$+C(K_c) - C(\geq K_c + w)$	credit call vertical	wing gap	premium from the call side alone, capped
call (put) butterfly, w	short butterfly $-C(K_0 - w) + 2C(K_0) - C(K_0 + w)$; K_0 the straddle strike nearer S	long butterfly (debit; pays on a pin at K_0)	w	defined-risk selling with a pin payoff; one wing in the money
put (call) ratio 1×2, w	backspread $-P(K_p)+2P(\leq K_p - w)$	front ratio $+P(K_p) - 2P(\leq K_p - w)$, one leg uncovered	none	tail convexity when bought, tail premium when sold
straddle + wing hedge, w	the straddle	the short iron butterfly	none (two units)	cap the selling side's tail, keep the buying side's convexity

Table 23 scores every variation under $\text{sign}(s)$ on the block-diagonal ridge, the paper's forecast, and Table 24 under always short.

Table 23: Strategy variations under sign(s) with the block-diagonal ridge (the paper's forecast, plain), 866 days (861 for the butterflies at $w = 25, 50$). Sharpe ratios annualized by $\sqrt{252}$ per trade day, return per straddle premium; ΔSharpe: the variation minus the straddle, paired, a point estimate. Worst day: mid fill, in straddle premiums. Cost share: the median crossed-minus-mid entry cost over the structure's own absolute midpoint premium, averaged over the buying and the selling side.

	mid fill		crossed fill		worst	cost
variation	Sharpe	ΔSharpe	Sharpe	ΔSharpe	day	share
straddle	1.28	—	0.80	—	−5.42	0.03
strangle, one strike wide	1.26	−0.02	0.80	0.01	−4.68	0.04
strangle 0.5% OTM	1.24	−0.04	0.36	−0.44	−2.87	0.25
strangle 1% OTM	0.25	−1.03	−1.22	−2.02	−1.77	0.67
strangle 2% OTM	0.58	−0.70	−2.13	−2.93	−1.43	1.00
iron butterfly, w10	0.90	−0.38	−0.19	−0.99	−3.88	0.06
iron butterfly, w25	1.13	−0.15	0.40	−0.39	−5.44	0.04
iron butterfly, w50	1.25	−0.03	0.60	−0.20	−5.44	0.04
iron condor, w10	0.98	−0.30	−0.16	−0.96	−2.53	0.10
iron condor, w25	1.16	−0.12	0.39	−0.41	−4.70	0.08
iron condor, w50	1.22	−0.06	0.54	−0.26	−4.70	0.07
put vertical, w10	0.54	−0.74	−0.11	−0.90	−3.79	0.07
put vertical, w25	0.88	−0.40	0.44	−0.36	−4.35	0.05
put vertical, w50	0.99	−0.29	0.60	−0.20	−4.33	0.04
call vertical, w10	0.50	−0.78	−0.11	−0.91	−4.63	0.06
call vertical, w25	0.46	−0.82	0.03	−0.76	−5.71	0.04
call vertical, w50	0.50	−0.78	0.11	−0.69	−5.71	0.04
call butterfly, w10	0.71	−0.57	−1.70	−2.49	−3.84	0.13
call butterfly, w25	0.93	−0.30	−1.24	−1.98	−5.37	0.04
call butterfly, w50	1.20	−0.18	−1.16	−2.06	−5.32	0.02
put butterfly, w10	0.68	−0.60	−1.86	−2.66	−3.84	0.14
put butterfly, w25	0.95	−0.27	−1.35	−2.10	−5.68	0.04
put butterfly, w50	1.23	−0.15	−1.64	−2.54	−5.68	0.02
put ratio 1×2, w10	0.36	−0.92	−0.49	−1.29	−4.79	0.14
put ratio 1×2, w25	−0.71	−1.99	−1.24	−2.04	−7.21	0.06
put ratio 1×2, w50	−0.96	−2.24	−1.42	−2.22	−10.72	0.05
call ratio 1×2, w10	−0.32	−1.60	−1.21	−2.01	−2.80	0.11
call ratio 1×2, w25	−0.39	−1.67	−0.96	−1.76	−3.97	0.06
call ratio 1×2, w50	−0.49	−1.77	−1.00	−1.79	−4.64	0.05
straddle + wing hedge, w10	0.84	−0.44	0.03	−0.77	−3.88	0.04
straddle + wing hedge, w25	1.04	−0.24	0.41	−0.39	−5.44	0.04
straddle + wing hedge, w50	1.19	−0.09	0.59	−0.21	−5.44	0.03

Table 24: As Table 23, always short (no forecast): every structure sold every day.

	mid fill		crossed fill		worst	cost
variation	Sharpe	ΔSharpe	Sharpe	ΔSharpe	day	share
straddle	0.20	—	−0.27	—	−10.32	0.03

Table 24: (continued)

variation	mid fill Sharpe	mid fill ΔSharpe	crossed fill Sharpe	crossed fill ΔSharpe	worst day	cost share
strangle, one strike wide	0.63	0.43	0.15	0.43	−9.47	0.04
strangle 0.5% OTM	1.78	1.57	0.83	1.10	−3.65	0.25
strangle 1% OTM	1.40	1.20	−0.20	0.08	−1.77	0.67
strangle 2% OTM	2.64	2.44	−0.55	−0.28	−1.43	1.00
iron butterfly, w10	−0.40	−0.61	−1.47	−1.19	−5.66	0.06
iron butterfly, w25	−0.11	−0.31	−0.82	−0.55	−6.87	0.04
iron butterfly, w50	0.06	−0.15	−0.59	−0.31	−10.33	0.04
iron condor, w10	0.07	−0.14	−1.07	−0.79	−4.38	0.10
iron condor, w25	0.32	0.12	−0.45	−0.18	−7.34	0.08
iron condor, w50	0.43	0.23	−0.26	0.01	−9.48	0.07
put vertical, w10	−0.79	−1.00	−1.41	−1.14	−5.98	0.07
put vertical, w25	−0.48	−0.69	−0.90	−0.63	−7.20	0.05
put vertical, w50	−0.29	−0.50	−0.67	−0.40	−10.71	0.04
call vertical, w10	0.35	0.15	−0.27	0.01	−4.63	0.06
call vertical, w25	0.38	0.18	−0.05	0.22	−5.71	0.04
call vertical, w50	0.40	0.19	−0.01	0.27	−5.71	0.04
call butterfly, w10	−0.29	−0.50	−2.63	−2.36	−4.23	0.13
call butterfly, w25	0.02	−0.22	−2.12	−1.88	−5.95	0.04
call butterfly, w50	0.24	0.01	−2.07	−1.83	−9.61	0.02
put butterfly, w10	−0.30	−0.51	−2.79	−2.51	−4.31	0.14
put butterfly, w25	0.02	−0.22	−2.25	−2.01	−5.95	0.04
put butterfly, w50	0.28	0.04	−2.53	−2.28	−9.69	0.02
put ratio 1×2, w10	1.42	1.21	0.51	0.79	−5.64	0.14
put ratio 1×2, w25	0.81	0.61	0.24	0.52	−1.22	0.06
put ratio 1×2, w50	0.42	0.21	−0.05	0.23	−0.85	0.05
call ratio 1×2, w10	−0.10	−0.31	−1.00	−0.72	−2.49	0.11
call ratio 1×2, w25	−0.30	−0.50	−0.88	−0.60	−1.74	0.06
call ratio 1×2, w50	−0.34	−0.55	−0.85	−0.58	−0.83	0.05
straddle + wing hedge, w10	−0.40	−0.61	−1.47	−1.19	−5.66	0.04
straddle + wing hedge, w25	−0.11	−0.31	−0.82	−0.55	−6.87	0.04
straddle + wing hedge, w50	0.06	−0.15	−0.59	−0.31	−10.33	0.03

B.12 Parked from Section 5.4: The Paper's Forecast under the Session-Bar Recalibration

Why it is parked. Until 29 September 2026, Section 5.4 read the paper's eight forecasts under the session-bar recalibration (a Mincer–Zarnowitz map fitted on the regular-session bars, since removed from Section 4.5) and qualified the block-diagonal ridge's sign(s) portfolio with the paragraphs below. Section 5.4 now uses every forecast as its plain back-transform f^2B (Equation (7)), with no recalibration, in Table 6. These paragraphs belong to the session-bar recalibration and to the paper's forecast rather than the headline; none of their numbers is comparable with Table 6 or set beside it. They are kept as they stood, because three of them answer questions a reader will ask of any same-day trade — whether the forecast is dated where it claims to be (the shift test), how much of the trade's sign a

variance forecast can reach at all, and why QLIKE and the trade rank forecasts differently — and they have not been re-run with the plain forecast.

Where the level comes from. Two sample splits qualify the level rather than the sign. Dropping the 39 expiration days through 31 March 2020 — the February–March 2020 spike — *raises* the block-diagonal ridge portfolio to Sharpe 1.51 ($t = 2.73$ on the remaining 827 days): the full-sample figure is dragged, not flattered, by that quarter. Splitting instead at 16 May 2022, when SPXW listings became daily — a market-structure date, not a threshold chosen on the portfolio — the same portfolio scores Sharpe 0.98 ($t = 1.20$) on the 377 days before and 1.63 ($t = 2.27$) on the 489 after: the direction edge is positive on both tapes but earns its significance on the daily-0DTE era. The split is not uniform across columns — the two tree forecasts score higher before the date than after — which is one more reason to read the panel as one rule rather than as eight results.

Settlement pins carry a disproportionate share. The straddle expires worthless — $R = -1$ exactly — on 193 of the 866 days, 22.3 percent, whenever the close settles inside the one-strike window. The always-short control collects $+1$ on those days against -0.27 on all others; the sign(s) portfolio averages $+0.18$ on them against $+0.07$ elsewhere, so pin days supply 43 percent of its total return from 22.3 percent of the days. The portfolio monetizes a direction call and the pin premium together, and its P&L remains tilted toward days the index settles inside the strikes.

Eight rows are one or two bets. The eight daily sign(s) portfolios are correlated (mean pairwise correlation 0.70, from 0.58 to 0.93; their buy-day sets overlap at a mean pairwise Jaccard index of 0.70), an effective count of 1.8 independent portfolios by the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ of their correlation matrix. Agreement down the eight columns is evidence that the rule is insensitive to which forecast drives it — not eight independent confirmations of the rule.

What the forecasts are scored on, and what the trade uses. The eight columns cannot be ordered by the trade, and the paper's accuracy measure does not order them on it either. The reason is structural. QLIKE (Equation (4)) is consistent for the conditional mean of the variance [Patton, 2011], and every consistent scoring function for a mean can be taken apart by threshold. For a threshold $\theta > 0$ let

$$S_\theta(f, y) \;=\; |y - \theta|\, \mathbf{1}\{\min(f, y) \le \theta < \max(f, y)\}, \tag{12}$$

the loss of someone who uses the forecast f to decide whether the outcome y will exceed θ: nothing when f and y fall on the same side of θ, and the distance by which y landed from θ when they do not. Every consistent scoring function for the mean is, up to a term that does not involve the forecast, a mixture of these elementary scores over θ with a nonnegative weight [Ehm et al., 2016]; for a Bregman loss generated by a convex function ϕ the weight is $\phi''(\theta)\, d\theta$ in the normalization of Equation (12). QLIKE is the Bregman loss of $\phi(u) = -\log u$,

so for every forecast and outcome

$$\frac{RV}{\widehat{RV}} - \log\frac{RV}{\widehat{RV}} - 1 \;=\; \int_0^\infty S_\theta\big(\widehat{RV},\, RV\big)\,\theta^{-2}\,d\theta. \tag{13}$$

The identity is a two-line calculation — when $RV > \widehat{RV}$ the integral is $\int_{\widehat{RV}}^{RV}(RV-\theta)\,\theta^{-2}\,d\theta = RV/\widehat{RV} - 1 - \log(RV/\widehat{RV})$, and the other case is its mirror image — and it was also checked numerically, day by day, for the block-diagonal ridge on the scored days. QLIKE is therefore an average of threshold decisions, weighted by θ^{-2} toward small thresholds.

The sign(s) rule takes one of those decisions and no other: it is long exactly when $\widehat{RV}_t > \mathrm{IV}^2_{30,t}$, the decision at $\theta = \mathrm{IV}^2_{30,t}$. With the threshold measured in units of the day's implied variance, $\theta = c\,\mathrm{IV}^2_{30,t}$, the right-hand side of Equation (13) becomes $\int_0^\infty e_t(c)\,c^{-2}\,dc$, where $e_t(c)$ is the elementary score of $\widehat{RV}_t/\mathrm{IV}^2_{30,t}$ against $RV_t/\mathrm{IV}^2_{30,t}$ at the threshold c; the trade reads that score at $c = 1$ only,

$$e_t(1) \;=\; \left|\frac{RV_t}{\mathrm{IV}^2_{30,t}} - 1\right| \mathbf{1}\big\{\widehat{RV}_t \text{ and } RV_t \text{ fall on opposite sides of } \mathrm{IV}^2_{30,t}\big\}. \tag{14}$$

A single elementary score is itself consistent for the mean, though not strictly, so it ranks variance forecasts in the same sense QLIKE does; it spends all of its weight at the threshold the trade uses. Table 25 reports it beside QLIKE. QLIKE remains the paper's measure of accuracy everywhere else.

Table 25: The 15:30 forecasts scored two ways, on the 866 expiration days of Section 5.4. QLIKE is Equation (4) applied to the 16:00 row; the score at the threshold is Equation (14), averaged over days (lower is better for both). "Same side as RV" is the percentage of days on which $\widehat{RV}_t$ and RV_t fall on the same side of $\mathrm{IV}^2_{30,t}$; "same side as R" the percentage on which the sign(s) position has the sign of the settlement return. The last column is the annualized Sharpe ratio of the sign(s) portfolio at the crossed spread. The last row is a rule told RV_t in advance. †: the four columns still fitted on the earlier design panel (Section 4.5).

	QLIKE	score at $\theta = \mathrm{IV}^2_{30}$	same side as RV	same side as R	$\mathrm{Sharpe}_{\mathrm{ann}}$, crossed
baseline (HAR + calendar OLS)	0.108	0.102	68.5	55.2	0.49
block-diagonal ridge	0.110	0.109	65.9	54.5	0.87
block-diagonal ridge, without the FOMC columns	0.110	0.111	64.7	54.2	0.80
LightGBM†	0.101	0.094	68.7	54.7	0.98
XGBoost†	0.103	0.097	68.8	55.3	1.02
lasso (causally tuned)†	0.104	0.106	66.3	55.1	0.93
lasso ($\alpha = 10^{-4}$)	0.111	0.107	65.7	54.0	1.04
elastic net (causally tuned)†	0.104	0.109	66.5	54.6	0.54
realized variance known in advance	0	0	100.0	62.9	4.07

The Murphy diagram. Averaging $e_t(c)$ over the scored days and plotting it against c, one line per forecast, gives the Murphy diagram of Ehm et al. [2016] (Figure 3). QLIKE is the area under each line weighted by c^{-2}; the trade reads each line at $c = 1$. The lines cross within a factor of two of the implied variance. At every threshold on the grid up to 0.71 times it the lowest score belongs to a penalized linear forecast — the block-diagonal ridge,

one of the two lassos or the elastic net. From 0.77 to 1.41 times it the two tree forecasts lead, LightGBM at $c = 1$ itself with a score of 0.094. Between 1.54 and 2.18 times it the baseline leads at three of the five thresholds, and from 2.38 times upward the best score is shared exactly by forecasts of different families, because few forecasts ever reach so high a threshold and those that do not are scored identically. At the trade's own threshold the block-diagonal ridge ranks sixth of the eight, with 0.109, and the same estimator without the FOMC columns ranks eighth, with 0.111.

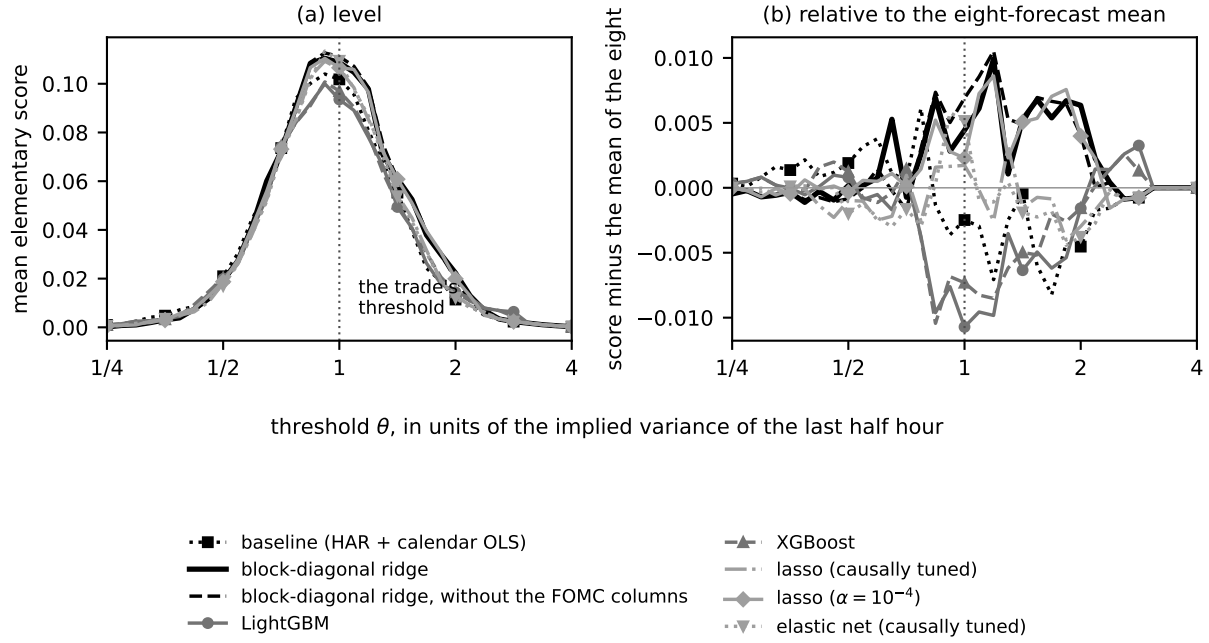


Figure 3: Murphy diagram of the eight 15:30 forecasts on the 866 expiration days. The horizontal axis is the threshold θ in units of the day's implied variance of the last half hour, $\mathrm{IV}^2_{30,t}$, on a logarithmic scale from 0.25 to 4; the dotted vertical line at 1 is the threshold the sign(s) rule uses. (a) The mean of the elementary score $e_t(c)$ (lower is better); QLIKE is the area under each line weighted by c^{-2}. (b) Each line minus the mean of the eight at the same threshold, which makes the crossings visible.

What a variance forecast can reach. Even the right threshold decision does not settle the trade. The straddle's settlement value depends on where the index closes relative to the two strikes — on the net move of the last half hour — and not on its realized variance, which counts every swing whether or not it is undone before the close. A rule told RV_t in advance and trading $\mathrm{sign}(RV_t - \mathrm{IV}^2_{30,t})$ takes the side of the payoff's sign on only 62.9 percent of days, for an annualized Sharpe ratio of 4.53 at the mid and 4.07 across the spread, against 17.3 across the spread for a rule told the sign of the payoff itself. On 37.1 percent of days — about a third — the trade's sign is therefore beyond any variance forecast, however accurate. The eight forecasts take the side of the payoff's sign on 54.0 to 55.3 percent of days, and the side of $RV_t - \mathrm{IV}^2_{30,t}$ on 64.7 to 68.8 percent.

Neither score orders the forecasts on this trade. The fixed lasso ranks eighth of the eight on QLIKE and fifth at the trade's threshold, and has the highest crossed-spread Sharpe ratio of the eight (1.04). Across the eight columns the rank correlation between the crossed-spread Sharpe ratio and minus the loss is 0.12 for QLIKE (95% interval from a day-block bootstrap −0.69 to 0.76) and 0.40 for the score at the threshold (−0.62 to 0.90); both intervals contain zero. Adding fifteen deliberately improved and degraded copies of the ridge forecast — blends toward the realized value, and multiplicative noise — lifts both correlations, to 0.95 and 0.97, but that says only that a much better or much worse forecast trades better or worse: every loss orders those copies perfectly, so they cannot choose between losses. The score at the threshold is reported because it isolates the one decision the trade takes, not because it ranks the eight columns here; with eight forecasts that take the same side on most days, and a payoff whose sign a variance forecast can reach only in part, no variance score does.

The signal's content is a same-day sign, and the right test is the same-day one. The signal is fixed at 15:30 and the return settles at 16:00 on the same expiration day, so pairing s_t with R_t involves nothing observed after the decision. On that pairing the content is a shift in the conditional mean by sign: when the forecast exceeds the implied variance the straddle returns +0.10 on average, and −0.09 when it does not (difference 0.19, heteroskedasticity-robust $t = 2.32$ for the block-diagonal ridge; t between 1.63 and 2.61 across the eight columns). That is the block-diagonal ridge's sign(s) portfolio under the session-bar recalibration, read as a regression, and Figure 4 shows it. A straight line is the wrong instrument for the same relation: fitted by least squares the slope of R_t on the raw signal comes out *negative* for every one of the eight columns (heteroskedasticity-robust t from −3.9 to −0.2) — the opposite sign to the step the split finds — because R has a point mass at −1 on 22.3 percent of days and a fat right tail that a few extreme values of s_t can lever. The effect lives in the conditional means of the two sides, which a sign split sees and a slope does not.

Magnitude carries information; sizing on it does not. Within either side of the split the binned means of the figure show no trend in the level of the signal, but a line fitted *inside* a side does find one, and it runs the wrong way for a trader. Restricted to the sell side the heteroskedasticity-robust slope of R_t on s_t is negative and resolved for all eight columns ($|t|$ from 2.2 to 5.7; 5.4 for the block-diagonal ridge): the further below zero the signal, the larger the settlement return, so the short earns *less* on precisely the days its signal is most extreme. On the buy side the slope is negative too, but resolved for only two of the eight. The precise claim is therefore narrower than "magnitude adds nothing to the sign": the conditional mean does vary with $|s|$ inside a side. What does not follow is a sizing rule, and none is retained — scaling the position by the rank of $|s_t|$ raised no column's Sharpe ratio, and that overlay is parked (Section 4.5). Only the sign is used.

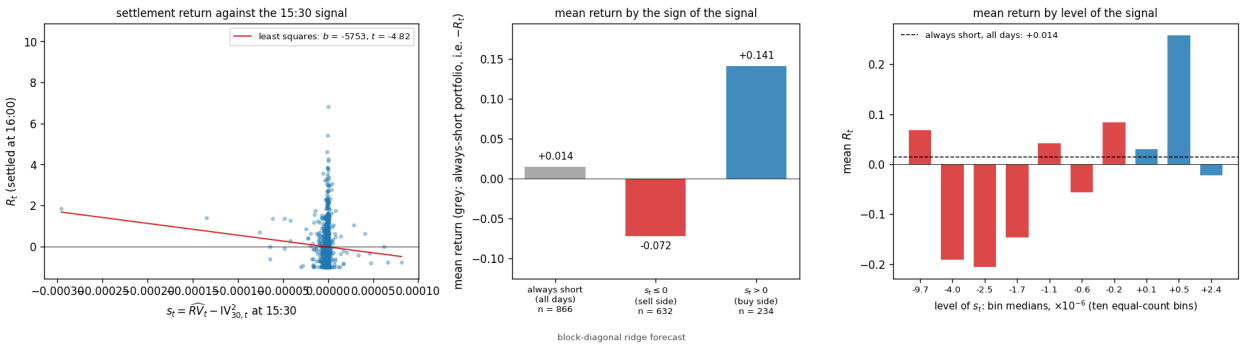


Figure 4: The 15:30 signal against the 16:00 settlement return, block-diagonal ridge forecast, 866 expiration days. Left: the settled return R_t against the raw signal s_t with the least-squares line. Middle: the mean of R_t on days with $s_t \leq 0$ and $s_t > 0$ (means only, no error bars) beside the always-short portfolio's mean over all days (grey bar), the benchmark throughout. Right: the mean of R_t in ten equal-count bins of the signal's level, each labelled by its median value of s_t in units of 10^{-6} (thirty-minute variance); the dashed line is the always-short mean. The relation is a step at the sign, and the binned means inside either half show no trend in the level: what the signal carries is which side of the quoted variance the forecast stands on.

The forecast is dated where it claims to be. A same-day trade driven by a same-day forecast invites the obvious suspicion, so the shift test is reported rather than asserted. Re-running the whole portfolio on the forecast row stamped k bars away from the one the rule uses gives an annualized Sharpe of 0.48 at $k = -1$ (a forecast issued one bar early, for a window that has already closed), 1.34 at $k = 0$ (the row actually used), and 3.11 at $k = +1$ — one bar of genuine look-ahead — against a ceiling of 4.53 for a portfolio told the last half hour's realized variance in advance. The profile is a step exactly at $k = 0$: flat to its left and far higher to its right. A leak would have lifted the bars left of $k = 0$ or flattened the step, and it does neither. One column is worth flagging: the elastic net's best stale shift, at $k = -2$, is 1.04 against its own 1.02 at $k = 0$, so that one forecast is not separated from its own stale copies by this test. Every other column's best stale shift sits below its trade.

Compounded at a fixed three percent of wealth, the $\operatorname{sign}(s)$ **portfolio grows; the control does not.** Betting the fixed fraction $f = 0.03$ of Section 4.5, the block-diagonal ridge $\operatorname{sign}(s)$ portfolio turns one unit of wealth into 7.2 over the sample (annualized log-growth +0.57), with a maximum wealth drawdown of 42 percent and a worst single day that removes 16 percent of wealth; the ruin bound $1/|\min_t R'_t|$ is 0.18 for this portfolio and 0.10 for the control, so three percent sits well inside both. All eight $\operatorname{sign}(s)$ portfolios compound positively under the same fraction (terminal wealth 3.6 to 9.9), and the headline portfolio's annualized log-growth is positive in four of the five calendar years (2022 is −0.04). The always-short control ends at 0.86 — its premium does not compound. A defined-risk variant is parked and not reported here.

Table 26: The shift (dating) test across the paper's model set: the 15:30 sign(s) rule with each forecast's row shifted by k half-hour bars and placed on the traded bar's scale, $\hat{y}_k^2 B_{16:00}$ (the plain forecast, no recalibration), for the 9 forecasts whose tables carry the session rows, on the 865 deck days on which every shifted row exists. $k < 0$: the same session's earlier row; $k = 0$: the trade; $k > 0$: the next session's row. Per k: the median Sharpe ratio across the shifted forecasts (and its range), the number of forecasts whose Sharpe ratio falls below its own $k = 0$ value, the median change and the median share of days on which the position agrees with $k = 0$ (columns: median mid, range mid, median crossed, below $k = 0$, median Δ mid, agree).

row	read	mid	range mid	crossed	below k=0 (of 9)	Δ mid	agree %
$k = -12$	10:00, same	0.71	0.39 to 1.09	0.23	9	-0.51	66
$k = -11$	10:30, same	0.81	0.55 to 1.02	0.33	7	-0.57	66
$k = -10$	11:00, same	0.00	-0.16 to 0.30	-0.47	9	-1.40	68
$k = -9$	11:30, same	0.53	0.31 to 0.86	0.05	9	-0.65	68
$k = -8$	12:00, same	0.20	-0.15 to 0.47	-0.28	9	-1.16	69
$k = -7$	12:30, same	0.67	0.32 to 0.96	0.19	8	-0.54	71
$k = -6$	13:00, same	0.99	0.39 to 1.27	0.51	9	-0.38	71
$k = -5$	13:30, same	1.16	0.94 to 1.33	0.67	7	-0.12	72
$k = -4$	14:00, same	0.84	0.50 to 1.28	0.36	7	-0.25	73
$k = -3$	14:30, same	1.40	0.91 to 1.55	0.91	3	+0.09	73
$k = -2$	15:00, same	1.00	0.23 to 1.19	0.51	8	-0.34	77
$k = -1$	15:30, same	0.53	0.31 to 0.74	0.04	8	-0.94	81
$k = +0$	16:00, same	1.35	0.49 to 1.58	0.87	0	+0.00	100
$k = +1$	10:00, next	0.41	0.29 to 0.68	-0.06	8	-0.94	67
$k = +2$	10:30, next	1.03	0.76 to 1.10	0.56	7	-0.32	68
$k = +3$	11:00, next	1.24	1.05 to 1.38	0.76	6	-0.09	68
$k = +4$	11:30, next	0.93	0.86 to 1.15	0.45	7	-0.42	66
$k = +5$	12:00, next	1.48	1.19 to 1.57	1.00	4	+0.12	67
$k = +6$	12:30, next	1.58	1.39 to 1.78	1.10	0	+0.32	67
realized (target)	16:00	4.48		4.02			
realized (unclipped)	16:00	4.50		4.03			
always short		0.27		-0.22			

B.13 The Shift (Dating) Test across the Model Set

What is tested. The last-30-min trade reads each forecast's row labelled 16:00 — the forecast of the 15:30–16:00 bar, issued at 15:30. The test re-scores the sign(s) rule with the forecast *row* shifted by k half-hour bars and everything else unchanged, every shifted row placed on the traded bar's scale as $\hat{y}_k^2 B_{16:00}$, so that only what the model had seen changes with k. A forecast can be shifted only if its table carries rows for the whole session, and of the 112 forecasts of the model set only the 9 paper's forecasts do. The other 103 carry only the 16:00 row and cannot be shifted: the 7 per-bar linear forecasts and 90 per-bar trees, fitted on the 16:00 bar alone, and the 6 pre-tuned pooled twins, whose tables hold the 16:00 row only. The test does not speak for them. For $k < 0$ the rule reads the same session's earlier row (the 15:30 row at $k = -1$, down to the 10:00 row at $k = -12$): a stale forecast, still free of look-ahead. For $k > 0$ it reads the next session's row, whose inputs contain the traded bar. Scorer, trade and fills are those of Table 6, on the 865 days on which every shifted row exists.

Reading. Table 26 gives every k. At $k = 0$ the 9 forecasts trade at a median Sharpe ratio of 1.35 (0.49 to 1.58; 0.87 crossed). One bar stale the median falls to 0.53 (median change

-0.94), 8 of the 9 fall below their own $k = 0$ value, and the median position agrees with the trade's on 81 % of days. Over the twelve stale rows between 3 and 9 forecasts fall below $k = 0$, all of them at 3 rows; the best stale row by median is $k = -3$ at 1.40. The next session's rows score medians of 0.41 to 1.58, with between 0 and 8 forecasts below $k = 0$ (all of them at 0 rows); at $k = +1$ the median is 0.41 (-0.06 crossed), 8 below $k = 0$, at a median agreement of 67 %. The realized variance itself, in place of the forecast, scores 4.48 mid and 4.02 crossed, against always short at 0.27 (-0.22 crossed) on the same days.

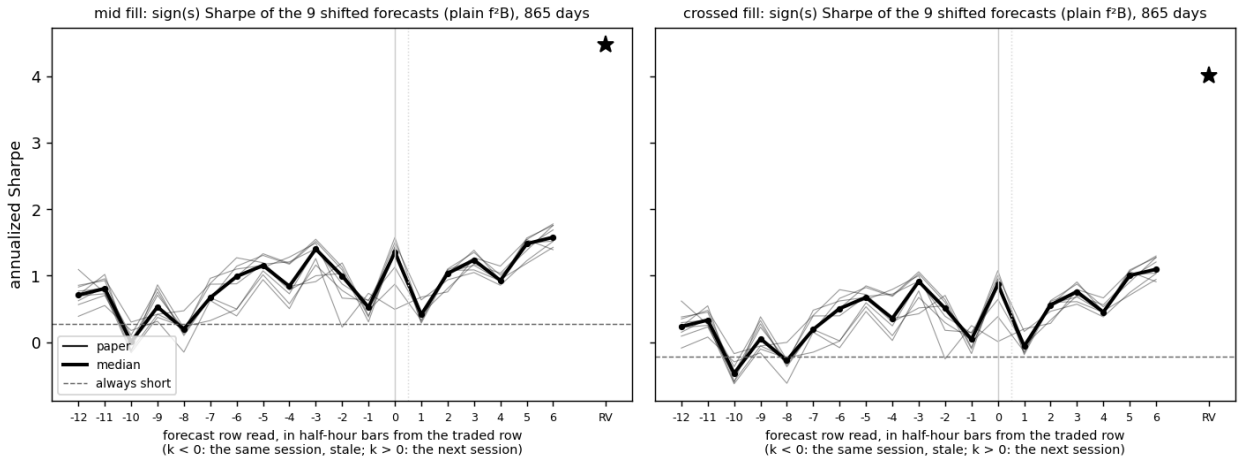


Figure 5: The sign(s) Sharpe ratio of the 9 shiftable forecasts of the model set as the forecast row is shifted, on the same 865 days, at both fills (source: `results/close_shift_test/shift_test_by_k.csv`).

Table 27: The paper's model set (112 forecasts, each its plain forecast $\hat{y}^2 B$, no recalibration) scored two ways on the 866 deck days, by group: the median QLIKE against the scorer's target and its range, the median elementary score at the trade's threshold (the 15:30 implied variance), the median economic loss $|R|\,\mathbf{1}\{q \neq \text{sign}(R)\}$, the median shares of days on the realized variance's side of the threshold and on the payoff's side, and the median sign(s) Sharpe ratio at both fills. The last two rows are rules told the realized variance, and the payoff's sign, in advance.

group	n	QLIKE	range	ES(c=1)	EL	RV side %	R side %	Sharpe mid/crossed
paper	9	0.107	0.105 to 0.123	0.092	0.371	70.2	57.3	1.28 / 0.80
pooled linear	6	0.105	0.102 to 0.107	0.092	0.378	71.2	56.6	1.06 / 0.57
per-bar linear	7	0.095	0.091 to 0.096	0.098	0.367	68.8	56.8	1.37 / 0.89
per-bar trees	90	0.100	0.093 to 0.107	0.098	0.382	69.6	55.7	0.96 / 0.48
model set	112	0.100	0.091 to 0.123	0.097	0.378	69.6	56.0	1.08 / 0.59
oracle: told RV		–		–	0.262	100.0	62.8	4.52 / 4.06
oracle: told the payoff's sign		–		–	0.000	62.8	100.0	17.35 / 17.28

Table 28: Sample splits of the sign(s) return (midpoint fill) across the paper's model set (112 forecasts, plain forecast): per split the number of forecasts with a positive Sharpe ratio, the median, smallest and largest Sharpe ratio, the median $t = \sqrt{n}\,\bar{x}/s_x$ and the number of forecasts with $t > 1.96$, and always short on the same days.

split	days	Sharpe > 0	median	range	median t	$t > 1.96$	always short
all days	866	112	1.08	0.29 to 1.92	1.99	58	0.20
through 2020-03-31 (the COVID quarter)	39	61	0.31	-3.96 to 3.67	0.12	0	-3.25
after 2020-03-31	827	112	1.13	0.21 to 2.20	2.06	59	0.42
before 2022-05-16	377	110	0.87	-0.51 to 2.12	1.07	13	-0.45
from 2022-05-16 (daily SPXW listings)	489	112	1.14	0.22 to 2.34	1.59	40	0.73
settlement pins (R = -1)	193	112	7.54	4.34 to 14.63	6.60	112	–
off pins	673	25	-0.33	-1.94 to 0.68	-0.53	0	-3.77

B.14 The Model Set Scored at the Trade's Threshold, and Its Sample Splits

The score at the threshold. QLIKE averages a forecast's elementary scores over every threshold (Equations (12)–(14)); the trade reads one threshold, the day's implied variance of the last half hour. Table 27 scores the 112 forecasts of the model set both ways, by group: the 9 paper's forecasts, the 6 pooled twins, the 7 per-bar linear forecasts and the 90 per-bar trees. The median QLIKE is 0.100 (0.091 to 0.123) and the median score at the threshold 0.097 (0.087 to 0.105). The median forecast falls on the realized variance's side of the threshold on 70 % of days and on the payoff's side on 56 % (54 to 60 %), and is long on 7 of the 20 best long-straddle days.

The Murphy diagram. Figure 6 plots each forecast's mean elementary score against the threshold, in units of the implied variance, from 0.25 to 4 on a logarithmic grid of 33 points. The forecast with the lowest score at a threshold is one of the paper's forecasts at 14 of the 33 thresholds, a pooled twin at 3, a per-bar linear forecast at 7 and a tree at 9; no group leads everywhere.

What a variance forecast can reach. A rule told the realized variance in advance takes the payoff's side on 63 % of days (Sharpe 4.52 at the midpoint, 4.06 across the spread), against 17.28 across the spread for a rule told the payoff's sign; on 37 % of days the trade's sign is beyond any variance forecast.

Which score orders the forecasts on the trade. Across the 112 forecasts the rank correlation of minus the loss with the midpoint Sharpe ratio is +0.18 for QLIKE and +0.50 for the score at the threshold (+0.18 and +0.50 with the crossed-spread Sharpe ratio), point estimates: the score at the trade's threshold tracks the trade more closely than QLIKE does. The economic loss is affine in the midpoint P&L and correlates perfectly by construction; it is reported as the ceiling.

Sample splits. Table 28 splits the sign(s) return of every forecast. Over all 866 days 112 of the 112 forecasts have a positive Sharpe ratio (median 1.08, 0.29 to 1.92; median t 1.99, 58 with $t > 1.96$), against 0.20 for always short. Dropping the 39 expiration days through 31 March 2020 leaves a median of 1.13 on the remaining 827 days (59 with $t > 1.96$); on those 39 days alone the median is 0.31 (61 positive), against -3.25 for always short. Split at 16 May 2022, when SPXW listings became daily, the median is 0.87 on the 377 days before (110 positive, 13 with $t > 1.96$) and 1.14 on the 489 days after (112 positive, 40 with $t > 1.96$). The straddle expires worthless on 193 of the days (22.3 %): always short collects +1.00 on them against -0.27 elsewhere, and the median forecast averages +0.43 on them against -0.02 off them, so pin days supply a median 123 % of a forecast's total return. Read as a same-day sign split, the settlement return is higher on the days a forecast buys than on the days it sells for 112 of the forecasts (median difference 0.17, median heteroskedasticity-robust t 1.82, 52 with $t > 1.96$).

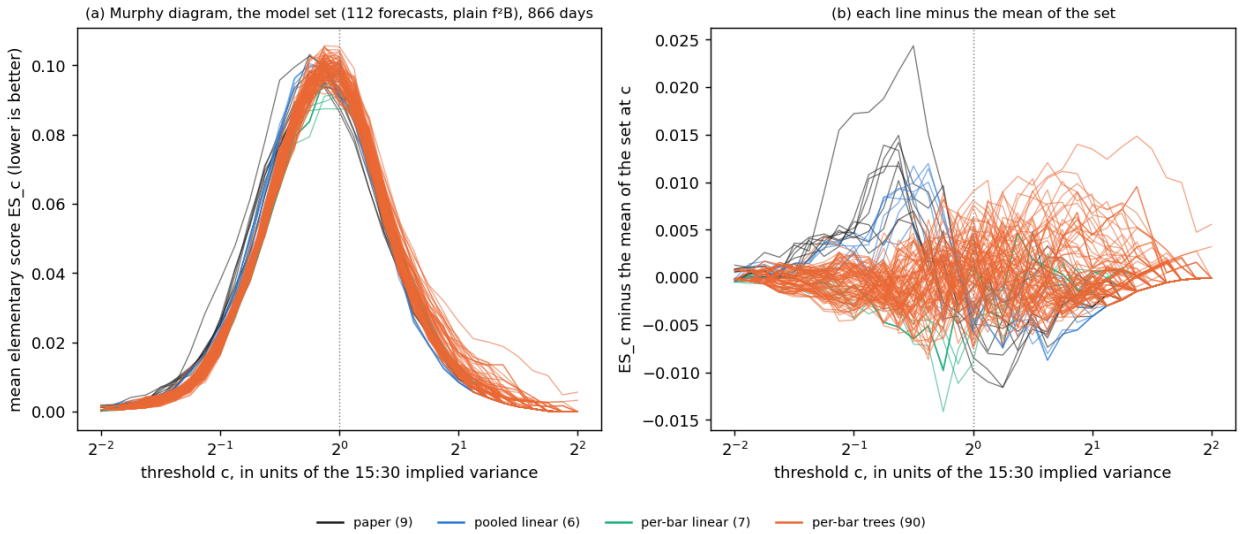


Figure 6: Murphy diagram of the model set on the 866 deck days, every forecast its plain back-transform. The horizontal axis is the threshold in units of the day's implied variance of the last half hour, on a logarithmic scale from 0.25 to 4; the threshold 1 is the one the sign(s) rule uses. (a) The mean elementary score (lower is better); QLIKE is the area under each line weighted by c^{-2}. (b) Each line minus the mean of the set at the same threshold (source: `results/close_murphy_splits/murphy_grid_modelset.csv`).

Table 29: Compounding at a fixed $f = 0.03$ of wealth per day across the paper's model set (112 forecasts, each its plain forecast $\hat{y}^2 B$, no recalibration), 866 days, by group: how many end above one unit of wealth, the median terminal wealth $W_T = \prod_t (1 + f\, q_t R_t)$ and its range, the median annualized log-growth g, the median and the largest fall of wealth from its running peak, and the smallest ruin bound $1/|\min_t R'_t|$ (the sample's worst day). Always short on the same days.

group	fill	above 1	median W_T	range	median g	median DD	largest DD	min ruin bound
paper	mid	9 of 9	6.48	1.32 to 9.64	+0.54	41 %	64 %	0.10
paper	crossed	8 of 9	2.67	0.53 to 3.91	+0.29	51 %	78 %	0.09
pooled linear	mid	6 of 6	4.43	2.20 to 9.60	+0.42	45 %	48 %	0.15
pooled linear	crossed	5 of 6	1.81	0.90 to 3.94	+0.16	52 %	59 %	0.14
per-bar linear	mid	7 of 7	7.67	7.67 to 20.62	+0.59	39 %	39 %	0.15
per-bar linear	crossed	7 of 7	3.16	3.16 to 8.54	+0.33	55 %	55 %	0.14
per-bar trees	mid	90 of 90	3.56	1.02 to 17.02	+0.37	45 %	64 %	0.10
per-bar trees	crossed	62 of 90	1.47	0.41 to 7.12	+0.11	51 %	74 %	0.09
model set	mid	112 of 112	4.38	1.02 to 20.62	+0.43	43 %	64 %	0.10
model set	crossed	82 of 112	1.80	0.41 to 8.54	+0.17	51 %	78 %	0.09
always short	mid		0.86		-0.04	55 %	55 %	0.10
always short	crossed		0.33		-0.32	74 %	74 %	0.09

B.15 The Model Set Compounded at a Fixed Fraction of Wealth

What is computed. The fixed-fraction exercise of Section 4.5 — a share $f = 0.03$ of current wealth deployed as straddle premium every day, the same on every day and for every rule, wealth $W_T = \prod_t (1 + f\, q_t R_t)$, the annualized mean of $\log(1 + f R'_t)$ as the growth rate, every wealth factor asserted positive — run for the $\text{sign}(s)$ positions of each of the 112 forecasts of the model set, on the 866 days and at both fills, beside always short (Table 29, Figure 7). The crossed frame pays the ask and hits the bid on every day.

Reading. At the midpoint 112 of the 112 forecasts end above one unit of wealth (median terminal wealth 4.4, from 1.0 to 20.6; median annualized log-growth +0.43, from +0.00 to +0.88), and 82 across the spread (median 1.8, from 0.4 to 8.5; median log-growth +0.17). Always short ends at 0.86 (-0.04) at the midpoint and 0.33 across the spread. The median fall of wealth from its running peak is 43 percent at the midpoint and 51 across the spread (largest 64 and 78), and the worst single day removes up to 33 percent of a forecast's wealth. The ruin bound $1/|\min_t R'_t|$ is at least 0.09 over the model set (always short 0.10), and $f = 0.03$ sits below it for 112 of the 112 forecasts. By calendar year, annualized from the days traded in each, the number of forecasts with a positive midpoint log-growth is 2020: 94, 2021: 70, 2022: 112, 2023: 99, 2024: 103 (medians 2020: +0.38, 2021: +0.24, 2022: +0.54, 2023: +0.37, 2024: +0.65) and across the spread 2020: 47, 2021: 50, 2022: 111, 2023: 86, 2024: 89 (medians 2020: −0.06, 2021: −0.09, 2022: +0.36, 2023: +0.19, 2024: +0.45).

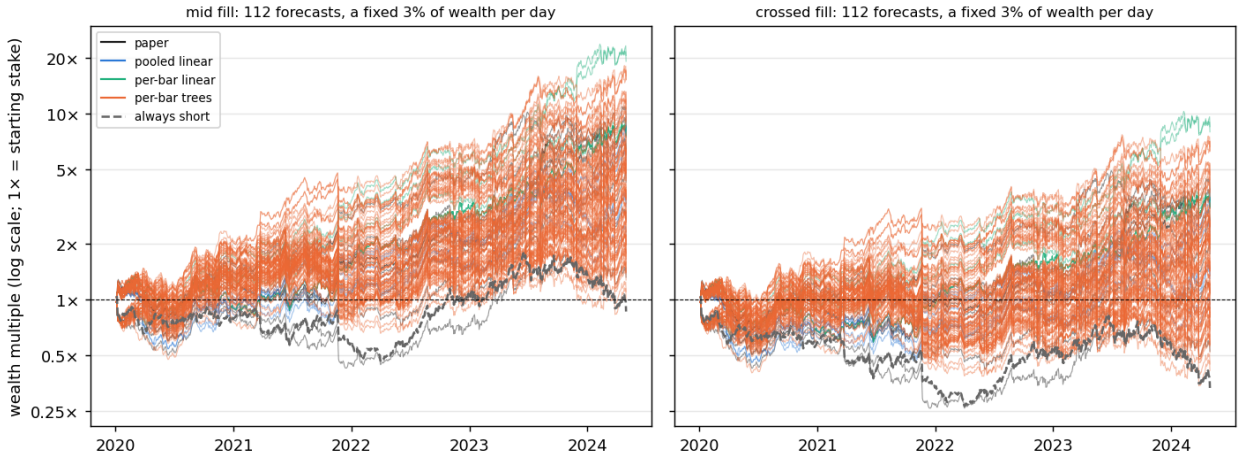


Figure 7: Wealth at a fixed 0.03 of wealth per day, 866 days, for every forecast of the model set (112, each its plain back-transform) and always short; midpoint fill (left) and crossed spread (right), log scale (source: `results/close_compounding/fixedfrac_modelset.csv`).

B.16 The 16:00 Models Re-run: Design, Mask, Cadence, Tuning

Every per-bar model that forecasts the 16:00 bar (the 15:30–16:00 half hour, issued at 15:30, the bar the last-30-min trade is decided on) was re-run with one change at a time: the design (B.16.1), a per-window column mask for the trees (B.16.2), the refit cadence (B.16.3) and the tuning (B.16.4). The section also records what the re-run showed about reproducibility across CPUs (B.16.5), the master table before and after (B.16.6), the importance study on the new design (B.16.7) and the compute used (B.16.8). Nothing is discarded: a rung that did worse is reported beside one that did better. XGBoost and the random forest were later re-run, every rung, on their current settings (Section 4.5); every XGBoost and random-forest pair below is on those runs, except the CPU-class records of B.16.5 and the first-pass side of the importance comparison in B.16.7.

Every comparison uses every forecast as its plain back-transform f^2B (Equation (7)), the same 866 trade days, and the sign(s) rule on the straddle (nearest out-of-the-money call and put, same-day expiry, one position). A difference is $a - b$, and every difference is a point estimate; a QLIKE difference is given in percent of b's QLIKE (negative = a has the lower loss); a Sharpe difference is the midpoint-fill sign(s) Sharpe ratio of a minus that of b. A count of differences below or above zero counts point estimates, and "significant" means significant at 5 %: by the Diebold–Mariano statistic for QLIKE and by the heteroskedasticity- and autocorrelation-robust (Newey–West) t of the daily midpoint-return difference for the Sharpe ratio. In the grid tables each cell holds the QLIKE difference on its first line and the Sharpe difference on its second. Input sets: `baseline` (the HAR ladder and the calendar columns) and `all_features`. Key to the design columns named below: `har_ma_*` is the bar's realized variance averaged over the last 1, 5, 25, 125, 625 and 3125 bars (the HAR ladder, `har_ma_k` one rung); `har_ma_k_x_open` and `har_ma_k_x_close` are that rung times the session-open and session-close indicators (the session-edge interactions).

B.16.1 The design: the session-edge columns

The session-edge interactions were built for the pooled 48-bar models. In a one-bar-per-session series at the 16:00 bar the 6 `har_ma_k_x_open` columns are identically zero and the 6 `har_ma_k_x_close` columns are exact copies of `har_ma_k`. The 12 columns were dropped from the per-bar design and kept in the pooled models: `baseline` goes from 34 to 22 columns and `all_features` from 640 to 628 (Table 30).

The linear arms' identifiability mask — at every penalty tune, drop the columns constant on the tuning window and the exact copies of an earlier kept column — had already removed all of them: in 45 of the 45 masked linear forecasts the number of masked columns per tune falls by exactly the number of dropped columns, and the columns each fit keeps are the same before and after (last column of Table 30). The fits solve the same problem, and the 46 linear forecasts of the master table move by rounding only (Table 31): OLS by at most 8×10^{-15} and ridge by at most 4×10^{-10} (relative change of the 16:00 forecast). The lasso and the elastic net move by up to 2×10^{-2} and 2×10^{-2}, on the rows of one penalty-tune period, in 3 of 15 and 5 of 15 forecasts; positions change on 1 of the 46×866 forecast-days, and no Sharpe ratio moves by more than 0.02. A control on the 8 arms that moved attributes this to the solvers' floating-point path: the pre-change code re-run on the same CPU architecture reproduces its first run bit for bit in 8 of 8; the new code against the pre-change code on one architecture moves 8 of them, by 1.1×10^{-3} to 1.7×10^{-2}; the pre-change code alone, moved to another machine, moves 7, by 7.3×10^{-4} to 1.2×10^{-2}.

The rest-of-day model's 78 arms were re-run on the same design; their mask fell by exactly 12 columns per tune in 78 of them. The master table's 6 check rows (the rest-of-day model at 15:30, which should equal its per-bar 16:00 twin) had differed from their twins by up to 1.8×10^{-2} (the same position on as few as 866 of 866 days); they are now bit-identical to their twins in 6 of 6, on every one of the 1,469 16:00 rows.

Table 30: The per-bar design before and after the session-edge columns were dropped, and the linear arms' identifiability mask per penalty tune (range over tunes) for the per-bar ridge: the columns it masked fall by exactly the columns removed, so the columns kept are unchanged.

input set	columns before	columns after	removed	masked per tune, before	masked per tune, after	kept per tune
`baseline`	34	22	12	17	5	17
`all_features`	640	628	12	262–280	250–268	360–378

Table 31: The master table's per-bar linear forecasts, new design minus old, by estimator: the largest and the median (over forecasts) of the per-forecast maximum and median relative change of the 16:00 forecast, the forecasts whose maximum exceeds the reproduction gate's bound, the trade positions changed (forecast-days, of 866 per forecast) and the largest change of a sign(s) Sharpe ratio (midpoint).

estimator	forecasts	max rel. change	median rel. change	moved ($> 1 \times 10^{-8}$)	positions changed	max $\lvert\Delta\text{Sharpe}\rvert$ mid
OLS	1	8.4×10^{-15}	4.4×10^{-16}	0	0	0.00
ridge	15	3.9×10^{-10}	1.1×10^{-14}	0	0	0.00
lasso	15	1.7×10^{-2}	2.8×10^{-15}	3	0	0.00
elastic net	15	1.5×10^{-2}	2.7×10^{-15}	5	1	0.02

B.16.2 One mask for every model

The trees were then given the linear arms' rule. At every fit — a refit on its training window, or a tuning point on the window it tunes on — every column constant on that window and every exact copy of an earlier kept column is dropped, first occurrences kept in column order. Importance and TreeSHAP are mapped back to all columns with zero for a dropped column. The rule looks at the window only, so every model of an input set keeps the same columns at the same refit: over the 1,469 daily refits `baseline` keeps 17 of 22 columns at every refit and `all_features` 369 of 628 (median; range 360–378); the Optuna runs below keep the same counts.

Table 32 sets each masked rung against the same rung unmasked (both on the new design). Rungs: T10 and T1 are the shipped tree configurations refit every 10 sessions and every session; RS10 and RS1 the random search, refit every 10 sessions and every session. Over the 16 tree rows the median absolute QLIKE change is 0.4 %; 10 QLIKE differences lie above zero (the masked rung has the higher loss; 0 significant) and 6 below (0 significant); 14 Sharpe differences lie below zero and 2 above (1 and 0 significant). The masked rungs ran on one CPU class; XGBoost and the random forest on their current settings were re-run unmasked at T10 and T1 only, so their random-search cells are empty (—, 8 pairs); the campaign's unmasked rungs did not pin the class (the class of their chunks was not recorded), so a masked-minus-unmasked difference also contains any difference of CPU class, whose size alone is recorded in B.16.5.

Table 32: The per-window mask, masked minus unmasked, per rung, model and input set, same design, 866 days. Each cell: QLIKE difference in % of the unmasked QLIKE over the sign(s) Sharpe difference at the midpoint, point estimates. The campaign's unmasked rungs were not pinned to a CPU class. —: no unmasked run of XGBoost or the random forest on their current settings.

model	T10 masked − unmasked	T1 masked − unmasked	RS10 masked − unmasked	RS1 masked − unmasked
baseline				
	+0.4	−0.1	0.0	−0.1
LightGBM	−0.17	−0.12	−0.13	+0.06
	−0.6	+0.7		
XGBoost	−0.04	−0.08	—	—
	+0.3	0.0		
random forest	+0.05	−0.10	—	—
all_features				
	−1.0	+1.4	+2.6	+0.1
LightGBM	−0.21	−0.23	−0.08	−0.05
	+1.0	+0.8		
XGBoost	−0.22	−0.30	—	—
	+0.1	−0.4		
random forest	−0.18	−0.34	—	—

B.16.3 Refit cadence

With the mask in place, the trees refit every session, as the per-bar linear arms do, and every 10th session as before (first two columns of Table 33). For the shipped configuration, T1 minus T10 ranges from −3.6 % to +0.1 % in QLIKE, with 5 of 6 differences below zero (1 significant) and 1 above; 1 Sharpe differences lie above zero and 5 below (0 and 0 significant). For the random search, RS1 minus RS10 ranges from −3.7 % to +0.2 %, with 4 of 6 QLIKE differences below zero (1 significant) and 2 above; 3 Sharpe differences lie below zero and 3 above (0 and 0 significant).

Against the per-bar ridge, 2 of the 24 masked tree rows have a QLIKE difference below zero (0 significant) and 22 above (7 significant); the lowest relative QLIKE is T1 XGBoost `all_features` at −2.4 %; 19 Sharpe differences lie below zero and 5 above (0 and 0 significant).

Table 33: The masked ladder: refit every session minus every 10th (shipped configuration; random search), and random search minus the shipped configuration, both refit every session. Cells as in Table 32, the second forecast of each pair being the comparator.

model	T1 − T10 (refit 1 vs 10)	RS1 − RS10 (refit 1 vs 10)	RS1 − T1 (search vs shipped)
baseline			
	−0.8	+0.2	+5.9
LightGBM	+0.06	−0.31	−0.15
	+0.1	−0.2	+9.3
XGBoost	−0.25	−0.42	−0.38
	−1.5	+0.1	+9.4
random forest	−0.31	+0.03	+0.01
all_features			
	−2.4	−1.6	+2.9
LightGBM	−0.04	−0.20	+0.28
	−3.6	−2.2	+5.3
XGBoost	−0.43	+0.09	+0.65
	−1.6	−3.7	+5.6
random forest	−0.43	+0.06	+0.33

B.16.4 Tuning: random search and Optuna

Random search against the shipped configuration. The random search re-chooses each tree's configuration from a fixed set of candidates on the validation tail of its window. Both refit every session, RS1 minus T1 ranges from +2.9 % to +9.4 % in QLIKE, with the difference above zero in 6 of 6 rows (1 significant) and below zero in 0; 2 Sharpe differences lie below zero and 4 above (0 and 0 significant; last column of Table 33).

Optuna. The random search was replaced by Optuna's TPE sampler: at every tuning point a fresh study of 50 sequential trials (trial 1 the shipped configuration, trials 1–10 TPE's random start-up) minimises validation MSE on the tail of the tuning window after an embargo; the trees refit every session with the configuration in force; a new tuning point is drawn every TUNE_PER sessions (TUNE_PER 1, 5, 25 or 250), and at TUNE_PER 1 and 25 the best of the first 10 and the first 25 trials is kept beside the best of all 50. Every Optuna run is masked and ran on one CPU class.

Are 50 trials enough? Over the 1,469 tuning points of each arm (Table 34), the best of the 50 trials falls in trials 41–50 at 27–37 % of the points, against 20 % if the trials were exchangeable draws, and the shipped configuration is the best at no more than 1.0 % of them. The last 25 trials lower the best validation MSE by 0.6–2.1 % on average (medians 0.3–1.2 %) and the last 10 by 0.2–0.7 % (medians at most 0.0 %); the pick of the first 25 differs from the pick of all 50 at 63–72 % of the points. Out of sample (Table 35), of the 36 budget comparisons (best of 50 against 10, against 25, and 25 against 10, at TUNE_PER 1 and 25) 14 have a QLIKE difference below zero and 22 above (1 and 3 significant); 18 Sharpe differences lie below zero and 18 above (3 and 1 significant).

How often to tune. Against a new tuning point every 250 sessions, tuning every 1, 5 or 25 sessions (every path refit every session; Table 36) has 12 of 18 QLIKE differences below zero and 6 above (0 and 0 significant); 16 Sharpe differences lie below zero and 2 above

(0 and 0 significant). Choosing the configuration by validation QLIKE instead of MSE (at TUNE_PER 25) has 4 of 6 QLIKE differences below zero and 2 above (1 and 0 significant), 3 Sharpe differences below zero and 3 above (1 and 0 significant).

Optuna against the other rungs. Table 37 counts the differences. Against the shipped configuration refit every session (T1), 24 of 24 Optuna QLIKE differences lie above zero (12 significant) and 0 below (0), with 11 Sharpe differences below zero (0 significant) and 13 above (0). Against the random search refit every session (RS1), 16 lie below zero (0 significant) and 8 above (1). Against the per-bar ridge, 0 lie below zero (0 significant) and 24 above (11), with 22 Sharpe differences below zero (1 significant) and 2 above (0).

Where the picks sit. Table 38 gives, per axis, the share of the best-of-50 picks in the outermost twentieth of the search space at either end (on the log scale for a log axis; for the forest's depth, the smallest and the largest choice), beside the share of TPE's random start-up draws there (the prior). The penalties `lambda_l2` and `reg_lambda` are picked at their upper end in no more than 0.0 % of points against a prior of at least 5.1 %; LightGBM's `min_child_samples` sits at its lower end in 14–19 % of points, and the forest picks unbounded depth in 19–23 % (prior 17 %).

Table 34: Is the Optuna budget enough? Per arm, over all tuning points: the median trial that holds the best of 50, the share of points where it is the shipped configuration and where it falls in trials 41–50 (20 % if exchangeable), the relative gain in the best validation MSE from the first 10, 25 and 40 trials to all 50 (mean / median), and the share of points whose best of 25 differs from the best of 50.

arm	best trial (median)	best = shipped (%)	best in 41–50 (%)	val. MSE gain 10→50 (%)	val. MSE gain 25→50 (%)	val. MSE gain 40→50 (%)	pick of 25 ≠ of 50 (%)
LightGBM `baseline`	35	0.1	36.8	5.22 / 4.42	1.66 / 0.99	0.50 / 0.00	72
LightGBM `all_features`	33	0.7	31.9	5.96 / 4.86	2.15 / 1.13	0.69 / 0.00	66
XGBoost `baseline`	34	0.3	36.5	5.22 / 4.36	1.95 / 1.04	0.66 / 0.00	70
XGBoost `all_features`	33	0.2	33.6	5.91 / 4.90	2.15 / 1.18	0.71 / 0.00	68
random forest `baseline`	32	1.0	26.8	2.69 / 1.95	0.62 / 0.28	0.16 / 0.00	63
random forest `all_features`	33	1.0	31.2	3.22 / 2.32	0.81 / 0.44	0.23 / 0.00	68

Table 35: The budget out of sample: best of 50 trials minus best of 10 and of 25, at TUNE_PER 1 and 25. Cells as in Table 32.

arm	TUNE_PER 1: 50 vs 10	TUNE_PER 1: 50 vs 25	TUNE_PER 25: 50 vs 10	TUNE_PER 25: 50 vs 25
	+0.3	−0.1	−0.6	+1.1
LightGBM `baseline`	+0.21	+0.49	+0.12	+0.30
	+0.4	+0.6	+1.1	+3.9
LightGBM `all_features`	−0.06	−0.02	+0.28	+0.70
	+1.3	+0.1	+0.4	−0.2
XGBoost `baseline`	+0.21	−0.06	+0.19	−0.05
	−1.4	+1.7	−3.6	+0.6
XGBoost `all_features`	−0.30	+0.07	−0.25	0.00
	+0.1	+0.2	−0.3	+0.2
random forest `baseline`	−0.15	+0.10	+0.16	+0.12
	+0.4	+0.6	+3.1	+1.4
random forest `all_features`	−0.80	+0.02	−0.56	+0.22

Table 36: How often to tune: Optuna (best of 50) with a new tuning point every 1, 5 and 25 sessions minus every 250 sessions, every path refit every session; the second column is the QLIKE at TUNE_PER 250. Cells as in Table 32.

arm	QLIKE TUNE_PER 250	TUNE_PER 1 vs 250	TUNE_PER 5 vs 250	TUNE_PER 25 vs 250
		−3.8	−4.9	−3.9
LightGBM `baseline`	0.1068	−0.13	−0.62	−0.26
		+2.8	+1.2	+6.1
LightGBM `all_features`	0.0983	−0.32	−0.59	+0.23
		−4.8	−5.9	−5.2
XGBoost `baseline`	0.1062	−0.42	−0.42	−0.55
		+0.2	−0.9	+2.4
XGBoost `all_features`	0.0991	−0.03	−0.08	−0.01
		−4.4	−4.5	−4.2
random forest `baseline`	0.1036	0.00	−0.24	−0.20
		−1.6	−1.6	+0.8
random forest `all_features`	0.1001	−0.35	−0.08	+0.08

Table 37: Optuna (best of 50, each TUNE_PER) minus the shipped trees refit every session (T1), the random search refit every session (RS1) and the per-bar ridge: the number of model × input-set arms whose difference (a point estimate) lies below or above zero, and in parentheses how many of them are significant at 5 % (Diebold–Mariano for QLIKE, the Newey–West t of the daily midpoint-return difference for the Sharpe ratio).

Optuna −	TUNE_PER	arms	QLIKE below 0 (DM)	QLIKE above 0 (DM)	Sharpe mid below 0 (HAC t)	Sharpe mid above 0 (HAC t)
T1	1	6	0 (0)	6 (3)	4 (0)	2 (0)
T1	5	6	0 (0)	6 (2)	4 (0)	2 (0)
T1	25	6	0 (0)	6 (4)	2 (0)	4 (0)
T1	250	6	0 (0)	6 (3)	1 (0)	5 (0)
all		24	0 (0)	24 (12)	11 (0)	13 (0)
RS1	1	6	4 (0)	2 (0)	4 (0)	2 (0)
RS1	5	6	5 (0)	1 (0)	5 (0)	1 (0)
RS1	25	6	4 (0)	2 (1)	3 (0)	3 (0)
RS1	250	6	3 (0)	3 (0)	4 (0)	2 (1)
all		24	16 (0)	8 (1)	16 (0)	8 (1)
per-bar ridge	1	6	0 (0)	6 (2)	6 (1)	0 (0)
per-bar ridge	5	6	0 (0)	6 (3)	6 (0)	0 (0)
per-bar ridge	25	6	0 (0)	6 (3)	5 (0)	1 (0)
per-bar ridge	250	6	0 (0)	6 (3)	5 (0)	1 (0)
all		24	0 (0)	24 (11)	22 (1)	2 (0)

Table 38: Where the best-of-50 picks sit: share of picks in the outermost twentieth of each axis at its low and high end, per input set (`baseline` / `all_features`), beside the share of the random start-up draws there (the prior), and the median pick. Each tree's axes are its own run's search space. †A pseudo-axis: the early-stopped number of rounds at the cap set by the learning rate.

axis	search space	picks at low bound (%) base / all	picks at high bound (%) base / all	prior low (%)	prior high (%)	median pick base / all
LightGBM						
`num_leaves`	[2, 128] log int	20.6 / 5.7	2.0 / 3.3	11.1	4.8	6 / 15
`min_child_samples`	[1, 256] log int	19.1 / 14.2	0.7 / 0.8	18.3	4.2	5 / 12
`feature_fraction`	[0.05, 1]	1.4 / 5.0	7.9 / 5.3	5.2	5.3	0.646 / 0.553
`bagging_fraction`	[0.3, 1]	5.2 / 4.1	5.0 / 4.6	4.7	4.9	0.638 / 0.666
`lambda_l2`	[0.001, 10000] log	5.4 / 5.6	0.0 / 0.0	5.5	5.1	0.633 / 0.637
`learning_rate`	[0.005, 0.1] log	3.1 / 3.7	6.3 / 4.7	4.9	5.2	0.0256 / 0.0203
`rounds_cap`†	early-stopped rounds ≤ cap(lr)	— / —	1.1 / 0.6	—	16.9–17.5	272 / 294
XGBoost						
`max_depth`	[1, 8] int	18.4 / 6.1	5.3 / 10.6	12.2	12.5	3 / 5
`min_child_weight`	[0.1, 256] log	6.3 / 4.4	0.7 / 0.8	5.3	4.8	2.24 / 5.88
`subsample`	[0.15, 1]	7.5 / 3.6	3.8 / 5.1	5.2	5.3	0.516 / 0.598
`colsample_bytree`	[0.05, 1]	0.8 / 4.3	8.3 / 5.6	4.7	4.9	0.691 / 0.569
`reg_lambda`	[1e-05, 10000] log	4.6 / 4.2	0.0 / 0.0	5.5	5.1	0.122 / 0.0899
`learning_rate`	[0.005, 0.1] log	3.9 / 5.6	4.2 / 3.9	4.9	5.2	0.024 / 0.0189
`rounds_cap`†	early-stopped rounds ≤ cap(lr)	— / —	1.6 / 1.1	—	14.5–15.6	298 / 316
random forest						
`max_features`	[0.02, 1]	0.1 / 1.2	6.4 / 10.3	5.1	4.8	0.612 / 0.452
`min_samples_leaf`	[1, 256] log int	31.7 / 19.1	0.0 / 0.0	17.7	4.4	2 / 4
`max_depth`	{3, 5, 10, 20, 40, None}	4.2 / 3.9	19.5 / 22.6	17.0	16.5	20 / 20

B.16.5 Reproducibility across CPUs

The tree numbers depend on the CPU's vector class. The executor builds the design matrix X with numpy's runtime-dispatched loops, and X comes out as one of two byte strings: one on every machine with AVX-512 and another on every machine without it (7 machines probed on two clusters). The two differ in 94 of 628 `all_features` columns, by at most 2.6×10^{-11} in absolute value; the target and the kept-column sets are the same on every machine. Within a class the Optuna trials and forecasts reproduce bit for bit across clusters (900/900 trials, 1,620/1,620 forecasts). Across classes the fits part: from the same stage-1 records the random forest's forecasts are identical in 360/360, XGBoost's in 344/360 (largest relative difference 0.033) and LightGBM's in 5/360 (0.103); re-running 20 tuning points on the other class changes LightGBM's best-of-50 pick at 18 of them (XGBoost 0, random forest 0), while a re-run on the same class is bit-identical at all 20. These checks predate XGBoost's and the random forest's current settings (Section 4.5): their rows here are the two trees as they ran before them.

So every canonical chunk ran on one CPU class: the census finds 1 class among the masked ladder's 291 chunks and 1 among Optuna's 214 chunk-arms (the masked ladder's 6 class-gate chunks ran on the other class on purpose). Optuna's first pass had landed on several classes; it is kept as the class-mixed run beside the one-class run: 18 of its 54 forecast tables differ, with 4 QLIKE differences below zero and 6 above (one class minus mixed; 0 and 0 significant) and 5 Sharpe differences below zero and 3 above (0 and 0 significant).

Table 39: CPU-class checks: what was compared and what reproduced. "Bit-identical a/b": a of b items equal bit for bit; relative differences are of the forecast; QLIKE and Sharpe differences (other class minus the fleet's class, in % of QLIKE and Sharpe units) as point estimates.

check	result
design X, machines of the two vector classes	`all_features`: 94 of 628 columns differ; largest absolute difference 2.6×10^{-11}; target and kept-column sets identical
Optuna, same class on two clusters	stage-1 trials bit-identical 900/900 (replicate 900/900); stage-2 forecasts 1,620/1,620
Optuna, across classes on one cluster (same inputs)	stage-1 trials bit-identical: LightGBM 100/300, XGBoost 211/300, random forest 283/300; stage-2 forecasts from the same records: 5/360 (max rel. 0.103), 344/360 (0.033), 360/360
Optuna, first 20 tuning points re-run on the other class	best-of-50 pick changes: LightGBM 18, XGBoost 0, random forest 0; XGBoost forecast max rel. 7.3×10^{-3}; same-class re-run bit-identical at 20 of 20 points for all three
Optuna, one class vs class-mixed first pass	18 of 54 tables differ (max rel. LightGBM 0.65, XGBoost 0.00, random forest 0.0025); QLIKE differences below / above 0: 4 / 6; Sharpe mid differences with a significant HAC t (5 %): 0

B.16.6 The master table before and after

The master table of the closing strategy was rebuilt from the new tables, on the same 866 days. Table 40 summarises the 73 forecasts present before and after, by family; 147 forecasts are new (the daily-refit and random-search-daily trees and the Optuna rungs). The own midpoint return of 3 of the forecasts present before and after moved with a significant HAC t. The 12 XGBoost and random-forest forecasts present before and after are left out: they now carry those trees' current settings, so their change mixes the settings with the campaign.

Table 40: The master table before and after the campaign, per family of the forecasts present in both: what changed, the QLIKE change in % and the sign(s) Sharpe change (midpoint), median and range (minimum to maximum) over the family, the number of Sharpe changes above / below zero and of those with a significant HAC t, and the trade positions changed (of 866 days).

family	keys	what changed	QLIKE change (%) median [min, max]	ΔSharpe mid median [min, max]	ΔSharpe mid above / below 0	HAC t significant (5%)	positions changed median [max]
paper	9	nothing	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]
per-bar linear	7	design de-dup	0.00 [0.00, +0.05]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]
pooled twin	6	nothing	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]
VIX-only family	15	design de-dup	0.00 [0.00, +0.06]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]
per-bar tree (untuned)	2	design de-dup + per-window mask + CPU-class pinning	+0.84 [+0.67, +1.01]	−0.21 [−0.46, +0.04]	1 / 1	1	34 [36]
per-bar tree (tuned)	4	design de-dup + per-window mask + CPU-class pinning	+2.35 [−0.43, +4.12]	−0.42 [−0.69, +0.13]	1 / 3	2	47 [65]
direct rest-of-day at 15:30 (check)	6	design de-dup + CPU-architecture pinning to the per-bar twin	0.00 [0.00, +0.01]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]
implied-vol representations	15	design de-dup	0.00 [0.00, +0.06]	0.00 [0.00, +0.02]	1 / 0	0	0 [1]
chain-period buckets	9	design de-dup	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0	0 [0]

B.16.7 Feature importance on the new design

The importance study of the 16:00 forecast (MDI, split count, permutation, TreeSHAP) was re-run on the new design with every tree fit masked, with the first pass's refits, held-out tails, permutation scheme and seeds. In the trees (Table 41) the `har_ma_*` ladder still leads 17 of 18 input-set × model × measure rows at the series level. The table gives its MDI, TreeSHAP and permutation shares before and after, beside the part of the old ladder's MDI that sat on the `har_ma_k_x_close` copies (column five). For XGBoost and the random forest the new side runs their current settings (Section 4.5) and the first pass their earlier ones, so their change mixes the settings with the design and the mask. The rank correlation of the series rankings, old against new, has median 0.76 over the tree rows (lowest 0.47). No interval is computed for these share changes; the yardstick is the ridge, which solves the same problem in both runs, and whose permutation importance of the ladder moves by at most 0.8 points of the tail loss between the two sets of permutation draws.

Table 41: The `har_ma_*` ladder's importance in the trees, first pass (old design, unmasked) to the new design with the mask: MDI and TreeSHAP shares, permutation (P2, QLIKE) in % of the model's tail loss, the part of the old ladder's MDI that sat on the `_x_close` copies, and the ladder's MDI rank among the series.

input set	model	MDI share (%) old → new	TreeSHAP share (%) old → new	perm. P2 QLIKE (% of tail loss)	old `_x_close` share of the ladder's MDI (%)	MDI rank old → new
`all_features`	LightGBM	57 → 52	47 → 44	+237 → +200	22	1 → 1
`all_features`	XGBoost	69 → 54	55 → 47	+224 → +181	22	1 → 1
`all_features`	random forest	62 → 59	58 → 61	+290 → +271	54	1 → 1

B.16.8 Compute used

Table 42 lists the compute each stage reported. Cluster A is the batch cluster that ran the tree fleets; cluster B the one that ran the linear and rest-of-day arms (SGE). Allocated CPU-hours count every core a job held for its whole run; used CPU-hours the busy share. The largest allocation is the masked Optuna run's (1422.6 allocated CPU-hours, at up to 900 CPUs at once).

Table 42: Compute used by the campaign's stages, as each stage recorded it (— = not reported). For cluster B, slot-hours are wall-clock hours of the job slots.

stage	cluster	allocated CPU-h	used CPU-h	peak CPUs	wall-clock (first start–last end)
trees T10 / T1 / RS10 / RS1, no mask	A	107.3	54.0	360	09-29 19:39–09-29 21:02 (1 h 23 min)
Optuna, no mask (stopped; not used)	A	928.3	—	1260	09-29 20:15–09-29 22:28 (2 h 13 min)
trees with the mask, one CPU class	A	115.2	57.3	408	09-29 22:09–09-30 00:50 (2 h 41 min)
Optuna with the mask, all jobs	A	1422.6	—	900	09-29 22:07–09-30 04:05 (5 h 58 min)
of which the one-class re-run	A	274.3	—	860	09-30 02:44–09-30 04:05 (1 h 21 min)
cross-cluster check (Optuna)	A	18.2	—	73	not reported
cross-cluster check (Optuna)	B	29.1 slot-h	6.7	58 slots	not reported
rest-of-day arms, all entry clocks	B	2.06 slot-h	1.58	not reported	21:47–21:59 (11.2 min)
attribution control	B	0.21 slot-h	0.18	not reported	22:35–22:50 (14.7 min)
feature importance re-run	A	10.8	7.1	45	21:50–22:11
dense-vs-sparse re-run	A	4.1	—	20	18 min of job time
its local part	local	—	4.6	4 processes	69 min

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